

LECTURE NOTES ON INTEGRABLE SYSTEMS

ZIXUAN DENG

ABSTRACT. These notes are based on the course *Introduction to Integrable Systems*, delivered by Professor Liming Ling during the Fall semester of 2025–2026.

The full L^AT_EX source code of these notes is available at: <https://codeberg.org/kill11kill/courseNotes>.

Week 1: Algebro-Geometric Solutions for the mKdV Equation (1)

2025-09-16

Objectives: Derive the mKdV equation from the associated spectral problem and construct the AKNS hierarchy by explicit recursion.

The focusing modified Korteweg–de Vries (mKdV) equation is given by

$$(1.1) \quad u_t + 6u^2u_x + u_{xxx} = 0, \quad u(x, t) \in \mathbb{R}, \quad (x, t) \in \mathbb{R}^2.$$

A standard method for establishing integrability is via a Lax pair. We introduce a matrix function $\Phi(x, t)$ satisfying

$$\Phi_x = U\Phi, \quad \Phi_t = V\Phi,$$

where U and V are 2×2 matrices depending on u and its derivatives. The compatibility condition

$$\Phi_{xt} = \Phi_{tx} \iff U_t - V_x + [U, V] = 0,$$

with $[U, V] = UV - VU$, is equivalent to the zero-curvature equation.

Following the AKNS formalism, the spatial part of the Lax pair is taken as

$$(1.2) \quad \Phi_x = U\Phi, \quad U = -i\lambda\sigma_3 + Q,$$

where

$$(1.3) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad \sigma_3 = \text{diag}(1, -1).$$

We introduce the transformation

$$(1.4) \quad \Phi(x; \lambda) = m(x; \lambda)e^{-i\lambda x\sigma_3}, \quad m(x; \lambda) \rightarrow \mathbb{I} \text{ as } \lambda \rightarrow \infty.$$

Substituting into the spectral problem (1.2) gives

$$(1.5) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i\lambda \sigma_3 + Q.$$

Define

$$(1.6) \quad \Psi = m \sigma_3 m^{-1}.$$

Differentiating Ψ and using (1.5), we compute

$$\begin{aligned} \Psi_x &= m_x \sigma_3 m^{-1} - m \sigma_3 m^{-1} m_x m^{-1} \\ &= [m_x m^{-1}, m \sigma_3 m^{-1}] \\ &= [m_x m^{-1} - i\lambda m \sigma_3 m^{-1}, m \sigma_3 m^{-1}] \\ &= [U, \Psi]. \end{aligned}$$

Thus Ψ satisfies

$$(1.7) \quad \Psi_x = [U, \Psi],$$

which is the stationary zero-curvature equation.

The matrix $m(x; \lambda)$ admits the asymptotic expansion

$$(1.8) \quad m(x; \lambda) = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + \dots.$$

From (1.6) and (1.8) we obtain the formal expansion

$$(1.9) \quad \Psi = \Psi_0 + \Psi_1\lambda^{-1} + \Psi_2\lambda^{-2} + \dots, \quad \Psi_0 = \sigma_3.$$

Substituting this into (1.7) yields

$$\begin{aligned} \sum_{j=1}^{\infty} \Psi_{j,x} \lambda^{-j} &= -i\lambda \left[\sigma_3, \sum_{j=1}^{\infty} \Psi_j \lambda^{-j} \right] + \left[Q, \sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right] \\ &= -i \sum_{j=0}^{\infty} \left[\sigma_3, \Psi_{j+1} \right] \lambda^{-j} + \sum_{j=0}^{\infty} \left[Q, \Psi_j \right] \lambda^{-j}. \end{aligned}$$

At order $O(1)$ we find

$$-i[\sigma_3, \Psi_1] + [Q, \sigma_3] = 0 \iff [i\Psi_1 + Q, \sigma_3] = 0,$$

which implies

$$\Psi_1^{\text{off}} = iQ.$$

For general $j \geq 1$, the coefficient of λ^{-j} gives

$$(1.10) \quad \Psi_{j,x} = -i[\sigma_3, \Psi_{j+1}] + [Q, \Psi_j] = -2i\sigma_3\Psi_{j+1}^{\text{off}} + [Q, \Psi_j].$$

Hence

$$(1.11) \quad \Psi_{j,x}^{\text{off}} = -2i\sigma_3\Psi_{j+1}^{\text{off}} + [Q, \Psi_j^{\text{diag}}] \iff \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3(\Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}]),$$

and

$$(1.12) \quad \Psi_{j,x}^{\text{diag}} = [Q, \Psi_j^{\text{off}}] \iff \Psi_j^{\text{diag}} = \partial_x^{-1}([Q, \Psi_j^{\text{off}}]).$$

Thus, the recursion relation (1.11) provides a direct way to compute the off-diagonal coefficients algebraically in terms of lower-order terms, while the diagonal recursion (1.12) requires performing an integration. In practice, this makes the off-diagonal part more convenient to obtain explicitly, whereas the diagonal part is generally harder to compute in closed form.

By (1.6), the additional constraint $\Psi^2 = \mathbb{I}$ provides algebraic relations among the coefficients. Substituting the expansion (1.9) yields

$$\mathbb{I} = \left(\sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right)^2 = \sum_{j=0}^{\infty} \left(\sum_{k=0}^j \Psi_k \Psi_{j-k} \right) \lambda^{-j},$$

so that

$$(1.13) \quad \sum_{k=0}^j \Psi_k \Psi_{j-k} = 0, \quad j \geq 1.$$

Separating diagonal and off-diagonal parts, for $j \geq 1$ we obtain

$$(1.14) \quad \sum_{k=0}^j (\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}}) = 0,$$

$$(1.15) \quad \sum_{k=0}^j (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Since $\Psi_0 = \sigma_3$, the off-diagonal constraint (1.15) for $j \geq 2$ takes the form

$$\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 + \sum_{k=1}^{j-1} (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Noting that $\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 = 0$, this simplifies to

$$(1.16) \quad \sum_{k=1}^{j-1} \left(\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}} \right) = 0.$$

It should be noted that from the off-diagonal constraint (1.15) one can only deduce the reduced form (1.16), which merely expresses a compatibility condition between lower-order coefficients. In particular, this relation does not provide a practical means to compute the diagonal parts explicitly.

For the diagonal part (1.14) we have

$$\sigma_3 \Psi_j^{\text{diag}} + \Psi_j^{\text{diag}} \sigma_3 + \sum_{k=1}^{j-1} \left(\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}} \right) = 0, \quad j \geq 2.$$

Since $\sigma_3 \Psi_j^{\text{diag}} = \Psi_j^{\text{diag}} \sigma_3$, this simplifies to

$$(1.17) \quad \Psi_j^{\text{diag}} = -\frac{1}{2} \sigma_3 \sum_{k=1}^{j-1} \left(\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}} \right), \quad j \geq 2.$$

We will use this reduced relation (1.17) to compute the diagonal part Ψ_j^{diag} recursively at each order. This provides an explicit algebraic formula for the diagonal coefficients in terms of lower-order diagonal and off-diagonal terms.

For $j = 1$, the condition (1.13) reads

$$\sigma_3 \Psi_1 + \Psi_1 \sigma_3 = 2\sigma_3 \Psi_1^{\text{diag}} = 0,$$

so that $\Psi_1 = \Psi_1^{\text{off}} = iQ$.

Using (1.11) and (1.17) recursively, we find

$$\Psi_2^{\text{off}} = -\frac{1}{2} \sigma_3 Q_x, \quad \Psi_2^{\text{diag}} = \frac{1}{2} \sigma_3 Q^2,$$

and hence

$$\Psi_2 = \frac{1}{2} \sigma_3 (Q^2 - Q_x).$$

For the next coefficient, we compute

$$\begin{aligned} \Psi_3^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{2,x}^{\text{off}} - [Q, \Psi_2^{\text{diag}}]) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{1}{2} \sigma_3 Q_{xx} - [Q, \frac{1}{2} \sigma_3 Q^2] \right) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{1}{2} \sigma_3 Q_{xx} - \frac{1}{2} (Q \sigma_3 Q^2 - \sigma_3 Q^2 Q) \right) \\ &= -\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3. \end{aligned}$$

For the diagonal part,

$$\begin{aligned} \Psi_3^{\text{diag}} &= -\frac{1}{2} \sigma_3 \left(\Psi_1^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{off}} \Psi_1^{\text{off}} + \Psi_1^{\text{diag}} \Psi_2^{\text{diag}} + \Psi_2^{\text{diag}} \Psi_1^{\text{diag}} \right) \\ &= -\frac{1}{2} \sigma_3 \left((iQ) \left(-\frac{1}{2} \sigma_3 Q_x \right) + \left(-\frac{1}{2} \sigma_3 Q_x \right) (iQ) \right) \\ &= -\frac{1}{2} \sigma_3 \left(-\frac{i}{2} (Q \sigma_3 Q_x + \sigma_3 Q_x Q) \right) \\ &= \frac{i}{4} (Q_x Q - Q Q_x) = 0. \end{aligned}$$

Proceeding to Ψ_4 , the off-diagonal recursion (1.11) with $\Psi_3^{\text{diag}} = 0$ gives

$$\begin{aligned} \Psi_4^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{3,x}^{\text{off}} - [Q, \Psi_3^{\text{diag}}]) = \frac{i}{2} \sigma_3 \Psi_{3,x}^{\text{off}} \\ &= \frac{i}{2} \sigma_3 \partial_x \left(-\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3 \right) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{i}{4} Q_{xxx} + \frac{i}{2} (Q^3)_x \right) \\ &= \frac{1}{8} \sigma_3 Q_{xxx} - \frac{1}{4} \sigma_3 (Q^3)_x. \end{aligned}$$

Since $Q_x Q = Q Q_x$ for the present Q given in (1.3), we have

$$(Q^3)_x = Q_x Q^2 + Q Q_x Q + Q^2 Q_x = 3Q_x Q^2,$$

and therefore

$$\Psi_4^{\text{off}} = \frac{1}{8}\sigma_3 Q_{xxx} - \frac{3}{4}\sigma_3 Q_x Q^2.$$

For the diagonal part, only the nonvanishing combinations contribute; terms involving Ψ_1^{diag} or Ψ_3^{diag} are zero and omitted. Hence

$$\begin{aligned} \Psi_4^{\text{diag}} &= -\frac{1}{2}\sigma_3 \left(\Psi_1^{\text{off}} \Psi_3^{\text{off}} + \Psi_3^{\text{off}} \Psi_1^{\text{off}} + \Psi_2^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{diag}} \Psi_2^{\text{diag}} \right) \\ &= -\frac{1}{2}\sigma_3 \left((iQ) \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right) + \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right) (iQ) \right. \\ &\quad \left. + \left(-\frac{1}{2}\sigma_3 Q_x \right)^2 + \left(\frac{1}{2}\sigma_3 Q^2 \right)^2 \right) \\ &= -\frac{1}{8}\sigma_3 (Q Q_{xx} + Q_{xx} Q - 3Q^4 - Q_x^2). \end{aligned}$$

Since $(-i\lambda\Psi)_+ = -i\lambda\sigma_3 + Q$, we obtain

$$\Phi_x \Phi^{-1} = U = (-i\lambda\Psi)_+.$$

Here the subscript $(\cdot)_+$ denotes the projection onto the nonnegative powers of λ in the Laurent expansion. This motivates the definition of the AKNS hierarchy by prescribing the time evolution

$$(1.18) \quad \Phi_{t_n} = (-i\lambda^n \Psi)_+ \Phi \equiv V_n \Phi.$$

The first few members of the hierarchy are given by

$$\begin{aligned} n = 2 &\Rightarrow \text{nonlinear Schrödinger (NLS) flow,} \\ n = 3 &\Rightarrow \text{complex mKdV flow,} \Rightarrow \text{mKdV flow.} \end{aligned}$$

In terms of the expansion coefficients Ψ_j , the operators V_n take the form

$$\begin{aligned} V_2 &= -i(\lambda^2 \Psi_0 + \lambda \Psi_1 + \Psi_2), \\ V_3 &= -i(\lambda^3 \Psi_0 + \lambda^2 \Psi_1 + \lambda \Psi_2 + \Psi_3). \end{aligned}$$

The zero-curvature equation $U_t - V_x + [U, V] = 0$ with $V_{\text{mKdV}} = 4V_3$ gives, by collecting the constant terms in λ (i.e., at order $O(1)$),

$$Q_t + 4i\Psi_{3,x} + [Q, -4i\Psi_3] = 0.$$

Taking the off-diagonal part gives $Q_t + 4i\Psi_{3,x}^{\text{off}} = 0$, that is,

$$(1.19) \quad Q_t + 4i \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right)_x = Q_t + Q_{xxx} - 2(Q^3)_x = 0.$$

Since

$$Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad Q^2 = -u^2 \mathbb{I},$$

this reduces to the scalar equation

$$u_t + u_{xxx} + 2(u^3)_x = 0,$$

which is precisely the mKdV equation (1.1).

Supplementary computations:

Using the definitions (1.4) and (1.6) we have

$$(1.20) \quad \Psi = \Phi \sigma_3 \Phi^{-1}.$$

Differentiating Ψ with respect to t_n and using (1.18) gives

$$\begin{aligned} \Psi_{t_n} &= \Phi_{t_n} \sigma_3 \Phi^{-1} - \Phi \sigma_3 \Phi^{-1} \Phi_{t_n} \Phi^{-1} \\ &= [\Phi_{t_n} \Phi^{-1}, \Psi] = [V_n, \Psi]. \end{aligned}$$

Thus

$$(1.21) \quad \Psi_{t_n} = [V_n, \Psi].$$

Next we simplify the zero-curvature equation for the pair (U, V_n) . The zero-curvature equation is

$$U_{t_n} - (V_n)_x + [U, V_n] = 0,$$

and since $U = -i\lambda\sigma_3 + Q$ we have $U_{t_n} = Q_{t_n}$. With the normalization

$$V_n = \alpha_n (-i\lambda^n \Psi)_+$$

we obtain

$$Q_{t_n} = (V_n)_x - [U, V_n] = \alpha_n (-i\lambda^n \Psi_x)_+ - [U, \alpha_n (-i\lambda^n \Psi)_+].$$

Pulling out constants and using $\Psi_x = [U, \Psi]$ gives

$$(1.22) \quad Q_{t_n} = -i\alpha_n \left((\lambda^n [U, \Psi])_+ - [U, (\lambda^n \Psi)_+] \right).$$

Write $U = -i\lambda\sigma_3 + Q$ and split the first term in (1.22):

$$\lambda^n [U, \Psi] = -i\lambda^{n+1} [\sigma_3, \Psi] + \lambda^n [Q, \Psi].$$

Because Q does not depend on λ we have

$$(\lambda^n [Q, \Psi])_+ = [Q, (\lambda^n \Psi)_+],$$

so the Q -commutator contributions cancel in (1.22). The remaining terms therefore come only from the σ_3 -commutator:

$$\begin{aligned} Q_{t_n} &= -i\alpha_n \left(-i(\lambda^{n+1} [\sigma_3, \Psi])_+ + i\lambda [\sigma_3, (\lambda^n \Psi)_+] \right) \\ &= \alpha_n \left(\lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ \right). \end{aligned}$$

To simplify the difference of finite sums, use the expansions

$$\Psi = \sum_{j \geq 0} \Psi_j \lambda^{-j}, \quad (\lambda^n \Psi)_+ = \sum_{j=0}^n \lambda^{n-j} \Psi_j.$$

A direct comparison of coefficients shows

$$\begin{aligned} \lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ &= \sum_{j=0}^n \lambda^{n+1-j} [\sigma_3, \Psi_j] - \sum_{j=0}^{n+1} \lambda^{n+1-j} [\sigma_3, \Psi_j] \\ &= -\lambda^{n+1-(n+1)} [\sigma_3, \Psi_{n+1}] \\ &= -[\sigma_3, \Psi_{n+1}]. \end{aligned}$$

Hence the evolution of Q is

$$(1.23) \quad Q_{t_n} = -\alpha_n [\sigma_3, \Psi_{n+1}].$$

Since diagonal parts commute with σ_3 and the commutator doubles the off-diagonal part, we may equivalently write

$$(1.24) \quad Q_{t_n} = -2\alpha_n \sigma_3 \Psi_{n+1}^{\text{off}}.$$

Objectives: Construct the L matrix for mKdV, establish its symmetries and conserved determinant, and relate the eigen-data to the associated hyperelliptic curve and Riemann–Theta functions.

Recall the AKNS spectral problem (1974)

$$(2.1) \quad \Phi_x = U(\lambda; Q) \Phi, \quad U(\lambda; Q) = -i\lambda\sigma_3 + Q,$$

where $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and Q is off-diagonal. In what follows we set

$$(2.2) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}.$$

Introduce the gauge

$$(2.3) \quad \Phi = m(x; \lambda) e^{-i\lambda x \sigma_3}, \quad m = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + O(\lambda^{-3}),$$

$$(2.4) \quad \Psi := m \sigma_3 m^{-1} = \Psi_0 + \Psi_1 \lambda^{-1} + \Psi_2 \lambda^{-2} + \dots,$$

so that $\Psi^2 = \mathbb{I}$ and Ψ satisfies the stationary zero-curvature equation

$$(2.5) \quad \Psi_x = [U, \Psi].$$

Decompose a matrix A into diagonal and off-diagonal parts $A = A^{\text{diag}} + A^{\text{off}}$. From (2.5) and $\Psi^2 = \mathbb{I}$ one obtains the recursion

$$(2.6) \quad \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3 \left(\Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}] \right), \quad j \geq 0,$$

$$(2.7) \quad \Psi_{j+1}^{\text{diag}} = -\frac{1}{2}\sigma_3 \sum_{k=1}^j \left(\Psi_k^{\text{diag}} \Psi_{j-k+1}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k+1}^{\text{off}} \right), \quad j \geq 1.$$

Hence each Ψ_j is a differential polynomial in Q and its x -derivatives.

One checks that

$$(2.8) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i(\lambda \Psi)_+.$$

Let $\Phi(x, t_n; \lambda) = m(x, t_n; \lambda) e^{-i(\lambda x + \lambda^n t_n) \sigma_3}$, then

$$(2.9) \quad \Phi_{t_n} \Phi^{-1} = m_{t_n} m^{-1} - i\lambda^n m \sigma_3 m^{-1} = -i(\lambda^n \Psi)_+ \equiv V_n,$$

so that $V_n = -i(\lambda^n \Psi)_+$ and

$$(2.10) \quad \Psi_{t_n} = [V_n, \Psi].$$

The compatibility $\Phi_{xt_n} = \Phi_{t_n x}$ is equivalent to the zero curvature equation

$$(2.11) \quad U_{t_n} - V_{n,x} + [U, V_n] = 0.$$

Definition 1. Let $g \geq 0$ be a fixed integer, and let $\{\alpha_j\}_{j=0}^{g+1}$ be constants. Define

$$(2.12) \quad L := -i \sum_{j=0}^{g+1} \alpha_j (\lambda^j \Psi(x, t, \lambda))_+,$$

where $(\cdot)_+$ denotes the projection onto nonnegative powers of λ .

It is convenient to rewrite L by expanding the nonnegative part explicitly. Using $\Psi = \sum_{k \geq 0} \Psi_k \lambda^{-k}$ we have

$$(\lambda^j \Psi)_+ = \sum_{k=0}^j \Psi_k \lambda^{j-k},$$

hence

$$(2.13) \quad L(\lambda) = -i \sum_{j=0}^{g+1} \alpha_j \sum_{k=0}^j \Psi_k \lambda^{j-k} = -i \sum_{l=0}^{g+1} \left(\sum_{j=l}^{g+1} \alpha_j \Psi_{j-l} \right) \lambda^l,$$

where we set $l = j - k$ to obtain the last form.

Proposition 1. Suppose L as defined in (2.12) satisfies the stationary zero-curvature equations

$$(2.14) \quad L_x = [U, L], \quad L_{t_n} = [V_n, L].$$

Then the following additional constraint holds:

$$(2.15) \quad \sum_{j=1}^{g+1} \alpha_j \Psi_{j+1}^{\text{off}}(x, t_n) + i \alpha_0 Q = 0.$$

Proof. Set

$$(2.16) \quad \widehat{L}(\lambda) := -i \left(\sum_{j=0}^{g+1} \alpha_j \lambda^j \right) \Psi(\lambda).$$

By construction $\widehat{L}_x = [U, \widehat{L}]$. Expand $\Psi = \sum_{k \geq 0} \Psi_k \lambda^{-k}$ and substitute into $\widehat{L}$:

$$(2.17) \quad \widehat{L}(\lambda) = -i \sum_{j=0}^{g+1} \sum_{k \geq 0} \alpha_j \Psi_k \lambda^{j-k} = -i \sum_{l=-\infty}^{g+1} \left(\sum_{j=0}^{g+1} \alpha_j \Psi_{j-l} \right) \lambda^l,$$

where we set $l = j - k$. Decompose $\widehat{L} = \widehat{L}_+ + \widehat{L}_-$ into nonnegative and negative powers of λ . From $\widehat{L}_x = [U, \widehat{L}]$ we obtain

$$\left(\widehat{L}_+ + \widehat{L}_- \right)_x = [U, \widehat{L}_+ + \widehat{L}_-].$$

After extracting nonnegative powers of λ , we have

$$(\widehat{L}_+)_x - [U, \widehat{L}_+] = \left([-i\lambda\sigma_3, \widehat{L}_-] \right)_+ = -i \left[\sigma_3, \sum_{j=0}^{g+1} \alpha_j \Psi_{j+1} \right] = -2i\sigma_3 \sum_{j=0}^{g+1} \alpha_j \Psi_{j+1}^{\text{off}} = 0,$$

which yields the constraint (2.15).

Next, to ensure compatibility, we require to prove $L_{xt_n} = L_{t_n x}$. Observe that

$$\begin{aligned} L_{xt_n} &= [U_{t_n}, L] + [U, [V_n, L]], \\ L_{t_n x} &= [V_{n,x}, L] + [V_n, [U, L]]. \end{aligned}$$

Applying the Jacobi identity,

$$[[A, B], C] + [[C, A], B] + [[B, C], A] = 0,$$

we find

$$[U, [V_n, L]] - [V_n, [U, L]] = [[L, V_n], U] + [[U, L], V_n] = [[U, V_n], L].$$

Therefore, it follows from the zero-curvature condition (2.11) that

$$L_{xt_n} - L_{t_n x} = [U_{t_n} - V_{n,x} + [U, V_n], L] = 0.$$

□

Specializing to genus $g = 2$, define

$$(2.18) \quad L(x, t; \lambda) = \frac{1}{4} V(\lambda; Q) + \alpha_1 U(\lambda; Q),$$

$$(2.19) \quad V(\lambda; Q) = 4\lambda^2 U(\lambda; Q) + 2i\lambda\sigma_3(Q_x - Q^2) - (Q_{xx} - 2Q^3).$$

The zero curvature condition for the mKdV equation is

$$U_t - V_x + [U, V] = 0,$$

which yields the scalar modified KdV equation

$$(2.20) \quad u_t + u_{xxx} + 6u^2 u_x = 0, \quad u(x, t) \in \mathbb{R}.$$

The matrices U and V satisfy the following symmetries:

$$(2.21) \quad \begin{aligned} U^\dagger(\lambda^*; Q) &= -U(\lambda; Q), & U^\top(-\lambda; Q) &= -U(\lambda; Q), \\ V^\dagger(\lambda^*; Q) &= -V(\lambda; Q), & V^\top(-\lambda; Q) &= -V(\lambda; Q). \end{aligned}$$

As a consequence the matrix L inherits analogous symmetries:

$$(2.22) \quad L^\dagger(x, t; \lambda^*) = -L(x, t; \lambda), \quad L^\top(x, t; -\lambda) = -L(x, t; \lambda).$$

Proposition 2. *The determinant of L is independent of x and t :*

$$\partial_x \det L = \partial_t \det L = 0.$$

Proof. By Jacobi's formula (or Abel's theorem applying to $L_x = (L_x L^{-1})L$), for any differentiable matrix $L(x, t; \lambda)$

$$\partial_x \det L = \det L \operatorname{tr}(L_x L^{-1}).$$

Using the stationary zero-curvature equation $L_x = [U, L] = UL - LU$ we obtain

$$\partial_x \det L = \det L \operatorname{tr}((UL - LU)L^{-1}) = \det L (\operatorname{tr} U - \operatorname{tr}(L^{-1}LU)) = 0,$$

since $\operatorname{tr}(LUL^{-1}) = \operatorname{tr}(L^{-1}LU)$. The identical argument with $\partial_t L = [V, L]$ yields $\partial_t \det L = 0$. Hence $\det L$ does not depend on x or t . $\square$

Therefore the determinant of L is constant in x and t . Write its factorization as

$$(2.23) \quad \det L(\lambda) = \prod_{j=1}^6 (\lambda - \hat{\lambda}_j).$$

By the symmetry relations (2.22) each root $\hat{\lambda}_j$ forces $\hat{\lambda}_j^*$ and $-\hat{\lambda}_j$ to be roots as well. Two possible root structures follow.

Case 1. Three purely imaginary pairs:

$$(2.24) \quad \det L(\lambda) = (\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2), \quad \lambda_j \in i\mathbb{R}.$$

Case 2. One nonreal quadruple together with one purely imaginary pair:

$$(2.25) \quad \det L(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_1^*)(\lambda + \lambda_1)(\lambda + \lambda_1^*)(\lambda - \lambda_2)(\lambda + \lambda_2),$$

with $\lambda_1 \in \mathbb{C} \setminus (\mathbb{R} \cup i\mathbb{R})$ and $\lambda_2 \in i\mathbb{R}$.

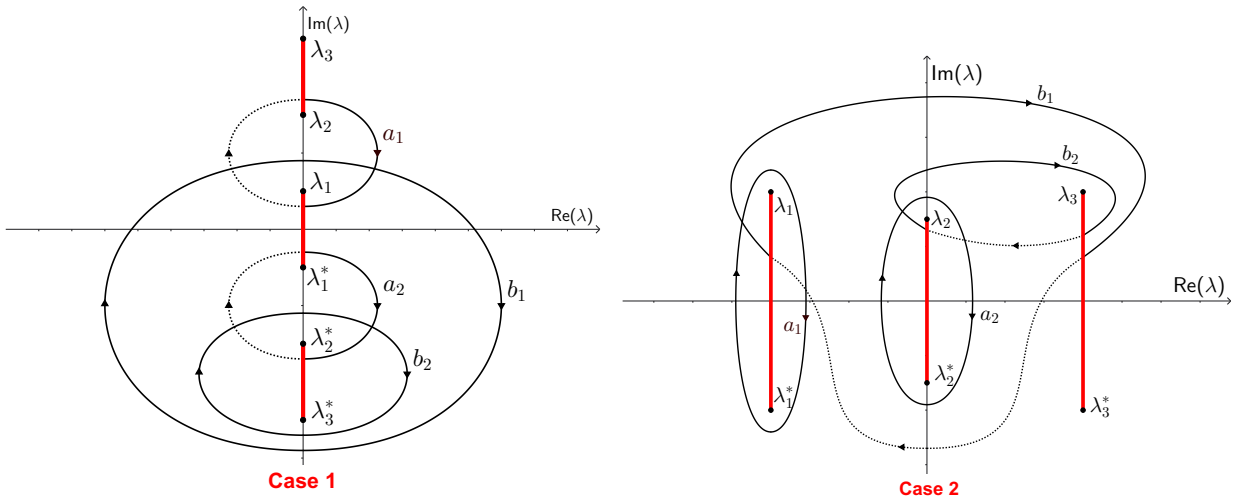


FIGURE 1. The homology basis for the curve $y^2 = \prod_{i=1}^3 (\lambda^2 - \lambda_i^2)$.

The matrix elements of L take the explicit form

$$\begin{aligned}
(2.26) \quad L_{11} &= -i\lambda^3 - i\lambda\left(\alpha_1 - \frac{1}{2}u^2\right), \quad L_{22} = -L_{11}, \\
L_{12} &= u\left(\lambda^2 + i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2\right) = u(\lambda - \mu_1)(\lambda - \mu_2), \\
L_{21} &= -u\left(\lambda^2 - i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2\right) = -u(\lambda - \mu_1^*)(\lambda - \mu_2^*),
\end{aligned}$$

where μ_1, μ_2 denote the (in general complex) roots of the quadratic factor in L_{12}/u . The conjugation in L_{21} is consistent with the reality of u and the symmetry relations (2.22).

Recall that

$$\begin{cases} \Phi_x = U\Phi, \\ \Phi_t = V\Phi, \end{cases} \quad \begin{cases} L_x = [U, L], \\ L_t = [V, L], \end{cases}$$

It follows that $L\Phi$ also satisfies the same Lax pair, since

$$\begin{aligned}
(L\Phi)_x &= L_x\Phi + L\Phi_x = (UL - LU)\Phi + LU\Phi = U(L\Phi), \\
(L\Phi)_t &= V(L\Phi).
\end{aligned}$$

This observation is also a direct consequence of the identity $L = \Psi = \Phi\sigma_3\Phi^{-1}$ which implies $L\Phi = \Phi\sigma_3$.

Assume $\Phi(x, t; \lambda)$ is normalized by $\Phi(0, 0; \lambda) = \mathbb{I}$. By uniqueness of solutions of the linear ODEs we obtain

$$(2.27) \quad L(x, t; \lambda) \Phi(x, t; \lambda) = \Phi(x, t; \lambda) L(0, 0; \lambda).$$

In particular, evaluation at $(x, t) = (0, 0)$ yields $L(0, 0; \lambda)$. So knowledge of $L(0, 0; \lambda)$ determines the action of L on the columns of Φ for all (x, t) .

Set

$$(2.28) \quad \det(L(x, t; \lambda)) = \det(L(0, 0; \lambda)) = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2) = y^2, \quad y = \pm \sqrt{\prod_{j=1}^3 (\lambda^2 - \lambda_j^2)}.$$

A short calculation shows that $\pm iy$ are eigenvalues of $L(0, 0; \lambda)$:

$$\det(\pm iy \mathbb{I} - L(0, 0; \lambda)) = -y^2 - L_{11}^2 - L_{12}L_{21} = -y^2 + \det(L(0, 0; \lambda)) = 0,$$

where L_{ij} denotes the (i, j) -entry of $L(0, 0; \lambda)$.

Hence there exists an invertible matrix M , independent of x and t , such that

$$(2.29) \quad L(0, 0; \lambda) = M \begin{pmatrix} iy & 0 \\ 0 & -iy \end{pmatrix} M^{-1}.$$

Combining (2.27) and (2.29) yields

$$L(x, t; \lambda) \Phi(x, t; \lambda) M = \Phi(x, t; \lambda) M \begin{pmatrix} iy & 0 \\ 0 & -iy \end{pmatrix}.$$

Write the two columns of $\Phi(x, t; \lambda)M$ as

$$\Phi(x, t; \lambda)M = (\Phi_1, \Phi_2),$$

so that

$$L(x, t; \lambda)\Phi_1 = iy\Phi_1, \quad L(x, t; \lambda)\Phi_2 = -iy\Phi_2.$$

Equivalently,

$$(L - iy)\Phi_1 = 0, \quad (L + iy)\Phi_2 = 0,$$

and therefore

$$\ker(L \mp iy) = \text{span}\{\Phi_{1,2}\}.$$

Since each eigenvector may be rescaled, we may write

$$\Phi_1 = \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \Phi_{11} \begin{pmatrix} 1 \\ r_1 \end{pmatrix}, \quad \Phi_2 = \Phi_{12} \begin{pmatrix} 1 \\ r_2 \end{pmatrix},$$

so

$$\ker(L \mp iy) = \text{span} \begin{pmatrix} 1 \\ r_{1,2} \end{pmatrix}.$$

Writing the eigenvalue equation explicitly gives

$$\begin{pmatrix} L_{11} \mp iy & L_{12} \\ L_{21} & -L_{11} \mp iy \end{pmatrix} \begin{pmatrix} 1 \\ r_{1,2} \end{pmatrix} = 0,$$

from which one obtains the two equivalent expressions for the ratio $r_{1,2}$:

$$(2.30) \quad r_{1,2} = \frac{\pm iy - L_{11}}{L_{12}} = \frac{L_{21}}{L_{11} \pm iy}.$$

The point $P = (\lambda, y)$ lies on the hyperelliptic curve (2.28), which is a compact Riemann surface of genus $g = 2$ with punctures $\infty^\pm$. Here $P \rightarrow \infty^\pm \Leftrightarrow \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3$. Denote

$$\begin{aligned} \mu &= \{\hat{\mu}_1, \hat{\mu}_2\}, & \hat{\mu}_j &= (\mu_j, iL_{11}(\mu_j)), & j &= 1, 2, \\ \mu^* &= \{\hat{\mu}_1^*, \hat{\mu}_2^*\}, & \hat{\mu}_j^* &= (\mu_j^*, -iL_{11}(\mu_j^*)), & j &= 1, 2. \end{aligned}$$

The divisor of r_1 is then given by

$$\text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda).$$

That is, $r_1(x, t; P)$ has zeros at the points of μ^* and ∞^- , and poles at the points of μ and ∞^+ . This description of the divisor is central in constructing Baker–Akhiezer functions on the hyperelliptic Riemann surface defined by

$$y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2).$$

Definition 2 (Riemann–Theta function). *Let R_g be a compact Riemann surface of genus g . Fix a canonical homology basis and let B denote the associated Riemann period matrix, which is symmetric with negative definite real part. The Riemann–Theta function is the entire function*

$$\Theta(B; z) = \sum_{n \in \mathbb{Z}^g} \exp \left\{ \frac{1}{2} n^\top B n + n^\top z \right\},$$

where

$$n = \begin{pmatrix} n_1 \\ \vdots \\ n_g \end{pmatrix}, \quad n_j \in \mathbb{Z}, \quad z = \begin{pmatrix} z_1 \\ \vdots \\ z_g \end{pmatrix}, \quad z_j \in \mathbb{C}.$$

Proposition 3. *The Riemann–Theta function satisfies:*

(1) *Periodicity in the standard lattice directions:*

$$\Theta(z + 2\pi i e_k) = \Theta(z), \quad k = 1, \dots, g.$$

(2) *Quasi-periodicity in the period matrix directions:*

$$\Theta(z + B e_k) = \Theta(z) e^{-z_k - B_{kk}/2}, \quad k = 1, \dots, g.$$

(3) *Evenness:*

$$\Theta(-z) = \Theta(z).$$

Proof. (1) For $z \mapsto z + 2\pi i e_k$, the exponential in the definition acquires a factor

$$\exp(n^\top (2\pi i e_k)) = \exp(2\pi i n_k) = 1,$$

since $n_k \in \mathbb{Z}$. Thus Θ is periodic in this direction.

(2) For $z \mapsto z + B e_k$, we compute

$$\begin{aligned}\Theta(z + B e_k) &= \sum_{n \in \mathbb{Z}^g} \exp\left\{\frac{1}{2} n^\top B n + n^\top (z + B e_k)\right\} \\ &= \sum_{n \in \mathbb{Z}^g} \exp\left\{\frac{1}{2} n^\top B n + n^\top z + n^\top B e_k\right\}.\end{aligned}$$

Write $m = n + e_k$. Then $n = m - e_k$, and the exponent becomes

$$\begin{aligned}&\frac{1}{2}(m - e_k)^\top B(m - e_k) + (m - e_k)^\top z + (m - e_k)^\top B e_k \\ &= \frac{1}{2}(m^\top B m - m^\top B e_k - e_k^\top B m + e_k^\top B e_k) + m^\top z - e_k^\top z + m^\top B e_k - e_k^\top B e_k \\ &= \frac{1}{2}m^\top B m + m^\top z - \frac{1}{2}(m^\top B e_k + e_k^\top B m) + m^\top B e_k - \frac{1}{2}e_k^\top B e_k - e_k^\top z.\end{aligned}$$

Since $B = B^\top$, we have $m^\top B e_k = e_k^\top B m$; moreover, $e_k^\top z = z_k$ and $e_k^\top B e_k = B_{kk}$. Hence

$$\frac{1}{2}(m - e_k)^\top B(m - e_k) + (m - e_k)^\top z + (m - e_k)^\top B e_k = \frac{1}{2}m^\top B m + m^\top z - z_k - \frac{1}{2}B_{kk}.$$

Therefore,

$$\Theta(z + B e_k) = e^{-z_k - B_{kk}/2} \Theta(z).$$

(3) Finally, since the defining series is invariant under $n \mapsto -n$, it follows that

$$\Theta(-z) = \Theta(z).$$

□

Supplementary computations:

Since

$$U(\lambda; Q) = -i\lambda\sigma_3 + Q = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix},$$

we have

$$Q^2 = -u^2 \mathbb{I}, \quad Q^3 = -u^2 Q.$$

Compute each term in

$$V(\lambda; Q) = 4\lambda^2 U + 2i\lambda\sigma_3(Q_x - Q^2) - (Q_{xx} - 2Q^3).$$

First,

$$4\lambda^2 U = \begin{pmatrix} -4i\lambda^3 & 4\lambda^2 u \\ -4\lambda^2 u & 4i\lambda^3 \end{pmatrix}, \quad Q_x - Q^2 = \begin{pmatrix} u^2 & u_x \\ -u_x & u^2 \end{pmatrix}.$$

Hence

$$2i\lambda\sigma_3(Q_x - Q^2) = \begin{pmatrix} 2i\lambda u^2 & 2i\lambda u_x \\ 2i\lambda u_x & -2i\lambda u^2 \end{pmatrix}.$$

Since $Q_{xx} - 2Q^3 = Q_{xx} + 2u^2 Q$,

$$-(Q_{xx} - 2Q^3) = \begin{pmatrix} 0 & -(u_{xx} + 2u^3) \\ u_{xx} + 2u^3 & 0 \end{pmatrix}.$$

Adding all contributions gives

$$V(\lambda; Q) = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix},$$

where

$$(2.31) \quad V_{11} = -4i\lambda^3 + 2i\lambda u^2, \quad V_{22} = -V_{11},$$

$$(2.32) \quad V_{12} = 4\lambda^2 u + 2i\lambda u_x - (u_{xx} + 2u^3),$$

$$(2.33) \quad V_{21} = -4\lambda^2 u + 2i\lambda u_x + (u_{xx} + 2u^3).$$

Week 3: Algebro-Geometric Solutions for the mKdV Equation (3)

2025-09-28

Objectives: Construct algebro-geometric mKdV solutions on a genus-two curve by expressing r_1 and u in Riemann theta functions and matching their asymptotics at $\infty^\pm$, poles and zeroes.

The Riemann–Roch theorem provides a fundamental relation between the divisor of a meromorphic function and the corresponding Abel map data on a Riemann surface. In the present setting, for the meromorphic function $r_1(x, t; P)$ we have

$$(3.1) \quad \text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda),$$

which identifies the divisor of r_1 as the difference between two effective divisors supported at the points $\infty^\pm$ and at the divisor of spectral data μ, μ^* .

A concrete representation of the function r_1 is given in terms of Riemann theta functions:

$$(3.2) \quad r_1(x, t; P) = \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu^*}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_\mu))} \exp(\Omega_1(P) - \ln(\omega_1)), \quad P = (\lambda, y).$$

Here Θ denotes the Riemann theta function, $\mathcal{C}$ is the vector of Riemann constants, $\mathcal{A}_{P_0}$ is the Abel map based at P_0 , and α_{P_0} is its linear extension to divisors. The exponential factor is determined by an Abelian integral Ω_1 , while $\hat{r}_1(x, t)$ is a normalizing prefactor independent of λ .

The underlying algebraic curve R_2 associated with this construction is the genus-two hyperelliptic Riemann surface

$$(3.3) \quad R_2 := \{(\lambda, y) \mid y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)\}, \quad P \rightarrow \infty^\pm \Leftrightarrow \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3 + O(\lambda).$$

Thus each point $P \in R_2$ is represented by (λ, y) , where

$$y = \pm \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

To construct the Jacobi variety one chooses a canonical homology basis $\{a_1, a_2; b_1, b_2\}$ on R_2 , satisfying

$$(3.4) \quad a_i \circ a_j = 0, \quad b_i \circ b_j = 0, \quad a_i \circ b_j = -b_j \circ a_i = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad i, j = 1, 2.$$

A basis of holomorphic differentials is given by

$$(3.5) \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}, \quad \oint_{b_j} w_i d\lambda = B_{ij}, \quad w_i(\lambda) = \sum_{k=0}^1 d_{ik} \lambda^k y^{-1}, \quad d_{ik} \in \mathbb{C},$$

where the normalization guarantees the a -periods form the identity matrix and the b -periods form the Riemann period matrix $B = (B_{ij})$.

The Abel map is defined using these normalized differentials. First, introduce the period lattice

$$(3.6) \quad L_2 = \{z \in \mathbb{C}^2 \mid z = 2\pi i(n + Bm), \quad n, m \in \mathbb{Z}^2\},$$

where B is the period matrix and n, m, z are column vectors in $\mathbb{C}^2$. The Jacobian of R_2 is then

$$(3.7) \quad J(R_2) = \mathbb{C}^2 / L_2.$$

For a base point $P_0 \in R_2$, the Abel map sends a point $P \in R_2$ to

$$(3.8) \quad \mathcal{A}_{P_0} : R_2 \rightarrow J(R_2),$$

$$(3.9) \quad P \mapsto \mathcal{A}_{P_0}(P) = (\mathcal{A}_{P_0,1}(P), \mathcal{A}_{P_0,2}(P)) = \left(\int_{P_0}^P w_1 d\lambda, \int_{P_0}^P w_2 d\lambda \right).$$

This definition extends linearly to divisors. A divisor $D = \sum_{P \in R_2} D(P) \cdot P$ is a finite formal sum of points P on R_2 with integer coefficients $D(P) \in \mathbb{Z}$, and we set

$$(3.10) \quad \alpha_{P_0}(D) = \sum_{P \in R_2} D(P) \mathcal{A}_{P_0}(P).$$

Finally, we introduce the Abelian integral

$$(3.11) \quad \Omega_1(P) = \int_{P_0}^P d\Omega_1, \quad \oint_{a_i} d\Omega_1 = 0,$$

with the prescribed asymptotics at the points at infinity,

$$(3.12) \quad \Omega_1(P) = \pm \left(\ln(\lambda) + \ln(\omega_1) + o(1) \right), \quad P \rightarrow \infty^\pm.$$

This integral provides the exponential term appearing in (3.2) and ensures the correct divisor structure of r_1 .

Recall that the function $r_1(x, t; P)$ admits the representation

$$(3.13) \quad r_1(x, t; P) = \frac{L_{21}}{iy + L_{11}},$$

where

$$(3.14) \quad L_{21} = -u \left(\lambda^2 - \frac{i u_x}{2u} \lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2} \right), \quad L_{11} = -i\lambda^3 - i\lambda \left(\alpha_1 - \frac{u^2}{2} \right).$$

To analyze its asymptotic behavior, note first that on the hyperelliptic curve R_2 ,

$$(3.15) \quad y = \pm \left(\lambda^3 - \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)\lambda \right) + O(\lambda^{-1}), \quad P \rightarrow \infty^\pm.$$

As $P \rightarrow \infty^-$, the expansion of the denominator yields

$$(3.16) \quad L_{11} + iy = -2i\lambda^3 - i\lambda \left(\alpha_1 - \frac{u^2}{2} \right) + \frac{i}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)\lambda + O(\lambda^{-1}),$$

so that

$$(3.17) \quad r_1(x, t; P) = \frac{L_{21}}{iy + L_{11}} = -\frac{i u}{2\lambda} + O(\lambda^{-2}), \quad P \rightarrow \infty^-.$$

Similarly, near ∞^+ one obtains

$$(3.18) \quad r_1(x, t; P) = \frac{iy - L_{11}}{L_{12}} = \frac{2i}{u}\lambda + O(1), \quad P \rightarrow \infty^+.$$

On the other hand, from the Riemann theta function representation (3.2), the asymptotics at the two points at infinity are

$$(3.19) \quad r_1(x, t; P) = \begin{cases} \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))} \lambda + O(1), & P \rightarrow \infty^+, \\ \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} \lambda^{-1} + O(\lambda^{-2}), & P \rightarrow \infty^-. \end{cases}$$

By comparing these expansions with the direct computation, one obtains the following theta-function formula for the potential $u(x, t)$:

$$(3.20) \quad u(x, t) = 2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))} = 2i \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}.$$

This identity expresses the potential u in terms of theta functions associated with the Jacobian of the spectral curve, thereby linking the algebro-geometric spectral data with the physical field variable.

Recall that the column vector

$$\Phi_1 = \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}$$

satisfies both parts of the Lax pair:

$$(3.21) \quad \partial_x \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix} = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix} \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}, \quad \partial_t \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix} = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & -V_{11} \end{pmatrix} \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}.$$

Hence the scalar component Φ_{11} satisfies

$$(3.22) \quad \Phi_{11,x} = (-i\lambda + u r_1) \Phi_{11}, \quad \Phi_{11,t} = (V_{11} + V_{12} r_1) \Phi_{11}.$$

Integrating the differential 1-form in (3.22) from the base point (x_0, t_0) to (x, t) yields

$$(3.23) \quad \Phi_{11}(x, t; \lambda) = \Phi_{11}(x_0, t_0; \lambda) \exp \left(\int_{(x_0, t_0)}^{(x, t)} \left[(-i\lambda + u r_1) dx + (V_{11} + V_{12} r_1) dt \right] \right).$$

Hence $\Phi_{11} \in \mathcal{M}(R_2 \setminus \{\infty^\pm\})$, the space of functions meromorphic on R_2 away from $\infty^\pm$. The representation (3.23) also shows that Φ_{11} has an essential singularity at each of $\infty^\pm$. For convenience, we set the initial point as $(x_0, t_0) = (0, 0)$.

The large- λ asymptotics differ on the two sheets. On the sheet with $P \rightarrow \infty^-$ we fix the normalization

$$\Phi_{1,0}^- := \lim_{P \rightarrow \infty^-} \Phi_{11}(x_0, t_0; \lambda) = 1,$$

and obtain

$$(3.24) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \exp(-i\lambda(x + 4\lambda^2 t)) + O(1), & P \rightarrow \infty^-, \\ \Phi_{21}(x, t; \lambda) &= r_1 \Phi_{11} = O(\lambda^{-1}) \exp(-i\lambda(x + 4\lambda^2 t)), & P \rightarrow \infty^-. \end{aligned}$$

On the other sheet, $P \rightarrow \infty^+$, the expansion of r_1 is

$$(3.25) \quad r_1(x, t; P) = \frac{iy - L_{11}}{L_{12}} = \frac{2i}{u} \lambda + \frac{u_x}{u^2} + O(\lambda^{-1}), \quad P \rightarrow \infty^+.$$

Consequently

$$(3.26) \quad -i\lambda + u r_1 = i\lambda + (\ln u)_x + O(\lambda^{-1}), \quad P \rightarrow \infty^+.$$

Substituting (3.26) into (3.23) gives, with the normalization constant

$$(3.27) \quad \Phi_{1,0}^+ := \lim_{P \rightarrow \infty^+} \Phi_{11}(x_0, t_0; \lambda),$$

the asymptotic formulas

$$(3.28) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \Phi_{1,0}^+ u(x, t) \exp(i\lambda(x + 4\lambda^2 t)) (1 + O(\lambda^{-1})), & P \rightarrow \infty^+, \\ \Phi_{21}(x, t; \lambda) &= r_1 \Phi_{11} = 2i\lambda \Phi_{1,0}^+ \exp(i\lambda(x + 4\lambda^2 t)) (1 + O(\lambda^{-1})), & P \rightarrow \infty^+. \end{aligned}$$

Recall

$$(3.29) \quad r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{L_{12}(\lambda)} = \frac{iy - L_{11}}{u(\lambda - \mu_1)(\lambda - \mu_2)},$$

$$(3.30) \quad r_1(\lambda) = \frac{L_{21}(\lambda)}{iy(\lambda) + L_{11}(\lambda)} = \frac{-u(\lambda - \mu_1^*)(\lambda - \mu_2^*)}{iy + L_{11}}.$$

We write

$$L(\lambda) = \begin{pmatrix} L_{11}(\lambda) & L_{12}(\lambda) \\ L_{21}(\lambda) & -L_{11}(\lambda) \end{pmatrix}, \quad U(\lambda) = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix},$$

and recall the stationary zero-curvature equation

$$(3.31) \quad L_x = [U, L].$$

With $\sigma_3 = \text{diag}(1, -1)$ and off-diagonal part $L^{\text{off}} = \begin{pmatrix} 0 & L_{12} \\ L_{21} & 0 \end{pmatrix}$, we compute

$$\begin{aligned} [U, L] &= -2i\lambda \sigma_3 L^{\text{off}} + [Q, L] \\ &= -2i\lambda \sigma_3 L^{\text{off}} + \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix} \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & -L_{11} \end{pmatrix} - \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & -L_{11} \end{pmatrix} \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix} \\ &= -2i\lambda \begin{pmatrix} 0 & L_{12} \\ -L_{21} & 0 \end{pmatrix} + \begin{pmatrix} u(L_{21} + L_{12}) & -2uL_{11} \\ -2uL_{11} & -u(L_{21} + L_{12}) \end{pmatrix} \\ (3.32) \quad &= \begin{pmatrix} u(L_{21} + L_{12}) & -2i\lambda L_{12} - 2uL_{11} \\ 2i\lambda L_{21} - 2uL_{11} & -u(L_{21} + L_{12}) \end{pmatrix}. \end{aligned}$$

Thus from (3.31) we obtain

$$\begin{aligned} (3.33) \quad \partial_x L_{11} &= u(L_{21} + L_{12}), \\ \partial_x L_{12} &= -2i\lambda L_{12} - 2uL_{11}, \\ \partial_x L_{21} &= 2i\lambda L_{21} - 2uL_{11}. \end{aligned}$$

By assumption

$$(3.34) \quad L_{12}(\lambda) = u(\lambda - \mu_1)(\lambda - \mu_2).$$

Differentiating with respect to x gives

$$(3.35) \quad \partial_x L_{12}(\lambda) = u_x(\lambda - \mu_1)(\lambda - \mu_2) - u \mu_{1,x}(\lambda - \mu_2) - u \mu_{2,x}(\lambda - \mu_1).$$

Evaluating (3.35) at $\lambda = \mu_1$ leaves only the middle term:

$$(3.36) \quad \partial_x L_{12}(\mu_1) = -u \mu_{1,x}(\mu_1 - \mu_2).$$

On the other hand, from (3.33) evaluated at $\lambda = \mu_1$ we have

$$(3.37) \quad \partial_x L_{12}(\mu_1) = -2u L_{11}(\mu_1).$$

Equating the two expressions yields

$$(3.38) \quad \mu_{1,x} = \frac{2L_{11}(\mu_1)}{\mu_1 - \mu_2}.$$

Next, the determinant relation $y^2 = \det L = -L_{11}^2 - L_{12}L_{21}$ at $\lambda = \mu_1$ reduces to

$$(3.39) \quad y(\mu_1)^2 = -L_{11}(\mu_1)^2,$$

since $L_{12}(\mu_1) = 0$. Hence $-iy(\mu_1) = \pm L_{11}(\mu_1)$. We fix the sign by setting

$$(3.40) \quad -iy(\mu_1) = L_{11}(\mu_1),$$

so that $P = (\mu_1, iL_{11}(\mu_1))$ represents a pole of the meromorphic function $r_1(P) = r_1(\lambda)$ on the Riemann surface R_2 .

Consider now the principal part of $u r_1$ near $\lambda = \mu_1$. From (3.29) and (3.34),

$$(3.41) \quad u r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{iy(\mu_1) - L_{11}(\mu_1)}{(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Using (3.40) and (3.38),

$$iy(\mu_1) - L_{11}(\mu_1) = -2L_{11}(\mu_1) = -(\mu_1 - \mu_2) \mu_{1,x}.$$

Thus (3.41) becomes

$$(3.42) \quad u r_1(\lambda) = -\frac{\mu_{1,x}}{\lambda - \mu_1} + O(1) = \partial_x(\ln(\lambda - \mu_1)) + O(1), \quad \lambda \rightarrow \mu_1.$$

Then by (3.23) and (3.42) we conclude that $\lambda = \mu_1$ is a simple zero of $\Phi_{11}(x, t; \lambda)$.

Supplementary computations:

Evaluate (3.35) at $\lambda = \mu_2$. Only the third term survives, so

$$(3.43) \quad \partial_x L_{12}(\mu_2) = -u \mu_{2,x}(\mu_2 - \mu_1).$$

From (3.33) at $\lambda = \mu_2$ we have

$$(3.44) \quad \partial_x L_{12}(\mu_2) = -2u L_{11}(\mu_2).$$

Equating these expressions yields

$$(3.45) \quad \mu_{2,x} = \frac{2L_{11}(\mu_2)}{\mu_2 - \mu_1}.$$

Adopting the same sheet convention as for μ_1 we take

$$(3.46) \quad -iy(\mu_2) = L_{11}(\mu_2).$$

Compute the principal part of $u r_1$ near $\lambda = \mu_2$. From (3.29) and (3.34),

$$(3.47) \quad u r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{iy(\mu_2) - L_{11}(\mu_2)}{(\mu_2 - \mu_1)} \cdot \frac{1}{\lambda - \mu_2} + O(1), \quad \lambda \rightarrow \mu_2.$$

Using (3.46) and (3.45) we obtain

$$iy(\mu_2) - L_{11}(\mu_2) = -2L_{11}(\mu_2) = -(\mu_2 - \mu_1) \mu_{2,x}.$$

Hence (3.47) becomes

$$(3.48) \quad u r_1(\lambda) = -\frac{\mu_{2,x}}{\lambda - \mu_2} + O(1) = \partial_x(\ln(\lambda - \mu_2)) + O(1), \quad \lambda \rightarrow \mu_2.$$

Therefore, by (3.23) and (3.48), $\lambda = \mu_2$ is a simple zero of $\Phi_{11}(x, t; \lambda)$.

We next consider the time evolution governed by

$$(3.49) \quad L_t = [V, L], \quad V(\lambda) = \begin{pmatrix} V_{11}(\lambda) & V_{12}(\lambda) \\ V_{21}(\lambda) & -V_{11}(\lambda) \end{pmatrix}.$$

Direct computation of the commutator yields

$$[V, L] = VL - LV = \begin{pmatrix} V_{12}L_{21} - L_{12}V_{21} & 2V_{11}L_{12} - 2L_{11}V_{12} \\ 2L_{11}V_{21} - 2V_{11}L_{21} & L_{12}V_{21} - V_{12}L_{21} \end{pmatrix},$$

since the trace vanishes and the anti-diagonal structure pairs terms. Consequently from (3.49) we obtain the componentwise equations

$$(3.50) \quad \begin{aligned} \partial_t L_{11} &= V_{12}L_{21} - L_{12}V_{21}, \\ \partial_t L_{12} &= 2V_{11}L_{12} - 2L_{11}V_{12}, \\ \partial_t L_{21} &= 2L_{11}V_{21} - 2V_{11}L_{21}. \end{aligned}$$

By assumption

$$(3.51) \quad L_{12}(\lambda) = u(\lambda - \mu_1)(\lambda - \mu_2),$$

where $u = u(x, t)$ and $\mu_j = \mu_j(x, t)$. Differentiating (3.51) with respect to t gives

$$(3.52) \quad \partial_t L_{12}(\lambda) = u_t(\lambda - \mu_1)(\lambda - \mu_2) - u \mu_{1,t}(\lambda - \mu_2) - u \mu_{2,t}(\lambda - \mu_1).$$

Evaluating (3.52) at $\lambda = \mu_1$ annihilates the first and third terms and leaves

$$(3.53) \quad \partial_t L_{12}(\mu_1) = -u \mu_{1,t}(\mu_1 - \mu_2).$$

On the other hand, evaluating the second equation in (3.50) at $\lambda = \mu_1$ yields

$$(3.54) \quad \partial_t L_{12}(\mu_1) = 2V_{11}(\mu_1) L_{12}(\mu_1) - 2L_{11}(\mu_1) V_{12}(\mu_1).$$

Since $L_{12}(\mu_1) = 0$ by (3.51), the first term on the right of (3.54) vanishes and we obtain

$$(3.55) \quad \partial_t L_{12}(\mu_1) = -2L_{11}(\mu_1) V_{12}(\mu_1).$$

Equating (3.53) and (3.55) gives the evolution law for μ_1 :

$$(3.56) \quad \mu_{1,t} = \frac{2L_{11}(\mu_1) V_{12}(\mu_1)}{u(\mu_1 - \mu_2)}.$$

Consider the principal part of $V_{12}(\lambda) r_1(\lambda)$ near $P = (\mu_1, iL_{11}(\mu_1))$. From (3.29) we have

$$(3.57) \quad V_{12}(\lambda) r_1(\lambda) = V_{12}(\lambda) \frac{iy(\lambda) - L_{11}(\lambda)}{u(\lambda - \mu_1)(\lambda - \mu_2)}.$$

Expanding the numerator and evaluating at $\lambda = \mu_1$ yields the simple pole

$$(3.58) \quad V_{12}(\lambda) r_1(\lambda) = \frac{-2L_{11}(\mu_1) V_{12}(\mu_1)}{u(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Using the evolution law (3.56) for μ_1 this principal part becomes

$$(3.59) \quad V_{12}(\lambda) r_1(\lambda) = -\frac{\mu_{1,t}}{\lambda - \mu_1} + O(1) = \partial_t(\ln(\lambda - \mu_1)) + O(1), \quad \lambda \rightarrow \mu_1.$$

Week 4: Algebraic-Geometric Solutions for the mKdV Equation (4)

2025-10-14

Objectives: Continue to construct algebraic-geometric mKdV solutions on a genus-two curve by expressing r_1 and u in Riemann theta functions and matching their asymptotics at $\infty^\pm$, poles and zeroes.

Recall $P = (\lambda, y)$. The genus-two hyperelliptic curve and its points at infinity are

$$(4.1) \quad R_2 = \{(\lambda, y) \mid y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)\}, \quad P \rightarrow \infty^\pm \iff \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3.$$

Define the points

$$(4.2) \quad \mu = \{\hat{\mu}_1, \hat{\mu}_2\}, \quad \hat{\mu}_j = (\mu_j, iL_{11}(\mu_j)), \quad \mu^* = \{\hat{\mu}_1^*, \hat{\mu}_2^*\}, \quad \hat{\mu}_j^* = (\mu_j^*, -iL_{11}(\mu_j^*)),$$

so that μ^* are the zeros and μ the poles of r_1 .

At a pole $\hat{\mu}_1 = (\mu_1, iL_{11}(\mu_1))$ the meromorphic function $r_1(\lambda)$ has the local expansion

$$(4.3) \quad r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{L_{12}(\lambda)} = \frac{iy - L_{11}}{u(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{-2L_{11}(\mu_1)}{u(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Recall that $r_1(x, t; P)$ is defined in (3.2) and that $\Phi_{11}(x, t; P) = \Phi_{11}(x, t; \lambda)$ is given by the integral representation (3.23). Choosing a continuous path from (x_0, t_0) to (x, t) we have

$$(4.4) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \Phi_{11}(x_0, t_0; \lambda) \exp \left(\int_{(x_0, t_0)}^{(x, t)} (-i\lambda + u r_1) dx_1 + (V_{11} + V_{12} r_1) dt_1 \right) \\ &= \Phi_{11}(x_0, t_0; \lambda) \exp \left(\int_{x_0}^x (-i\lambda + u r_1)(x_1, t_0) dx_1 + \int_{t_0}^t (V_{11} + V_{12} r_1)(x, t_1) dt_1 \right). \end{aligned}$$

The singularity structure of Φ_{11} follows from the pole structure of r_1 . From (4.3) and the explicit expressions for $u r_1$ and $V_{12} r_1$ (see (3.42) and (3.48)) one obtains, as $\lambda \rightarrow \mu_j$,

$$(4.5) \quad u r_1(x_1, t_0; \lambda) = \partial_{x_1}(\ln(\lambda - \mu_j(x_1, t_0))) + O(1),$$

and

$$(4.6) \quad V_{12} r_1(x, t_1; \lambda) = \partial_{t_1}(\ln(\lambda - \mu_j(x, t_1))) + O(1).$$

Therefore each moving point $\hat{\mu}_j(x, t) = (\mu_j(x, t), iL_{11}(\mu_j(x, t)))$ is a zero of $\Phi_{11}(x, t; P)$, while the initial points $\mu_0 := \mu(x_0, t_0)$ give the poles of Φ_{11} .

Matching the asymptotic forms (3.24) and (3.28) of Φ_{11} as $P \rightarrow \infty^\pm$ gives the theta-function representation

$$(4.7) \quad \Phi_{11}(x, t; P) = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{i(\omega_2 + \Omega_2(P))x + i(\omega_3 + \Omega_3(P))t}.$$

Here $P_0 = (\lambda_3, 0)$, and the Abelian integrals Ω_2, Ω_3 are defined by

$$(4.8) \quad \Omega_2(P) := \int_{P_0}^P d\Omega_2, \quad \oint_{a_i} d\Omega_2 = 0, \quad \Omega_2(P) = \pm(\lambda + \omega_2 + o(1)), \quad P \rightarrow \infty^\pm,$$

$$(4.9) \quad \Omega_3(P) := \int_{P_0}^P d\Omega_3, \quad \oint_{a_i} d\Omega_3 = 0, \quad \Omega_3(P) = \pm(4\lambda^3 + \omega_3 + o(1)), \quad P \rightarrow \infty^\pm,$$

where ω_2, ω_3 are constants independent of λ . As $P \rightarrow \infty^-$, (4.7) is consistent with (3.24).

As $P \rightarrow \infty^+$, comparing (3.28) with (4.7) yields

$$(4.10) \quad u(x, t) = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{2i\omega_2 x + 2i\omega_3 t}.$$

Combining (3.20) and (4.10), we start from the first expression in (3.20):

$$u(x, t) = 2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))}.$$

Equating this with (4.10) gives

$$2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))} = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{2i\omega_2 x + 2i\omega_3 t}.$$

Cancel the common factor $\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))$ and solve for $\hat{r}_1(x, t)$:

$$(4.11) \quad \hat{r}_1(x, t) = \frac{2i \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t}.$$

Substitute (4.11) into the second form of (3.20),

$$u(x, t) = 2i \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}.$$

Then

$$\begin{aligned} u(x, t) &= 2i \frac{2i \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t} \\ &\quad \times \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}. \end{aligned}$$

After cancelling $\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))$, we obtain

$$(4.12) \quad u(x, t) = \frac{(2i)^2 \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t}.$$

We need to determine these the parameters:

$$\alpha_{P_0}(D_\mu), \quad \alpha_{P_0}(D_{\mu^*}), \quad B, \quad w_1, w_2, \quad \omega_1, \omega_2, \omega_3, \quad \Phi_{1,0}^+.$$

The Riemann-Roch theorem gives

$$(4.13) \quad r(-D) = \deg(D) + i(D) - g + 1,$$

where

$$\mathcal{L}(D) = \{f \in \mathcal{M}(R_g) \mid (f) \geq D\}, \quad r(D) = \dim(\mathcal{L}(D)),$$

and

$$\mathcal{L}'(D) = \{\omega \in \mathcal{M}'(R_g) \mid (\omega) \geq D\}, \quad i(D) = \dim(\mathcal{L}'(D)),$$

with $\mathcal{M}(R_g)$ the field of meromorphic functions on the genus g curve R_g and $\mathcal{M}'(R_g)$ the space of meromorphic differentials.

Recall that

$$(4.14) \quad \text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda).$$

Take the (non special) divisor $D = D_{\infty^+, \mu}$. Assume $i(D) = 0$ and that $(r_1) \geq -D_{\infty^+, \mu}$. Then by (4.13) (with $g = 2$ and $\deg(D) = 3$ in the present setup)

$$(4.15) \quad r(-D_{\infty^+, \mu}) = \deg(D) - g + 1 = 3 - 2 + 1 = 2.$$

Since r_1 and $\tilde{r}_1$ have the same zeros they span a two dimensional space and hence r_1 is proportional to $\tilde{r}_1$, i.e.

$$r_1 = c\tilde{r}_1 + d \Rightarrow r_1 = c\tilde{r}_1 \quad (c, d \in \mathbb{C}).$$

Recall the Abel map for the degree two divisor $D_\mu = \hat{\mu}_1 + \hat{\mu}_2$ based at P_0 :

$$(4.16) \quad \alpha_{P_0}(D_\mu) = \mathcal{A}_{P_0}(\hat{\mu}_1) + \mathcal{A}_{P_0}(\hat{\mu}_2) = \left(\int_{P_0}^{\hat{\mu}_1} w_1 d\lambda + \int_{P_0}^{\hat{\mu}_2} w_1 d\lambda, \int_{P_0}^{\hat{\mu}_1} w_2 d\lambda + \int_{P_0}^{\hat{\mu}_2} w_2 d\lambda \right).$$

Let $\{w_1 d\lambda, w_2 d\lambda\}$ be a basis of holomorphic differentials normalized by

$$(4.17) \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}, \quad \oint_{b_j} w_i d\lambda = B_{ij}, \quad i, j = 1, 2.$$

On the affine chart write each differential as a rational function of λ divided by $y(\lambda)$:

$$(4.18) \quad w_i(\lambda) = (d_{i0} + d_{i1}\lambda) \frac{1}{y(\lambda)}, \quad d_{ik} \in \mathbb{C}, \quad i = 1, 2, \quad k = 0, 1.$$

Thus

$$w_1(\lambda) = \frac{d_{10} + d_{11}\lambda}{y(\lambda)}, \quad w_2(\lambda) = \frac{d_{20} + d_{21}\lambda}{y(\lambda)}.$$

Differentiate the components of $\alpha_{P_0}(D_\mu)$ with respect to x . By the chain rule, for $j = 1, 2$ and $i = 1, 2$,

$$(4.19) \quad \frac{d}{dx} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = w_i(\hat{\mu}_j) \mu_{j,x} = \frac{d_{i0} + d_{i1}\mu_j}{y(\mu_j)} \mu_{j,x}.$$

We compute the w_1 -component of $\alpha_{P_0}(D_\mu)$. Using (4.19) and the expressions (3.38) and (3.45) for $\mu_{j,x}$ in terms of the Lax matrix entries (so that $\mu_{j,x} = \frac{2L_{11}(\mu_j)}{\mu_j - \mu_k}$ with $k \neq j$ and $y(\mu_j) = iL_{11}(\mu_j)$), one obtains

$$\begin{aligned} \frac{d}{dx} \left(\int_{P_0}^{\hat{\mu}_1} (d_{10} + d_{11}\lambda) \frac{1}{y} d\lambda + \int_{P_0}^{\hat{\mu}_2} (d_{10} + d_{11}\lambda) \frac{1}{y} d\lambda \right) &= \frac{d_{10} + d_{11}\mu_1}{y(\mu_1)} \mu_{1,x} + \frac{d_{10} + d_{11}\mu_2}{y(\mu_2)} \mu_{2,x} \\ &= d_{10} \left(\frac{\mu_{1,x}}{y(\mu_1)} + \frac{\mu_{2,x}}{y(\mu_2)} \right) + d_{11} \left(\frac{\mu_1 \mu_{1,x}}{y(\mu_1)} + \frac{\mu_2 \mu_{2,x}}{y(\mu_2)} \right) \\ &= d_{10} \left(\frac{2L_{11}(\mu_1)}{\mu_1 - \mu_2} \frac{1}{iL_{11}(\mu_1)} + \frac{2L_{11}(\mu_2)}{\mu_2 - \mu_1} \frac{1}{iL_{11}(\mu_2)} \right) + d_{11} \left(\frac{2\mu_1}{i(\mu_1 - \mu_2)} + \frac{2\mu_2}{i(\mu_2 - \mu_1)} \right) \\ &= \frac{2}{i} d_{10} \left(\frac{1}{\mu_1 - \mu_2} + \frac{1}{\mu_2 - \mu_1} \right) + \frac{2}{i} d_{11} \left(\frac{\mu_1}{\mu_1 - \mu_2} + \frac{\mu_2}{\mu_2 - \mu_1} \right) \\ (4.20) \quad &= \frac{2}{i} d_{11} = -2i d_{11}. \end{aligned}$$

The w_2 -component is analogous. Repeating the same substitutions gives

$$(4.21) \quad \frac{d}{dx} \left(\int_{P_0}^{\hat{\mu}_1} (d_{20} + d_{21}\lambda) \frac{1}{y} d\lambda + \int_{P_0}^{\hat{\mu}_2} (d_{20} + d_{21}\lambda) \frac{1}{y} d\lambda \right) = -2i d_{21}.$$

Combining (4.20) and (4.21) we obtain

$$(4.22) \quad \frac{d}{dx} (\alpha_{P_0}(D_\mu)) = -2i (d_{11}, d_{21}) = -i \hat{U},$$

where $\hat{U} = (U_1, U_2)$ and $U_i = \oint_{b_i} d\Omega_2$.

For the time evolution induced by the higher flow one finds similarly

$$(4.23) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = -i \hat{V} = -i v \hat{U} = -2i v (d_{11}, d_{21}),$$

with

$$v = 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2), \quad \hat{V} = (V_1, V_2), \quad V_i = \oint_{b_i} d\Omega_3.$$

Supplementary computations:

Differentiate the components of $\alpha_{P_0}(D_\mu)$ with respect to t . By the chain rule, for $j = 1, 2$ and $i = 1, 2$,

$$(4.24) \quad \frac{d}{dt} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = w_i(\hat{\mu}_j) \mu_{j,t} = \frac{d_{i0} + d_{i1}\mu_j}{y(\mu_j)} \mu_{j,t}.$$

Use the evolution law (3.56) (and the analogous formula for $\mu_{2,t}$):

$$(4.25) \quad \mu_{j,t} = \frac{2L_{11}(\mu_j) V_{12}(\mu_j)}{u(\mu_j - \mu_k)}, \quad k \neq j.$$

Also use $y(\mu_j) = iL_{11}(\mu_j)$. Substituting these into (4.24) gives, for each j ,

$$\frac{d}{dt} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = \frac{d_{i0} + d_{i1}\mu_j}{iL_{11}(\mu_j)} \frac{2L_{11}(\mu_j) V_{12}(\mu_j)}{u(\mu_j - \mu_k)} = \frac{2}{iu} \frac{d_{i0} + d_{i1}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j).$$

Summing the contributions from $j = 1, 2$ we obtain the two components of the time derivative of $\alpha_{P_0}(D_\mu)$. Thus

$$(4.26) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = \frac{2}{iu} \left(\sum_{j=1}^2 \frac{d_{10} + d_{11}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j), \sum_{j=1}^2 \frac{d_{20} + d_{21}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j) \right),$$

with in each sum $k \neq j$.

Using the relation $L_{12} = \frac{1}{4}V_{12} + \alpha_1 u$ from (2.18), and noting that $L_{12}(\mu_j) = 0$ for $j = 1, 2$, we obtain

$$(4.27) \quad V_{12}(\mu_j) = 4(L_{12}(\mu_j) - \alpha_1 u) = -4\alpha_1 u, \quad j = 1, 2.$$

Hence for each component $i \in \{1, 2\}$,

$$(4.28) \quad \sum_{j=1}^2 \frac{d_{i0} + d_{i1}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j) = -4\alpha_1 u \left(\frac{d_{i0} + d_{i1}\mu_1}{\mu_1 - \mu_2} + \frac{d_{i0} + d_{i1}\mu_2}{\mu_2 - \mu_1} \right) = -4\alpha_1 u \cdot d_{i1}.$$

Substituting (4.28) into (4.26) gives

$$(4.29) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = \frac{2}{iu} (-4\alpha_1 u) (d_{11}, d_{21}) = 8i \alpha_1 (d_{11}, d_{21}).$$

From $\det L = -L_{11}^2 - L_{12}L_{21}$ and (2.26) write

$$L_{11} = -i(\lambda^3 + \lambda(\alpha_1 - \frac{1}{2}u^2)), \quad L_{12} = u(\lambda^2 + a\lambda + b), \quad L_{21} = -u(\lambda^2 - a\lambda + b),$$

with $a = i\frac{u_x}{2u}$ and $b = -\frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2$. Compute the leading terms in λ :

$$-L_{11}^2 = (\lambda^3 + \lambda(\alpha_1 - \frac{1}{2}u^2))^2 = \lambda^6 + 2(\alpha_1 - \frac{1}{2}u^2)\lambda^4 + \mathcal{O}(\lambda^2),$$

and

$$-L_{12}L_{21} = u^2(\lambda^4 + \mathcal{O}(\lambda^2)) = u^2\lambda^4 + \mathcal{O}(\lambda^2).$$

Hence

$$\det L = \lambda^6 + [2(\alpha_1 - \frac{1}{2}u^2) + u^2]\lambda^4 + \mathcal{O}(\lambda^2) = \lambda^6 + 2\alpha_1\lambda^4 + \mathcal{O}(\lambda^2).$$

On the other hand (2.28) gives

$$\det L = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2) = \lambda^6 - \left(\sum_{j=1}^3 \lambda_j^2 \right) \lambda^4 + \mathcal{O}(\lambda^2).$$

Equating the λ^4 -coefficients yields $\alpha_1 = -\frac{1}{2} \sum_{j=1}^3 \lambda_j^2$.

Week 5: Algebraic-Geometric Solutions for the mKdV Equation (5)

2025-10-21

Objectives: Express ω_2 , ω_3 , and $\hat{U}$ via the Riemann bilinear relations, and obtain the identities relating $\alpha_{P_0}(D_{\mu}^*)$ and $\alpha_{P_0}(D_{\mu})$ from Abel's theorem.

Riemann bilinear relation.

(i) Let $\hat{\Omega}$ and $\tilde{\Omega}$ be (multi-valued) Abelian integrals on the Riemann surface R_2 , and write their corresponding differentials as $d\hat{\Omega}$ and $d\tilde{\Omega}$. For the standard canonical homology basis $\{a_i, b_i\}_{i=1}^2$ the following identities hold by elementary contour rearrangements:

$$(5.1) \quad \oint_{a_i} \hat{\Omega} d\tilde{\Omega} + \oint_{a_i^{-1}} \hat{\Omega} d\tilde{\Omega} = \oint_{a_i} \left(\int_{z_0}^z d\hat{\Omega} \right) d\tilde{\Omega} + \oint_{a_i^{-1}} \left(\int_{z_0}^{z'} d\hat{\Omega} \right) d\tilde{\Omega}$$

$$(5.2) \quad = - \oint_{a_i} \left(\int_z^{z'} d\hat{\Omega} \right) d\tilde{\Omega} = - \left(\oint_{a_i} d\tilde{\Omega} \right) \left(\oint_{b_i} d\hat{\Omega} \right),$$

$$(5.3) \quad \oint_{b_i} \hat{\Omega} d\tilde{\Omega} + \oint_{b_i^{-1}} \hat{\Omega} d\tilde{\Omega} = \oint_{b_i} \left(\int_{z'}^z d\hat{\Omega} \right) d\tilde{\Omega} = \left(\oint_{a_i} d\hat{\Omega} \right) \left(\oint_{b_i} d\tilde{\Omega} \right).$$

Applying Stokes' theorem to the boundary of the canonical polygon for R_2 gives

$$(5.4) \quad 0 = \int_{R_2} d(\hat{\Omega} d\tilde{\Omega}) = \int_{\partial R_2} \hat{\Omega} d\tilde{\Omega} = \sum_{i=1}^2 \left(\oint_{a_i} + \oint_{b_i} + \oint_{a_i^{-1}} + \oint_{b_i^{-1}} \right) \hat{\Omega} d\tilde{\Omega},$$

and hence the usual bilinear identity

$$(5.5) \quad \sum_{i=1}^2 \left(\oint_{a_i} d\hat{\Omega} \oint_{b_i} d\tilde{\Omega} - \oint_{b_i} d\hat{\Omega} \oint_{a_i} d\tilde{\Omega} \right) = 0.$$

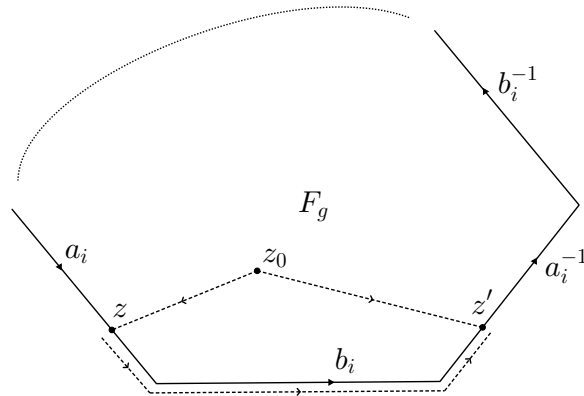


FIGURE 2. To the proof of the Riemann bilinear relations.

(ii) If the product $\hat{\Omega} d\tilde{\Omega}$ has poles (only) at the points $\infty^\pm$, then performing the same computation on $R_2 \setminus \{\infty^\pm\}$ introduces the boundary integral around small loops encircling the poles. One obtains

$$(5.6) \quad \sum_{i=1}^2 \left(\oint_{a_i} d\hat{\Omega} \oint_{b_i} d\tilde{\Omega} - \oint_{b_i} d\hat{\Omega} \oint_{a_i} d\tilde{\Omega} \right) + \oint_{l+\pi(l)} \hat{\Omega} d\tilde{\Omega} = 0,$$

where $l + \pi(l)$ denotes the small contours around the poles at $\infty^\pm$ with the orientation induced by the boundary of $R_2 \setminus \{\infty^\pm\}$.

Applications. Recall the Abelian integrals Ω_2, Ω_3 from (4.8)–(4.9). Define

$$\hat{U} = (U_1, U_2), \quad U_i = \oint_{b_i} d\Omega_2, \quad \hat{V} = (V_1, V_2), \quad V_i = \oint_{b_i} d\Omega_3.$$

(1) The constants ω_2, ω_3 . Choose

$$\hat{\Omega} = \Omega_2, \quad d\tilde{\Omega} = \frac{\lambda^2}{y} d\lambda.$$

Since $\oint_{a_i} d\Omega_2 = 0$ and $\oint_{b_i} d\Omega_2 = U_i$, the bilinear identity (5.6) becomes

$$(5.7) \quad - \sum_{i=1}^2 U_i \oint_{a_i} \frac{\lambda^2}{y} d\lambda + \oint_{l+\pi(l)} \Omega_2 \frac{\lambda^2}{y} d\lambda = 0.$$

The boundary integral is the sum of residues at $\infty^\pm$:

$$(5.8) \quad \oint_{l+\pi(l)} \Omega_2 \frac{\lambda^2}{y} d\lambda = 2\pi i \left(\text{Res}_{\lambda=\infty^+} \Omega_2 \frac{\lambda^2}{y} d\lambda + \text{Res}_{\lambda=\infty^-} \Omega_2 \frac{\lambda^2}{y} d\lambda \right).$$

Using (4.8) and the large- λ expansion of y , one obtains

$$(5.9) \quad \Omega_2 \frac{\lambda^2}{y} = (\lambda + \omega_2 + o(1)) \lambda^{-1} \left(1 + \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \lambda^{-2} + O(\lambda^{-4}) \right) = 1 + \frac{\omega_2}{\lambda} + \dots$$

as $P \rightarrow \infty^\pm$. Thus the residue at each of $\infty^\pm$ equals ω_2 .

Substitution of (5.8) and (5.9) into (5.7) yields

$$(5.10) \quad \omega_2 = -\frac{i}{4\pi} \sum_{j=1}^2 U_j \oint_{a_j} \frac{\lambda^2}{y} d\lambda.$$

Repeating the computation with $\hat{\Omega} = \Omega_3$ gives

$$(5.11) \quad \omega_3 = -\frac{i}{4\pi} \sum_{j=1}^2 V_j \oint_{a_j} \frac{\lambda^2}{y} d\lambda.$$

(2) The vector $\hat{U}$. We now verify that

$$\hat{U} = (U_1, U_2) = (2d_{11}, 2d_{21}), \quad U_1 = 2d_{11}.$$

Choose $\hat{\Omega} = \Omega_2$ and

$$\tilde{\Omega} = w_1 d\lambda = (d_{10} + d_{11}\lambda) y^{-1} d\lambda, \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}.$$

Applying the bilinear identity gives

$$\sum_{i=1}^2 \left(\oint_{a_i} d\Omega_2 \oint_{b_i} w_1 d\lambda - \oint_{b_i} d\Omega_2 \oint_{a_i} w_1 d\lambda \right) + 2\pi i (\text{Res}_{\infty^+} \Omega_2 w_1 d\lambda + \text{Res}_{\infty^-} \Omega_2 w_1 d\lambda) = 0.$$

To determine the residues, use the large- λ expansions of Ω_2 and y :

$$\Omega_2 w_1 = (\lambda + \omega_2 + o(1))(d_{10} + d_{11}\lambda) \lambda^{-3} \left(1 + \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \lambda^{-2} + O(\lambda^{-4}) \right) = \frac{d_{11}}{\lambda} + \dots$$

as $P \rightarrow \infty^\pm$. Hence each of the points $\infty^\pm$ contributes a residue d_{11} , and the total equals $2d_{11}$. Since

$$\oint_{a_i} d\Omega_2 = 0, \quad \oint_{b_i} d\Omega_2 = U_i, \quad \oint_{a_i} w_1 d\lambda = 2\pi i \delta_{1i},$$

the bilinear identity reduces to

$$-2\pi i U_1 + 2\pi i (2d_{11}) = 0, \quad \text{so} \quad U_1 = 2d_{11}.$$

The same computation with w_2 gives $U_2 = 2d_{21}$.

Abel's theorem. Let $M(R_2)$ denote the meromorphic functions on R_2 , and $M'(R_2)$ the meromorphic 1-forms. For $f \in M(R_2) \setminus \{0\}$ and $P \in R_2$, the order $\text{ord}_P(f)$ is the integer k such that, in a local coordinate z with $z(P) = 0$,

$$f(z) = z^k g(z), \quad g(0) \neq 0.$$

The divisor of f is

$$(f) = \sum_{P \in R_2} \text{ord}_P(f) P,$$

which is a finite sum because f has only finitely many zeros and poles. The divisor (w) of a meromorphic 1-form $w \in M'(R_2) \setminus \{0\}$ is defined analogously by using its local expression $w(z) = z^k h(z) dz$ with $h(0) \neq 0$.

A meromorphic function is holomorphic if and only if $(f) \geq 0$. A meromorphic 1-form is holomorphic (of the first kind) if and only if $(w) \geq 0$. A divisor of the form (f) is called principal; a divisor of the form (w) is canonical.

Theorem 1 (Abel's theorem). *A divisor D is principal if and only if $\deg D = 0$ and $\alpha_{P_0}(D) = 0$.*

As an example,

$$\text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*} - D_{\infty^+, \mu},$$

which has degree zero. Abel's theorem then gives

$$\alpha_{P_0}(D_{\mu^*}) + \mathcal{A}_{P_0}(\infty^-) = \alpha_{P_0}(D_{\infty^-, \mu^*}) = \alpha_{P_0}(D_{\infty^+, \mu}) = \alpha_{P_0}(D_\mu) + \mathcal{A}_{P_0}(\infty^+),$$

hence

$$\alpha_{P_0}(D_{\mu^*}) = \alpha_{P_0}(D_\mu) + \mathcal{A}_{\infty^-}(\infty^+),$$

where

$$\mathcal{A}_{P_0}(P) = \left(\int_{P_0}^P w_1 d\lambda, \int_{P_0}^P w_2 d\lambda \right).$$

Week 6: Algebraic-Geometric Solutions for the mKdV Equation (6)

2025-10-28

Objectives: Focus on Case 2. Determine the homology basis under the involution τ , and compute the conjugate quantities $\overline{\ln(\omega_1)}$, $\overline{\omega_2}$, $\overline{\omega_3}$, and $\overline{B_{ij}}$.

Recall the definition (3.3):

$$(6.1) \quad R_2 := \left\{ (\lambda, y) \mid y^2 = (\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2) \right\}, \quad P = (\lambda, y).$$

We work in **Case 2** (one nonreal quadruple and one purely imaginary pair). Introduce the involutions

$$(6.2) \quad \tau_1 : (\lambda, y) \mapsto (\lambda, -y), \quad \tau : (\lambda, y) \mapsto (\bar{\lambda}, \bar{y}),$$

where τ_1 is the hyperelliptic (sheet-exchange) involution and τ is the antiholomorphic involution; both satisfy

$$\tau_1^2 = \tau^2 = \mathbb{I}.$$

Fix a canonical homology basis $\{a_1, a_2; b_1, b_2\}$ on R_2 with the intersection properties

$$(6.3) \quad a_i \circ a_j = 0, \quad b_i \circ b_j = 0, \quad a_i \circ b_j = \delta_{ij}, \quad i, j = 1, 2.$$

The antiholomorphic involution τ acts by sign change on the a -cycles and acts on the b -cycles by a combination of a - and b -cycles. Write

$$(6.4) \quad \tau(a_i) = -a_i, \quad \tau(b_j) = \sum_{k=1}^2 \alpha_{jk} a_k + \sum_{k=1}^2 \beta_{jk} b_k, \quad j = 1, 2,$$

for some integers α_{jk}, β_{jk} . Using the intersection pairing (6.3) we determine these coefficients as follows.

Compute the intersections with a_1, a_2 :

$$a_1 \circ \tau(b_1) = \beta_{11}(a_1 \circ b_1) = \beta_{11}, \quad a_2 \circ \tau(b_1) = \beta_{12}(a_2 \circ b_2) = \beta_{12}.$$

But the geometry of the antiholomorphic involution in this configuration yields

$$a_1 \circ \tau(b_1) = 1, \quad a_2 \circ \tau(b_1) = 0,$$

hence

$$(6.5) \quad \beta_{11} = 1, \quad \beta_{12} = 0.$$

Similarly, intersecting $\tau(b_1)$ with the b -cycles gives

$$\tau(b_1) \circ b_1 = \alpha_{11} = 0, \quad \tau(b_1) \circ b_2 = \alpha_{12} = 1,$$

so that $\alpha_{11} = 0, \alpha_{12} = 1$. Collecting these values yields

$$(6.6) \quad \tau(b_1) = b_1 + a_2.$$

By the same type of computation one obtains

$$(6.7) \quad \tau(b_2) = b_2 + a_1.$$

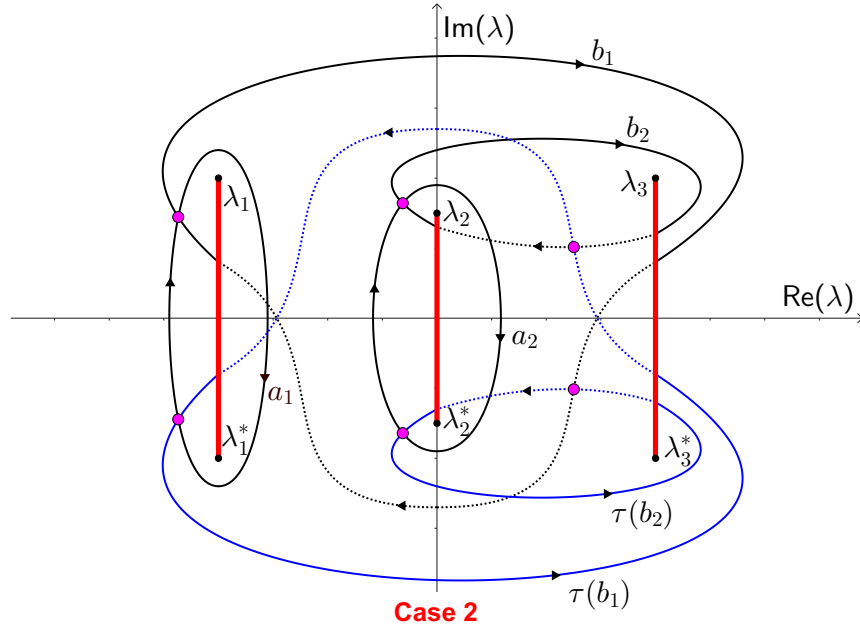


FIGURE 3. To the computations of $\tau(b_1), \tau(b_2)$ for Case 2.

Denote by $P_{\pm}$ the two points above the same λ -value on the two sheets of R_2 . Let C_P be the oriented path which starts at P_- (on the second sheet), crosses the cut joining the branch points λ_3, λ_3^* through their midpoint, then continues on the first sheet and winds counterclockwise along the cut until it reaches P_+ . (This is the path used below to connect ∞^- to ∞^+ in the indicated way.)

The path C_P is transformed by τ as

$$(6.8) \quad \tau(C_P) = C_P + a_3,$$

since $\tau(C_P) - C_P = a_3$. Computing the intersections $a_3 \circ b_j$ from the geometry one finds

$$a_3 \circ b_1 = -1, \quad a_3 \circ b_2 = -1,$$

and therefore

$$(6.9) \quad a_3 = -a_1 - a_2.$$

In fact one must include the small loop around ∞^+ . Thus the correct homology relation is

$$(6.10) \quad a_3 = -a_1 - a_2 + \ell,$$

where ℓ denotes a small positively oriented loop encircling ∞^+ .

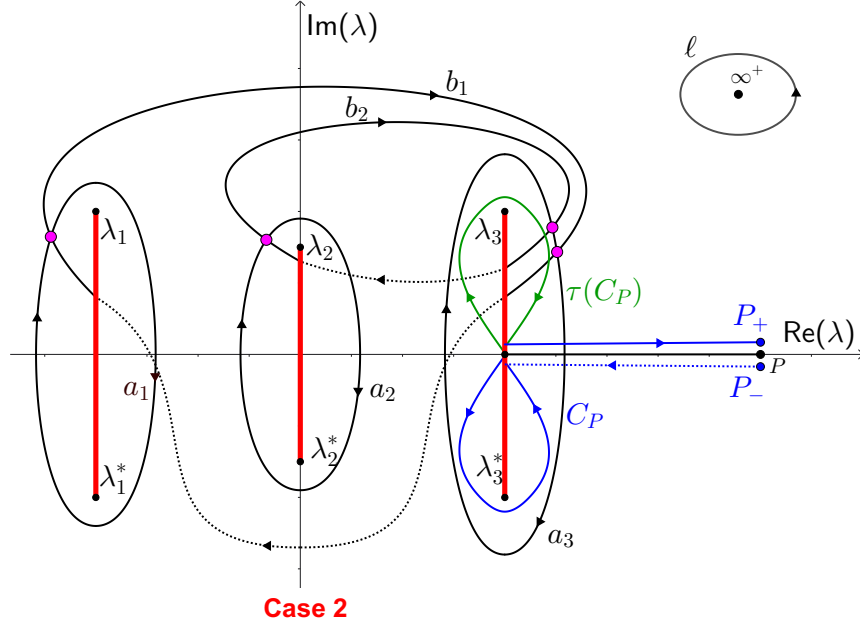


FIGURE 4. To the computation of $\tau(C_P)$ for Case 2.

Let $d\Omega_1$ be the normalized (third-kind) Abelian differential whose a -periods vanish:

$$(6.11) \quad \oint_{a_i} d\Omega_1 = 0, \quad i = 1, 2,$$

and which has the prescribed logarithmic behaviour at the points at infinity:

$$(6.12) \quad \Omega_1(P) = \int_{P_0}^P d\Omega_1 = \pm (\ln \lambda + \ln \omega_1 + o(1)), \quad P \rightarrow \infty^\pm,$$

where ω_1 is the (nonzero) constant appearing in the local expansion and w_i will denote holomorphic differentials as before.

Since $d\Omega_1$ is normalized by $\oint_{a_i} d\Omega_1 = 0$ and has residue 1 at ∞^+ , we obtain

$$(6.13) \quad \oint_{\ell} d\Omega_1 = 2\pi i.$$

Therefore

$$(6.14) \quad \oint_{a_3} d\Omega_1 = \oint_{-a_1-a_2+\ell} d\Omega_1 = -\sum_{i=1}^2 \oint_{a_i} d\Omega_1 + \oint_{\ell} d\Omega_1 = 0 + 2\pi i = 2\pi i.$$

Using the path C_P joining ∞^- to ∞^+ and the relation $\tau(C_P) = C_P + a_3$, the asymptotics (6.12) imply that

$$(6.15) \quad \lim_{P \rightarrow \infty^+} \left(\int_{\infty^-}^{\infty^+} d\Omega_1 - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left(\int_{C_P} d\Omega_1 - 2 \ln \lambda \right) = 2 \ln \omega_1.$$

We compute the conjugate quantity

$$(6.16) \quad \begin{aligned} 2 \overline{\ln(\omega_1)} &= \lim_{P \rightarrow \infty^+} \left(\int_{C_P} \overline{d\Omega_1(P)} - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left(\int_{\tau(C_P)} d\Omega_1 - 2 \ln \lambda \right) \\ &= \lim_{P \rightarrow \infty^+} \left(\int_{C_P + a_3} d\Omega_1 - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left(\int_{C_P} d\Omega_1 - 2 \ln \lambda \right) + \oint_{a_3} d\Omega_1 \\ &= 2 \ln(\omega_1) + 2\pi i, \end{aligned}$$

where in the second equality we used $\overline{d\Omega_1(P)} = \tau^* d\Omega_1$ and $\tau(C_P) = C_P + a_3$, and in the last step we used (6.14). From (6.16) we obtain

$$(6.17) \quad \overline{\ln(\omega_1)} = \ln(\omega_1) + \pi i, \quad \Rightarrow \quad \Im(\ln(\omega_1)) = -\frac{\pi}{2}.$$

Hence, writing ω_1 in modulus–argument form gives

$$(6.18) \quad \ln(\omega_1) = \ln|\omega_1| - i\beta, \quad \beta > 0,$$

(and in our normalization one finds $\beta = \frac{\pi}{2}$).

The other normalized Abelian integrals have the asymptotics

$$(6.19) \quad \Omega_2(P) = \pm(\lambda + \omega_2 + o(1)), \quad \Omega_3(P) = \pm(4\lambda^3 + \omega_3 + o(1)), \quad P \rightarrow \infty^\pm,$$

so that, using the same connecting path C_P ,

$$(6.20) \quad 2\omega_2 = \lim_{P \rightarrow \infty^+} \left(\int_{C_P} d\Omega_2 - 2\lambda \right), \quad 2\omega_3 = \lim_{P \rightarrow \infty^+} \left(\int_{C_P} d\Omega_3 - 8\lambda^3 \right).$$

Taking complex conjugates and using the action of τ on C_P one deduces

$$(6.21) \quad 2\overline{\omega_2} = 2\omega_2, \quad 2\overline{\omega_3} = 2\omega_3, \quad \Rightarrow \quad \omega_2, \omega_3 \in \mathbb{R}.$$

Recall the period matrix entries are defined by

$$(6.22) \quad B_{ij} = \oint_{b_j} w_i d\lambda,$$

where $\{w_1, w_2\}$ denote the normalized holomorphic differentials with

$$\oint_{a_k} w_i d\lambda = 2\pi i \delta_{ik}, \quad i, k = 1, 2.$$

Using the antiholomorphic involution τ and the relations $\tau(b_1) = b_1 + a_2$, $\tau(b_2) = b_2 + a_1$ we get

$$(6.23) \quad \begin{aligned} \overline{B_{ij}} &= \oint_{b_j} \overline{w_i d\lambda} = \oint_{\tau(b_j)} w_i d\lambda = \oint_{b_j} w_i d\lambda + \sum_{l \neq j} \oint_{a_l} w_i d\lambda \\ &= B_{ij} + 2\pi i \sum_{l \neq j} \delta_{il}. \end{aligned}$$

Therefore, in matrix form,

$$(6.24) \quad \overline{B} = B + 2\pi i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The Riemann theta function is defined by

$$(6.25) \quad \Theta(z | B) = \sum_{n \in \mathbb{Z}^2} \exp\left(\frac{1}{2} n^\top B n + n^\top z\right).$$

Using (6.24) one checks the conjugation property

$$(6.26) \quad \overline{\Theta(z \mid B)} = \Theta(\bar{z} \mid \bar{B}) = \sum_{n \in \mathbb{Z}^2} \exp\left(\frac{1}{2} n^\top B n + n^\top \bar{z} + 2\pi i n_1 n_2\right) = \Theta(\bar{z} \mid B),$$

since $\exp(2\pi i n_1 n_2) = 1$ for integers n_1, n_2 .

Notational conventions and auxiliary differentials. For later calculations we introduce the following integrals and differentials:

$$(6.27) \quad \mathcal{Y}_{a_i}^{[j]} = \oint_{a_i} \frac{\lambda^j}{y} d\lambda, \quad \mathcal{Y}_{b_i}^{[j]} = \oint_{b_i} \frac{\lambda^j}{y} d\lambda, \quad i = 1, 2, \quad j \geq 0,$$

$$(6.28) \quad \Omega_j(P) = \int_{(\lambda_3, 0)}^P d\Omega_j, \quad d\Omega_2^{(1)} = \sum_{i=0}^3 \frac{c_{2i} \lambda^i}{y(\lambda)} d\lambda, \quad d\Omega_3^{(1)} = \sum_{i=1}^5 \frac{c_{3i} \lambda^i}{y(\lambda)} d\lambda,$$

where $P = (\lambda, y) \in R_2$, the base point $(\lambda_3, 0)$ is fixed, and the constants c_{2i}, c_{3i} (as well as the coefficients d_{ik} appearing in the holomorphic differentials) will be determined later on.

Week 7: Algebro-Geometric Solutions for the mKdV Equation (7)

2025-11-04

Objectives: Focus on Case 1. Introduce two transformations used in the computation of $\mathcal{Y}_{a_i}^{[j]}$ and $\mathcal{Y}_{b_i}^{[j]}$; carry out the evaluation of $\mathcal{Y}_{a_i}^{[0]}, \mathcal{Y}_{a_i}^{[1]}$, and determine the coefficients d_{ij} .

Recall that

$$w_i(\lambda) = \frac{d_{i0} + d_{i1}\lambda}{y(\lambda)}, \quad y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

and

$$(7.1) \quad \oint_{a_j} w_i(\lambda) d\lambda = 2\pi i \delta_{ij}, \quad i, j = 1, 2.$$

Hence, writing the period integrals in matrix form, the normalization (7.1) is equivalent to

$$(7.2) \quad 2\pi i \mathbb{I}_2 = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix}.$$

In particular,

$$(7.3) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} = 2\pi i \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix}^{-1}.$$

Recall the intersection and period conventions for the canonical homology basis $\{a_i, b_i\}_{i=1}^2$:

$$(7.4) \quad a_i \circ a_j = b_i \circ b_j = 0, \quad a_i \circ b_j = \delta_{ij},$$

$$(7.5) \quad \oint_{b_j} w_i(\lambda) d\lambda = B_{ij}.$$

Case 1. $\lambda_1, \lambda_2, \lambda_3 \in i\mathbb{R}$.

Let τ denote complex conjugation on the curve,

$$\tau: (\lambda, y) \mapsto (\lambda^*, y^*).$$

The action of τ on the homology basis may be chosen (up to orientation) as

$$(7.6) \quad \tau(a_1) = -a_2, \quad \tau(a_2) = -a_1, \quad \tau(b_1) = b_2, \quad \tau(b_2) = b_1.$$

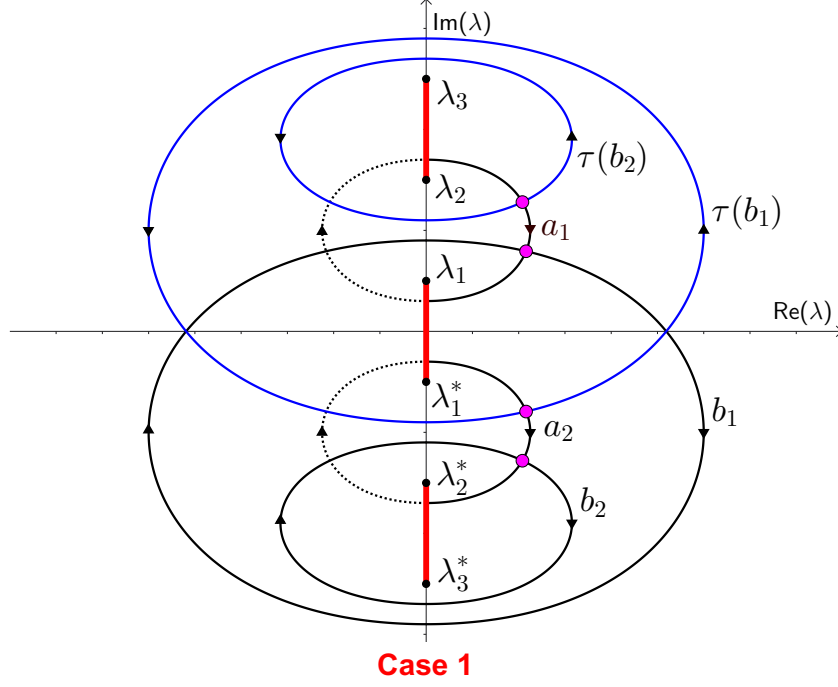


FIGURE 5. To the computations of $\tau(b_1), \tau(b_2)$ for Case 1.

With the choice of cycles described above, the a -periods reduce to integrals along paths on the first sheet:

$$(7.7) \quad \oint_{a_1} \frac{d\lambda}{y} = 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y}, \quad \oint_{a_1} \frac{\lambda d\lambda}{y} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda d\lambda}{y},$$

and

$$(7.8) \quad \oint_{a_2} \frac{d\lambda}{y} = 2 \int_{\lambda_1^*}^{\lambda_2^*} \frac{d\lambda}{y}, \quad \oint_{a_2} \frac{\lambda d\lambda}{y} = 2 \int_{\lambda_1^*}^{\lambda_2^*} \frac{\lambda d\lambda}{y}.$$

Recall the definition

$$(7.9) \quad y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}, \quad \Lambda = \lambda^2, \quad d\Lambda = 2\lambda d\lambda.$$

Applying the substitution $\Lambda = \lambda^2$ transforms the basic integral into a rational function of Λ :

$$(7.10) \quad \int \frac{d\lambda}{y(\lambda)} = \int \frac{d\lambda}{\sqrt{\prod_{i=1}^3 (\lambda^2 - \lambda_i^2)}} = \int \frac{d\Lambda}{2\sqrt{\Lambda \prod_{i=1}^3 (\Lambda - \lambda_i^2)}}.$$

Thus the problem reduces to integrals of the general form

$$(7.11) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}}, \quad \beta_1 < \beta_2 < \beta_3 < \beta_4.$$

To reduce (7.11) to the Legendre normal form we use the change of variable

$$(7.12) \quad z^2 = \frac{\beta_3 - \beta_1}{\beta_4 - \beta_1} \frac{\Lambda - \beta_4}{\Lambda - \beta_3}.$$

A direct computation yields the standard reduction

$$(7.13) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k^2 z^2)}},$$

with the elliptic modulus given by

$$(7.14) \quad k^2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)}{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}.$$

For reference, the incomplete elliptic integrals in Jacobi form are

$$F(z, k) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \quad E(z, k) = \int_0^z \sqrt{\frac{1-k^2t^2}{1-t^2}} dt, \quad 0 < k < 1,$$

and the elliptic integral of the third kind is

$$\Pi(z, \alpha^2, k) = \int_0^z \frac{dt}{(1-\alpha^2t^2)\sqrt{(1-t^2)(1-k^2t^2)}}.$$

These standard forms allow the evaluation of the a -period integrals (7.7)–(7.8) once the correspondence between the branch points $\{\beta_i\}$ and the original λ -branch points is fixed.

Applying (7.13)–(7.14) to our concrete problem: we may take

$$\beta_1 = \lambda_3^2, \quad \beta_2 = \lambda_2^2, \quad \beta_3 = \lambda_1^2, \quad \beta_4 = 0,$$

and (using (7.12)) obtain

$$(7.15) \quad z^2 = \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2} \frac{\Lambda}{\Lambda - \lambda_1^2} = \frac{(\lambda_1^2 - \lambda_3^2) \lambda^2}{\lambda_3^2 (\lambda_1^2 - \lambda^2)}.$$

For these choices the modulus becomes

$$(7.16) \quad k_1^{(1)2} = \frac{(\lambda_1^2 - \lambda_2^2)(0 - \lambda_3^2)}{(\lambda_1^2 - \lambda_3^2)(0 - \lambda_2^2)} = \frac{\lambda_3^2}{\lambda_2^2} \frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2},$$

so that

$$(7.17) \quad \int \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int \frac{dz}{\sqrt{(1-z^2)(1-k_1^{(1)2}z^2)}}, \quad k_1^{(1)} = \frac{\lambda_3}{\lambda_2} \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}$$

Defining the incomplete elliptic integral

$$(7.18) \quad v = F(z, k_1^{(1)}) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k_1^{(1)2}t^2)}}, \quad z = \operatorname{sn}(v, k_1^{(1)}),$$

we obtain the standard elliptic parametrisation of λ via (7.15) and (7.18).

Recall the relations:

$$\operatorname{sn}^2(u, k) + \operatorname{cn}^2(u, k) = 1, \quad \operatorname{dn}^2(u, k) + k^2 \operatorname{sn}^2(u, k) = 1,$$

and the complete elliptic integrals

$$K(k) = F(1, k), \quad K'(k) = F(1, k'), \quad k'^2 = 1 - k^2.$$

From (7.15) one checks the mapping of special Λ -values to z -values:

$$\Lambda = \lambda^2 = 0 \leftrightarrow z = 0 \leftrightarrow v = 0, \quad \Lambda = \lambda_1^2 \leftrightarrow z^2 \rightarrow \infty \leftrightarrow v = \pm iK_1^{(1)'}$$

and

$$\Lambda = \lambda^2 = \lambda_2^2 \leftrightarrow z^2 = \frac{\lambda_2^2}{\lambda_3^2} \frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda_2^2} = \frac{1}{k_1^{(1)2}} \leftrightarrow v = \pm (K_1^{(1)} + iK_1^{(1)'}).$$

With this choice, the a_1 -period can be expressed in the v -variable as an integral between two points in the complex v -plane. In particular, we have

$$(7.19) \quad 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1^{(1)} + iK_1^{(1)'}}^{iK_1^{(1)'}} dv = \frac{-2K_1^{(1)}}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} = \frac{-2iK_1^{(1)'}}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^-.$$

The symmetry under complex conjugation, $\tau : (\lambda, y) \mapsto (\lambda^*, y^*)$, implies the relations

$$(7.20) \quad \left(\oint_{a_2} \frac{d\lambda}{y} \right)^* = \oint_{a_2} \tau^* \frac{d\lambda}{y} = \oint_{\tau(a_2)} \frac{d\lambda}{y} = \oint_{-a_1} \frac{d\lambda}{y} = - \oint_{a_1} \frac{d\lambda}{y} \in \mathbb{R}^+.$$

Similarly, one finds

$$(7.21) \quad \oint_{a_2} \frac{\lambda d\lambda}{y} = \oint_{a_1} \frac{\lambda d\lambda}{y}.$$

Using the substitution $\Lambda = \lambda^2$, one rewrites

$$(7.22) \quad \int \frac{\lambda d\lambda}{y(\lambda)} = \int \frac{d\Lambda}{2\sqrt{\prod_{i=1}^3 (\Lambda - \lambda_i^2)}}.$$

Introducing the change of variables

$$(7.23) \quad z^2 = \frac{\beta_1 - \beta_3}{\Lambda - \beta_3},$$

the integral becomes

$$(7.24) \quad \int \frac{\lambda d\lambda}{y(\lambda)} = \frac{1}{-i\sqrt{\lambda_1^2 - \lambda_3^2}} \int \frac{dz}{\sqrt{(1-z^2)(1-k_2^{(1)2}z^2)}}, \quad k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}.$$

Hence,

$$(7.25) \quad \oint_{a_1} \frac{\lambda d\lambda}{y} = \frac{-2i K_2^{(1)}}{\sqrt{\lambda_1^2 - \lambda_3^2}} = \oint_{a_2} \frac{\lambda d\lambda}{y}.$$

Since

$$(7.26) \quad \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix} = \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & 0 \\ 0 & \oint_{a_1} \frac{\lambda d\lambda}{y} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

the relation (7.3) becomes

$$\begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} = \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & 0 \\ 0 & \oint_{a_1} \frac{\lambda d\lambda}{y} \end{pmatrix}^{-1}.$$

This yields the coefficients

$$(7.27) \quad \begin{aligned} d_{10} &= -\frac{\pi \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}}{2 K_1^{(1)}}, & d_{11} &= -\frac{\pi \sqrt{\lambda_1^2 - \lambda_3^2}}{2 K_2^{(1)}}, \\ d_{20} &= -d_{10}, & d_{21} &= d_{11}. \end{aligned}$$

Supplementary computations:

Define

$$(7.28) \quad F(z, k) = \int_0^z \frac{dt}{f(t, k)}, \quad f(t, k) = \sqrt{(1-t^2)(1-k^2t^2)}, \quad 0 < k < 1,$$

where the principal branch of the square root is chosen and we fix $f(0, k) = 1$.

Set

$$(7.29) \quad K(k) = F(1, k), \quad K'(k) = K(k'), \quad k' = \sqrt{1-k^2}.$$

We take the segments $[-k^{-1}, -1] \cup [1, k^{-1}]$ as branch cuts for $f(t, k)$ on the extended real axis. Starting at $t = 0$ and moving right along the real axis, each time one crosses a branch point an extra phase factor $e^{-\pi i/2} = -i$ is accumulated; moving left from 0 produces the conjugate factor $e^{\pi i/2} = i$. With these conventions, $f(t, k)$ on the first sheet admits the explicit real-axis description

$$(7.30) \quad f(t, k) = \begin{cases} \sqrt{(1-t^2)(1-k^2t^2)} \in \mathbb{R}^+, & t \in (-1, 1), \\ -i\sqrt{(t^2-1)(1-k^2t^2)} \in i\mathbb{R}^-, & t \in (1, k^{-1}), \\ i\sqrt{(t^2-1)(1-k^2t^2)} \in i\mathbb{R}^+, & t \in (-k^{-1}, -1), \\ -\sqrt{(t^2-1)(k^2t^2-1)} \in \mathbb{R}^-, & t \in (-\infty, -k^{-1}) \cup (k^{-1}, \infty). \end{cases}$$

We now evaluate several definite integrals that will be used below.

For the interval (k^{-1}, ∞) perform the substitution

$$(7.31) \quad t = k^{-1}s^{-1}, \quad dt = -k^{-1}s^{-2}ds,$$

which maps $t \in [k^{-1}, \infty)$ to $s \in (0, 1]$. Direct computation gives

$$(7.32) \quad \begin{aligned} \int_{k^{-1}}^{\infty} \frac{dt}{\sqrt{(t^2-1)(k^2t^2-1)}} &= \int_1^0 \frac{-k^{-1}s^{-2}ds}{\sqrt{(k^{-2}s^{-2}-1)(s^{-2}-1)}} \\ &= \int_0^1 \frac{ds}{\sqrt{(1-s^2)(1-k^2s^2)}} \\ &= F(1, k) = K(k). \end{aligned}$$

For the finite interval $(1, k^{-1})$ use the substitution

$$(7.33) \quad t = (1 - (k')^2s^2)^{-1/2}, \quad dt = (k')^2s(1 - (k')^2s^2)^{-3/2}ds,$$

which sends $s \in [0, 1]$ to $t \in [1, k^{-1}]$. Under this change of variables

$$(7.34) \quad \begin{aligned} \sqrt{(t^2-1)(1-k^2t^2)} &= \sqrt{\frac{(k's)^2}{1-(k's)^2} \cdot (k')^2 \frac{1-s^2}{1-(k's)^2}} \\ &= (k')^2s \frac{\sqrt{1-s^2}}{1-(k's)^2}. \end{aligned}$$

Hence

$$(7.35) \quad \begin{aligned} \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} &= \int_0^1 \frac{ds}{\sqrt{(1-s^2)(1-(k')^2s^2)}} \\ &= F(1, k') = K(k') = K'(k). \end{aligned}$$

Combining the evaluations on $[0, 1]$ and $(1, k^{-1})$, and recalling the phase $-i$ on $(1, k^{-1})$ from (7.30), we obtain

$$(7.36) \quad \begin{aligned} F(k^{-1}, k) &= \int_0^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}} + i \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= K(k) + iK'(k). \end{aligned}$$

Taking the limit $z \rightarrow +\infty$ along the positive real axis (passing through k^{-1}) yields

$$(7.37) \quad F(+\infty, k) = K(k) + iK'(k) - K(k) = iK'(k),$$

while symmetry across the origin gives

$$(7.38) \quad F(-1, k) = -F(1, k) = -K(k).$$

Similarly, evaluating F at $-k^{-1}$ (follow the path from 0 to -1 to $-k^{-1}$ and use the phase convention on $(-k^{-1}, -1)$ in (7.30)) yields

$$(7.39) \quad \begin{aligned} F(-k^{-1}, k) &= -K(k) + \int_{-1}^{-k^{-1}} \frac{dt}{i\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= -K(k) + i \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= -K(k) + iK'(k). \end{aligned}$$

Finally, taking $z \rightarrow -\infty$ along the negative real axis (passing through $-k^{-1}$) gives

$$(7.40) \quad F(-\infty, k) = -K(k) + iK'(k) + K(k) = iK'(k).$$

Since

$$(type\ 1) \quad \int \frac{\lambda^{2n}}{y(\lambda)} d\lambda = \int \frac{\Lambda^n d\Lambda}{2(\Lambda(\Lambda - \lambda_1^2)(\Lambda - \lambda_2^2)(\Lambda - \lambda_3^2))^{1/2}}, \quad \Lambda = \lambda^2,$$

and

$$(type\ 2) \quad \int \frac{\lambda^{2n+1}}{y(\lambda)} d\lambda = \int \frac{\Lambda^n d\Lambda}{2((\Lambda - \lambda_1^2)(\Lambda - \lambda_2^2)(\Lambda - \lambda_3^2))^{1/2}}, \quad \Lambda = \lambda^2,$$

we treat the two types of integrals separately.

Computation of Type 1 Integrals $(\mathcal{Y}_{a_i}^{[0]}, \mathcal{Y}_{b_i}^{[0]})$. Introduce a change of variable from Λ to z by

$$(7.41) \quad z^2 = \frac{\beta_3 - \beta_1}{\beta_4 - \beta_1} \frac{\Lambda - \beta_4}{\Lambda - \beta_3}, \quad \beta_1 < \beta_2 < \beta_3 < \beta_4.$$

Solving (7.41) for Λ yields

$$(7.42) \quad \Lambda = \frac{z^2\beta_3(\beta_4 - \beta_1) - (\beta_3 - \beta_1)\beta_4}{z^2(\beta_4 - \beta_1) - (\beta_3 - \beta_1)} = \beta_3 + \frac{(\beta_3 - \beta_1)(\beta_3 - \beta_4)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)}.$$

Differentiation gives

$$(7.43) \quad d\Lambda = \frac{2(\beta_3 - \beta_1)(\beta_4 - \beta_3)(\beta_4 - \beta_1)z}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} dz.$$

The shifted factors $\Lambda - \beta_i$ become rational functions of z . In particular

$$(7.44) \quad \Lambda - \beta_1 = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(z^2 - 1)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)},$$

$$(7.45) \quad \Lambda - \beta_2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)(\beta_4 - \beta_2)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)},$$

$$(7.46) \quad \Lambda - \beta_4 = \frac{(\beta_3 - \beta_4)(\beta_4 - \beta_1)z^2}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)}.$$

Hence the full quartic factor factors as

$$(7.47) \quad \prod_{i=1}^4 (\Lambda - \beta_i) = \frac{(\beta_3 - \beta_1)^2 (\beta_4 - \beta_1)^2 (\beta_4 - \beta_3)^2 z^2 (z^2 - 1)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^4} \\ \times \left((\beta_3 - \beta_2)(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)(\beta_4 - \beta_2) \right).$$

Define the modulus

$$(7.48) \quad k_1^2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)}{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}.$$

Taking a principal square root of (7.47) (and grouping algebraic constants) yields

$$(7.49) \quad \left(\prod_{i=1}^4 (\Lambda - \beta_i) \right)^{1/2} = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(\beta_4 - \beta_3)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} \sqrt{(\beta_3 - \beta_1)(\beta_4 - \beta_2)} z \sqrt{(1 - z^2)(1 - k_1^2 z^2)}.$$

Combining (7.43) and (7.49) produces the standard elliptic reduction

$$(7.50) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{(\beta_4 - \beta_2)(\beta_3 - \beta_1)}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k_1^2 z^2)}},$$

valid on the appropriate choice of branches for the square roots.

Specializing the parameters by

$$(7.51) \quad \beta_1 = \lambda_3^2, \quad \beta_2 = \lambda_2^2, \quad \beta_3 = \lambda_1^2, \quad \beta_4 = 0,$$

the change of variable (7.41) becomes

$$(7.52) \quad z^2 = \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2} \frac{\lambda^2}{\lambda^2 - \lambda_1^2}, \quad z = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\lambda}{\sqrt{\lambda^2 - \lambda_1^2}},$$

where the positive square root is taken for the principal branch. The modulus (7.48) simplifies to

$$(7.53) \quad k_1^2 = \frac{(\lambda_1^2 - \lambda_2^2)(\lambda_2^2 - \lambda_3^2)}{(\lambda_1^2 - \lambda_3^2)(0 - \lambda_2^2)} = \frac{(\lambda_1^2 - \lambda_2^2) \lambda_3^2}{(\lambda_1^2 - \lambda_3^2) \lambda_2^2}.$$

Choose the cut $[-\lambda_1, \lambda_1]$ for the function $\sqrt{\lambda^2 - \lambda_1^2}$ and fix the sheet by requiring

$$\sqrt{\lambda^2 - \lambda_1^2} \sim \lambda \quad (\lambda \rightarrow \infty).$$

For $\text{Re } \lambda \geq 0$ and $\lambda \in i\mathbb{R}$ write $\lambda = it$ with $t \in \mathbb{R}$. If $\lambda_j = it_j$ then

$$(7.54) \quad \sqrt{\lambda^2 - \lambda_1^2} = \begin{cases} i\sqrt{t^2 - t_1^2}, & t > t_1, \\ \sqrt{t_1^2 - t^2}, & -t_1 < t < t_1, \\ -i\sqrt{t^2 - t_1^2}, & t < -t_1. \end{cases}$$

Hence for $\lambda \in i\mathbb{R}$ on the imaginary axis with $\text{Re } \lambda \geq 0$,

$$(7.55) \quad z = \begin{cases} \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\lambda}{\sqrt{\lambda^2 - \lambda_1^2}} \in i\mathbb{R}, & \lambda \in (-\lambda_1, \lambda_1), \\ \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{|\lambda|}{\sqrt{\lambda_1^2 - \lambda^2}} \in \mathbb{R}^+, & \lambda \in (-i\infty, -\lambda_1) \cup (\lambda_1, +i\infty). \end{cases}$$

The oriented real/imaginary segments map under $\lambda \mapsto z$ as follows (the images are indicated in the z -plane):

$$(7.56) \quad \begin{aligned} (\lambda_3, \lambda_2) &\longrightarrow (1, k_1^{-1}), & (\lambda_2, \lambda_1) &\longrightarrow (k_1^{-1}, +\infty), \\ (\lambda_1, 0) &\longrightarrow (+i\infty, 0), & (0, -\lambda_1) &\longrightarrow (0, -i\infty), \\ (-\lambda_1, -\lambda_2) &\longrightarrow (+\infty, k_1^{-1}), & (-\lambda_2, -\lambda_3) &\longrightarrow (k_1^{-1}, 1). \end{aligned}$$

Since

$$(7.57) \quad \lambda y(\lambda) = \sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)} = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(\beta_4 - \beta_3)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} z \sqrt{(1 - z^2)(1 - k_1^2 z^2)},$$

for our chosen map between λ and z we may write

$$\lambda y(\lambda) = c_0 z \sqrt{(1 - z^2)(1 - k_1^2 z^2)}, \quad c_0 > 0.$$

For $\lambda = it \in i\mathbb{R}$, note that $z \in (0, 1)$ when $\lambda \in (\lambda_3, +i\infty)$; in this region $\lambda y(\lambda) \in \mathbb{R}^+$. The branch points $\pm\lambda_3, \pm\lambda_2$ of $y(\lambda)$ correspond to the branch points $1, k_1^{-1}$ of $f(z, k_1)$, where $f(z, k_1) = \sqrt{(1 - z^2)(1 - k_1^2 z^2)}$ and the branch cuts are chosen as in (7.28)–(7.30).

Thus, for $\lambda \in i\mathbb{R}$,

$$\lambda y(\lambda) \sim \begin{cases} z f(z, k_1), & \lambda \in (-\lambda_2, +i\infty) \cup (-\lambda_3, -i\infty), \\ -z f(z, k_1), & \lambda \in (-\lambda_2, -\lambda_3). \end{cases}$$

Hence for $\lambda \in i\mathbb{R} \setminus [-\lambda_2, -\lambda_3]$,

$$(7.58) \quad \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \frac{dz}{f(z, k_1)},$$

and on the interval $(-\lambda_2, -\lambda_3)$, where the sign of the square root changes,

$$(7.59) \quad \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \frac{dz}{-f(z, k_1)}.$$

Using (7.58)–(7.59) together with the standard elliptic integrals, and noting

$$(7.60) \quad \sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)} = -i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2} > 0,$$

one obtains the following closed forms

$$(7.61) \quad \begin{aligned} \mathcal{Y}_{a_1}^{[0]} &= 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{k_1^{-1}}^{+\infty} \frac{dz}{f(z, k_1)} \\ &= \frac{2}{-i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1 + iK_1'}^{iK_1'} dv = \frac{-2i K_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^-, \end{aligned}$$

$$(7.62) \quad \mathcal{Y}_{a_2}^{[0]} = 2 \int_{-\lambda_1}^{-\lambda_2} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{+\infty}^{k_1^{-1}} \frac{dz}{f(z, k_1)} = -\mathcal{Y}_{a_1}^{[0]} = \frac{2i K_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+,$$

$$(7.63) \quad \begin{aligned} \mathcal{Y}_{b_2}^{[0]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{k_1^{-1}}^1 \frac{dz}{-f(z, k_1)} \\ &= \frac{2}{-i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1}^{K_1 + iK_1'} dv = \frac{-2K_1'}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^+, \end{aligned}$$

and

$$\begin{aligned}
 \mathcal{Y}_{b_1}^{[0]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{d\lambda}{y(\lambda)} = -\frac{2K'_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} + \frac{2(-iK'_1 - iK'_1)}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \\
 (7.64) \qquad \qquad \qquad &= \frac{2K'_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-.
 \end{aligned}$$

Computation of Type 2 Integrals $(\mathcal{Y}_{a_i}^{[1]}, \mathcal{Y}_{b_i}^{[1]})$.

Let $\beta_1 < \beta_2 < \beta_3$ and introduce the substitution

$$(7.65) \qquad z^2 = \frac{\beta_1 - \beta_2}{\Lambda - \beta_3}, \quad \Lambda = \beta_3 + \frac{\beta_1 - \beta_3}{z^2}, \quad \Lambda = \lambda^2.$$

Differentiation yields

$$(7.66) \qquad d\Lambda = \frac{2(\beta_3 - \beta_1)}{z^3} dz.$$

Using (7.65) one checks

$$(7.67) \qquad \Lambda - \beta_1 = (\beta_3 - \beta_1) \frac{z^2 - 1}{z^2},$$

$$(7.68) \qquad \Lambda - \beta_2 = \frac{(\beta_3 - \beta_2)z^2 + (\beta_1 - \beta_3)}{z^2},$$

$$(7.69) \qquad \Lambda - \beta_3 = \frac{\beta_1 - \beta_3}{z^2}.$$

Consequently the cubic product factors as

$$(7.70) \qquad \prod_{i=1}^3 (\Lambda - \beta_i) = (\beta_3 - \beta_1)^3 \frac{(z^2 - 1)(1 - k_2^2 z^2)}{z^6},$$

with

$$(7.71) \qquad k_2^2 = \frac{\beta_3 - \beta_2}{\beta_3 - \beta_1} \in (0, 1).$$

Taking principal square roots gives

$$(7.72) \qquad \left(\prod_{i=1}^3 (\Lambda - \beta_i) \right)^{1/2} = (\beta_3 - \beta_1) \sqrt{\beta_1 - \beta_3} \frac{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}{z^3}.$$

Combining (7.66) and (7.72) we obtain the reduction

$$(7.73) \qquad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^3 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{\beta_1 - \beta_3}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}.$$

Specialize $\beta_1 = \lambda_3^2$, $\beta_2 = \lambda_2^2$, $\beta_3 = \lambda_1^2$. Then

$$(7.74) \qquad z^2 = \frac{\lambda_3^2 - \lambda_2^2}{\lambda^2 - \lambda_1^2}, \quad z = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{\sqrt{\lambda_1^2 - \lambda^2}},$$

and

$$(7.75) \qquad k_2^2 = \frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2} \in (0, 1).$$

Choose the cut $[-\lambda_1, \lambda_1]$ for $\sqrt{\lambda_1^2 - \lambda^2}$ and fix the sheet by the condition

$$\sqrt{\lambda_1^2 - \lambda^2} \sim -i\lambda \quad (\lambda \rightarrow \infty).$$

For $\lambda = it$ ($t \in \mathbb{R}$) and $\lambda_j = it_j$ one then has the pointwise values (for $\text{Re } \lambda \geq 0$)

$$(7.76) \quad \sqrt{\lambda_1^2 - \lambda^2} = \begin{cases} \sqrt{t^2 - t_1^2}, & t > t_1, \\ -i\sqrt{t_1^2 - t^2}, & -t_1 < t < t_1, \\ -\sqrt{t^2 - t_1^2}, & t < -t_1, \end{cases}$$

which are consistent with the chosen asymptotics.

Consequently, for $\lambda \in i\mathbb{R}$ with $\text{Re } \lambda \geq 0$, the substitution takes values as follows

$$(7.77) \quad z = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{\sqrt{\lambda_1^2 - \lambda^2}} \in \begin{cases} \mathbb{R}^+, & \lambda \in (\lambda_1, +i\infty), \\ i\mathbb{R}^+, & \lambda \in (-\lambda_1, \lambda_1), \\ \mathbb{R}^-, & \lambda \in (-i\infty, -\lambda_1). \end{cases}$$

The oriented segments in the λ -plane map to the z -plane under (7.77) as follows (principal images):

$$(7.78) \quad (+i\infty, \lambda_3) \longrightarrow (0, 1), \quad (\lambda_3, \lambda_2) \longrightarrow (1, k_2^{-1}), \quad (\lambda_2, \lambda_1) \longrightarrow (k_2^{-1}, +\infty),$$

$$(7.79) \quad (\lambda_1, 0) \longrightarrow \left(+i\infty, \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{-\lambda_1}\right), \quad (0, -\lambda_1) \longrightarrow \left(\frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{-\lambda_1}, +\infty\right),$$

$$(7.80) \quad (-\lambda_1, -\lambda_2) \longrightarrow (-\infty, -k_2^{-1}), \quad (-\lambda_2, -\lambda_3) \longrightarrow (-k_2^{-1}, -1), \quad (-\lambda_3, -i\infty) \longrightarrow (-1, 0).$$

By the Type 2 reduction (with $\Lambda = \lambda^2$) the cubic factor admits the representation

$$y(\lambda) = \prod_{i=1}^3 (\Lambda - \beta_i)^{1/2} = (\beta_3 - \beta_1) \sqrt{\beta_1 - \beta_3} \frac{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}{z^3}, \quad \beta_j = \lambda_{4-j}^2,$$

On the chosen sheet for $\lambda \in i\mathbb{R}$ along the imaginary axis, comparison of signs yields

$$y(\lambda) = -i(\beta_3 - \beta_1) \sqrt{\beta_3 - \beta_1} \frac{f(z, k_2)}{z^3}, \quad f(z, k_2) = \sqrt{(1 - z^2)(1 - k_2^2 z^2)}.$$

where the branch cuts of $f(z, k_2)$ are chosen as in (7.28)–(7.30). Hence, for $\lambda \in i\mathbb{R}$,

$$\frac{\lambda d\lambda}{y(\lambda)} = \frac{1}{-i\sqrt{\beta_3 - \beta_1}} \frac{dz}{f(z, k_2)} = \frac{i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \frac{dz}{f(z, k_2)},$$

Introducing the complete elliptic integrals $K_2 = K(k_2)$ and $K'_2 = K(k'_2)$, and using the standard substitutions along the indicated images, the period integrals (type 2 moments with a factor λ) evaluate as follows.

$$(7.81) \quad \mathcal{Y}_{a_1}^{[1]} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{k_2^{-1}}^{+\infty} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_2 + iK'_2}^{iK'_2} dv = \frac{-2i K_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-,$$

$$(7.82) \quad \mathcal{Y}_{a_2}^{[1]} = 2 \int_{-\lambda_1}^{-\lambda_2} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-\infty}^{-k_2^{-1}} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{iK'_2}^{-K_2 + iK'_2} dv = \frac{-2i K_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-,$$

$$(7.83) \quad \mathcal{Y}_{b_2}^{[1]} = 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-k_2^{-1}}^{-1} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-K_2 + iK'_2}^{-K_2} dv = \frac{2 K'_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+,$$

$$(7.84) \quad \mathcal{Y}_{b_1}^{[1]} = 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{\lambda d\lambda}{y(\lambda)} = \mathcal{Y}_{b_2}^{[1]} = \frac{2 K'_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+.$$

Week 8: Algebraic-Geometric Solutions for the mKdV Equation (8)

2025-11-11

Objectives: Focus on Case 1. Compute $\mathcal{Y}_{b_i}^{[0]}$, and discuss how to determine the value of a multivalued complex function with a prescribed branch cut, taking $y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}$ with the asymptotic condition $y(\lambda) \sim \lambda^3$ as $\lambda \rightarrow \infty$.

We have

$$(8.1) \quad B = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \oint_{b_1} \frac{d\lambda}{y} & \oint_{b_2} \frac{d\lambda}{y} \\ \oint_{b_1} \frac{\lambda d\lambda}{y} & \oint_{b_2} \frac{\lambda d\lambda}{y} \end{pmatrix} = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{b_1}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

We can obtain

$$(8.2) \quad \int_0^\lambda \frac{\chi d\chi}{\sqrt{\prod_{i=1}^3 (\chi^2 - \lambda_i^2)}} = \frac{1}{\sqrt{\lambda_3^2 - \lambda_1^2}} \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k_2^{(1)2}t^2)}} = \frac{1}{\sqrt{\lambda_3^2 - \lambda_1^2}} \int_0^v dv.$$

and

$$(8.3) \quad \lambda^2 = \frac{\lambda_3^2 - \lambda_1^2 \operatorname{cn}^2(v, k_2^{(1)})}{\operatorname{sn}^2(v, k_2^{(1)})}, \quad \operatorname{sn}(v, k_2^{(1)}) = z = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, \quad k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}$$

and

$$v = F(z, k_2^{(1)}) = F\left(\sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, k_2^{(1)}\right)$$

Supplementary computations:

Let $P = (\lambda, y)$ be a point on the Riemann surface defined by

$$y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

The first sheet is fixed by the asymptotic condition

$$y(\lambda) \sim \lambda^3 + O(\lambda), \quad \lambda \rightarrow \infty.$$

Set

$$(8.4) \quad \lambda_j = i t_j, \quad 0 < t_1 < t_2 < t_3, \quad j = 1, 2, 3.$$

For the chosen cut we rewrite $y(\lambda)$ for $\lambda = it \in i\mathbb{R}$ (keeping the factorised form with the fractional square roots):

$$(8.5) \quad \begin{aligned} y(\lambda) &= (\lambda - \lambda_3) \sqrt{\frac{\lambda - \lambda_2}{\lambda - \lambda_3}} (\lambda - \lambda_1) \sqrt{\frac{\lambda + \lambda_1}{\lambda - \lambda_1}} (\lambda + \lambda_2) \sqrt{\frac{\lambda + \lambda_3}{\lambda + \lambda_2}} \\ &= -i(t - t_3)(t - t_1)(t + t_2) \sqrt{\frac{t - t_2}{t - t_3}} \sqrt{\frac{t + t_1}{t - t_1}} \sqrt{\frac{t + t_3}{t + t_2}}. \end{aligned}$$

Here the square root is taken as the principal branch:

$$(8.6) \quad \sqrt{z} = \sqrt{|z|} e^{\frac{1}{2}i \arg z}, \quad \arg z \in (-\pi, \pi].$$

To determine the permutation of real/imaginary values and the signs of $y(it)$ as t varies along the imaginary axis, it is convenient to count

- the sign $s = \text{sign}((t - t_3)(t - t_1)(t + t_2))$ of the real linear prefactor, and
- the number $k \in \{0, 1, 2, 3\}$ of the three real ratios

$$\frac{t - t_2}{t - t_3}, \quad \frac{t + t_1}{t - t_1}, \quad \frac{t + t_3}{t + t_2}$$

that are negative (each negative ratio contributes a factor i from the principal square root).

With these conventions, from (8.5) one has

$$y(it) = (-i)^s (i)^k (\text{positive real}) = -s i^{k+1} (\text{positive real}),$$

so that $y(it)$ is imaginary when $k \equiv 0, 2 \pmod{2}$ and real when $k \equiv 1, 3 \pmod{2}$, with the sign determined by the combination above.

Evaluating s and k on the natural partition of the real axis by $\pm t_1, \pm t_2, \pm t_3$ gives the following classification:

$$(8.7) \quad y(it) \in \begin{cases} i\mathbb{R}^-, & t > t_3, \\ \mathbb{R}^-, & t_2 < t < t_3, \\ i\mathbb{R}^+, & t_1 < t < t_2, \\ \mathbb{R}^+, & -t_1 < t < -t_2, \\ i\mathbb{R}^-, & -t_2 < t < -t_1, \\ \mathbb{R}^-, & -t_3 < t < -t_2, \\ i\mathbb{R}^+, & t < -t_3. \end{cases}$$

For clarity, a short justification for each interval is as follows (sign of linear prefactor s and number k of negative ratios are given in parentheses):

- $t > t_3$: $s = +1, k = 0 \Rightarrow y = -i \cdot (+1) \cdot 1 \cdot (\text{pos}) \in i\mathbb{R}^-$.
- $t_2 < t < t_3$: $s = -1, k = 1 \Rightarrow y = (-1) \cdot (\text{pos}) \in \mathbb{R}^-$.
- $t_1 < t < t_2$: $s = -1, k = 0 \Rightarrow y = +i \cdot (\text{pos}) \in i\mathbb{R}^+$.
- $-t_1 < t < -t_2$: $s = +1, k = 1 \Rightarrow y = +(\text{pos}) \in \mathbb{R}^+$.
- $-t_2 < t < -t_1$: $s = +1, k = 0 \Rightarrow y = -i \cdot (\text{pos}) \in i\mathbb{R}^-$.
- $-t_3 < t < -t_2$: $s = -1, k = 1 \Rightarrow y = (-1) \cdot (\text{pos}) \in \mathbb{R}^-$.
- $t < -t_3$: $s = -1, k = 0 \Rightarrow y = +i \cdot (\text{pos}) \in i\mathbb{R}^+$.

All statements above are obtained without algebraically combining the fractional square-root factors; each square root is treated as the principal branch (8.6) applied to the linear fractional arguments appearing in (8.5).

Near each branch point λ_j or $\lambda_j^* = -\lambda_j$ the function behaves locally as $\sqrt{\lambda \mp \lambda_j}$ · (analytic part), and starting from $\lambda = -i\infty$ (where $y \in i\mathbb{R}^+$), moving upward along the right side of the imaginary axis one acquires at each branch point an additional phase $i\pi/2$, equivalently a multiplicative factor i , fully consistent with the above sign computation.

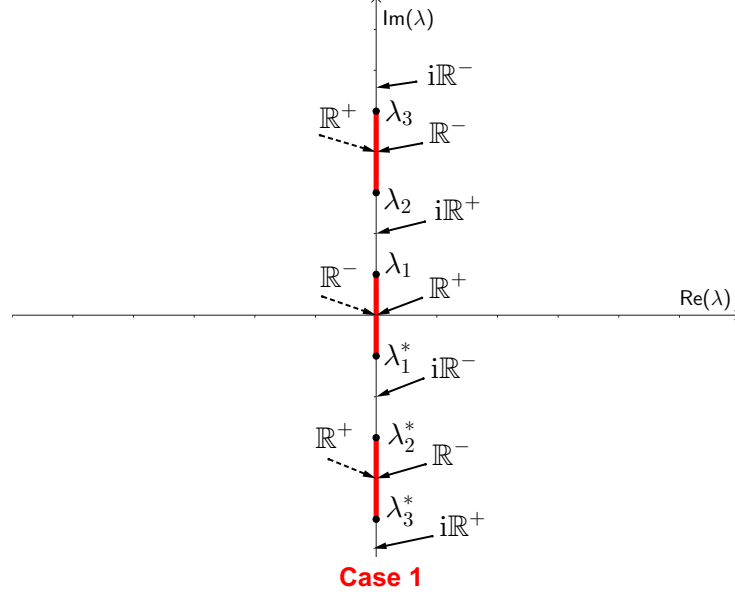


FIGURE 6. The sign values of $y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}$, with $\lambda \rightarrow \infty$, $y \rightarrow \lambda^3 + O(\lambda)$ on the first sheet for Case 1.

Week 9: Algebraic-Geometric Solutions for the mKdV Equation (9)

2025-11-18

Objectives: Focus on Case 1. Introduce the Joukowski function to parametrize the two transformations used in the computation of $\mathcal{Y}_{a_i}^{[j]}$; compute $\mathcal{Y}_{b_i}^{[1]}$ and B_{ij} ; and discuss how to determine a suitable ansatz for the differentials $d\Omega_j$ for $j = 1, 2, 3$.

Introduce the Joukowski transformation

$$(9.1) \quad \lambda = \frac{\lambda_1}{2} (\xi + \xi^{-1}).$$

A direct computation gives

$$(9.2) \quad \lambda_1^2 - \lambda^2 = \lambda_1^2 \left(1 - \left(\frac{1}{2} (\xi + \xi^{-1}) \right)^2 \right) = - \left(\frac{\lambda_1}{2} \right)^2 (\xi - \xi^{-1})^2.$$

For a type 2 integral (see (7.74)),

$$(9.3) \quad z = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{\sqrt{\lambda_1^2 - \lambda^2}} = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{i \frac{\lambda_1}{2}} \frac{1}{\xi - \xi^{-1}}.$$

For a type 1 integral (see (7.52)), using (9.1) one obtains

$$(9.4) \quad z = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\lambda}{\sqrt{\lambda^2 - \lambda_1^2}} = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\frac{\lambda_1}{2} (\xi + \xi^{-1})}{\frac{\lambda_1}{2} (\xi - \xi^{-1})} = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\xi^2 + 1}{\xi^2 - 1}.$$

As $\lambda \rightarrow \infty$, the function y admits the expansion

$$(9.5) \quad y = \lambda^3 \left(1 + \hat{v}_1 \lambda^{-2} + \hat{v}_2 \lambda^{-4} + \dots \right), \quad y^{-1} = \lambda^{-3} \left(1 - \hat{v}_1 \lambda^{-2} + (\hat{v}_1^2 - \hat{v}_2) \lambda^{-4} + \dots \right).$$

Decompose y as

$$(9.6) \quad y = y^{[n+]} + y^{[n-]} = \lambda^3 \left(1 + \hat{v}_1 \lambda^{-2} + \dots + \hat{v}_n \lambda^{-2n} \right) + y^{[n-]},$$

where $y^{[n-]} = \hat{v}_{n+1} \lambda^{-2(n+1)+3} + \dots$. Multiplying (9.6) by y^{-1} and using (9.5) gives

$$(9.7) \quad y^{[n+]} y^{-1} = (y - y^{[n-]}) y^{-1} = 1 - y^{[n-]} y^{-1}$$

$$(9.8) \quad = 1 - (\hat{v}_{n+1} \lambda^{-2(n+1)} + \dots) \lambda^3 \cdot \lambda^{-3} (1 - \hat{v}_1 \lambda^{-2} + \dots)$$

$$(9.9) \quad = 1 - \hat{v}_{n+1} \lambda^{-2(n+1)} + \dots.$$

In particular,

$$(9.10) \quad \lambda^3 y^{-1} = 1 - \hat{v}_1 \lambda^{-2} + \dots,$$

and

$$(9.11) \quad \lambda^3 (1 + \hat{v}_1 \lambda^{-2}) y^{-1} = 1 - \hat{v}_2 \lambda^{-4} + \dots,$$

with analogous identities holding for higher-order terms.

Since

$$(9.12) \quad \Omega_2(P) = \pm (\lambda + \omega_2 + o(1)), \quad p \rightarrow \infty^\pm,$$

differentiation shows that

$$(9.13) \quad d\Omega_2(P) = \pm (1 + O(\lambda^{-2})) d\lambda.$$

Thus $d\Omega_2$ can be written in the form

$$d\Omega_2(P) = \frac{1}{y} (\lambda^3 + c_{21} \lambda + c_{20}) d\lambda.$$

Supplementary computations:

The period normalization for the holomorphic differentials

$$w_i(\lambda) = \sum_{k=0}^1 d_{ik} \lambda^k y^{-1}$$

is expressed by

$$2\pi i \delta_{ij} = \oint_{a_j} w_i d\lambda = \sum_{k=0}^1 d_{ik} \mathcal{Y}_{a_j}^{[k]}, \quad B_{ij} = \oint_{b_j} w_i d\lambda = \sum_{k=0}^1 d_{ik} \mathcal{Y}_{b_j}^{[k]}.$$

The symmetry properties of the period integrals,

$$\mathcal{Y}_{a_1}^{[0]} = -\mathcal{Y}_{a_2}^{[0]}, \quad \mathcal{Y}_{b_1}^{[0]} = -\mathcal{Y}_{b_2}^{[0]}, \quad \mathcal{Y}_{a_1}^{[1]} = \mathcal{Y}_{a_2}^{[1]}, \quad \mathcal{Y}_{b_1}^{[1]} = \mathcal{Y}_{b_2}^{[1]},$$

allow the normalization conditions to be written as

$$(9.14) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{a_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{a_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = 2\pi i \mathbb{I}_2,$$

and

$$(9.15) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{b_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = B^{(1)}.$$

Eliminating the coefficients d_{ik} between (9.14) and (9.15) gives

$$\begin{aligned}
(9.16) \quad B^{(1)} &= \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} (\mathcal{Y}_{a_1}^{[0]})^{-1} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & (\mathcal{Y}_{a_2}^{[1]})^{-1} \mathcal{Y}_{b_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\
&= \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\
&= \pi i \begin{pmatrix} \tau_1^{(1)} & \tau_2^{(1)} \\ -\tau_1^{(1)} & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \pi i \begin{pmatrix} \tau_1^{(1)} + \tau_2^{(1)} & \tau_2^{(1)} - \tau_1^{(1)} \\ \tau_2^{(1)} - \tau_1^{(1)} & \tau_1^{(1)} + \tau_2^{(1)} \end{pmatrix}.
\end{aligned}$$

Here the reduced moduli are given by

$$(9.17) \quad \tau_1^{(1)} = (\mathcal{Y}_{a_1}^{[0]})^{-1} \mathcal{Y}_{b_1}^{[0]} = i \frac{K_1^{(1)'}}{K_1^{(1)}}, \quad \tau_2^{(1)} = (\mathcal{Y}_{a_2}^{[1]})^{-1} \mathcal{Y}_{b_2}^{[1]} = i \frac{K_2^{(1)'}}{K_2^{(1)}}.$$

By the definition of the Riemann theta function,

$$\begin{aligned}
(9.18) \quad \Theta(B^{(1)}; z) &= \sum_{n \in \mathbb{Z}^2} \exp \left\{ \frac{1}{2} n^\top B^{(1)} n + n^\top z \right\} \\
&= \sum_{n \in \mathbb{Z}^2} \exp \left\{ \frac{\pi i}{2} n^\top \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} n + n^\top z \right\},
\end{aligned}$$

where

$$(9.19) \quad n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}, \quad z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \quad n_1, n_2 \in \mathbb{Z}, \quad z_1, z_2 \in \mathbb{C}.$$

Decompose n as

$$(9.20) \quad n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \end{pmatrix} + \begin{pmatrix} t \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} m_1 + \frac{t}{2} \\ m_2 + \frac{t}{2} \end{pmatrix}, \quad t = 0, 1,$$

so that

$$(9.21) \quad n_1 = m_1 + m_2 + t = 2m_1 + n_2 + t, \quad n_2 = m_2 - m_1, \quad m_1, m_2 \in \mathbb{Z}.$$

As (m_1, m_2) range over $\mathbb{Z}^2$, so do (n_1, n_2) .

Substituting (9.20) into (9.18) yields

$$\begin{aligned}
(9.22) \quad \Theta(B^{(1)}; z) &= \sum_{t=0,1} \sum_{m_1, m_2 \in \mathbb{Z}} \exp \left\{ 2\pi i \left(m_1 + \frac{t}{2}, m_2 + \frac{t}{2} \right) \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} m_1 + \frac{t}{2} \\ m_2 + \frac{t}{2} \end{pmatrix} \right. \\
&\quad \left. + \left(m_1 + \frac{t}{2}, m_2 + \frac{t}{2} \right) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right\} \\
&= \sum_{t=0,1} \sum_{m_1, m_2 \in \mathbb{Z}} \exp \left\{ 2\pi i \left(m_1 + \frac{t}{2} \right)^2 \tau_1^{(1)} + 2\pi i \left(m_2 + \frac{t}{2} \right)^2 \tau_2^{(1)} \right. \\
&\quad \left. + \left(m_1 + \frac{t}{2} \right) (z_1 - z_2) + \left(m_2 + \frac{t}{2} \right) (z_1 + z_2) \right\} \\
&= \vartheta_3(z_1 - z_2, 2\tau_1^{(1)}) \vartheta_3(z_1 + z_2, 2\tau_2^{(1)}) + \vartheta_2(z_1 - z_2, 2\tau_1^{(1)}) \vartheta_2(z_1 + z_2, 2\tau_2^{(1)}).
\end{aligned}$$

Definition 3. The Jacobi theta functions are defined by

$$\begin{aligned}
(9.23) \quad \vartheta_1(u, \tau) &:= \Theta \begin{bmatrix} 1 \\ 1 \end{bmatrix} (u, \tau), \quad \vartheta_2(u, \tau) := \Theta \begin{bmatrix} 1 \\ 0 \end{bmatrix} (u, \tau), \\
\vartheta_3(u, \tau) &:= \Theta \begin{bmatrix} 0 \\ 0 \end{bmatrix} (u, \tau), \quad \vartheta_4(u, \tau) := \Theta \begin{bmatrix} 0 \\ 1 \end{bmatrix} (u, \tau),
\end{aligned}$$

with $\text{Im } \tau > 0$ and

$$(9.24) \quad \Theta \left[\begin{smallmatrix} \epsilon \\ \epsilon' \end{smallmatrix} \right] (u, \tau) := \sum_{n \in \mathbb{Z}} \exp \left\{ \pi i \left(n + \frac{\epsilon}{2} \right)^2 \tau + \left(n + \frac{\epsilon}{2} \right) (u + \pi i \epsilon') \right\}, \quad \epsilon, \epsilon' \in \{0, 1\}.$$

Assume

$$(9.25) \quad d\Omega_2(P) = \sum_{i=0}^3 c_{2i} \frac{\lambda^i}{y(\lambda)} d\lambda, \quad d\Omega_3(P) = \sum_{i=0}^5 c_{3i} \frac{\lambda^i}{y(\lambda)} d\lambda.$$

The Abelian integrals satisfy, as $P = (\lambda, y) \rightarrow \infty^\pm$,

$$(9.26) \quad \Omega_2(P) = \pm(\lambda + \omega_2 + o(1)), \quad \Omega_3(P) = \pm(4\lambda^3 + \omega_3 + o(1)),$$

and differentiation yields the asymptotics of the differentials

$$(9.27) \quad d\Omega_2(P) = \pm(1 + O(\lambda^{-2})) d\lambda, \quad d\Omega_3(P) = \pm(12\lambda^2 + O(\lambda^{-2})) d\lambda.$$

Two equivalent ways determine the coefficients c_{2i} and c_{3i} .

Method A — matching via the asymptotic identities. Substitute the ansatzes (9.25) into the asymptotic identities (9.27) and use (9.10)-(9.11).

For $d\Omega_2$ we require

$$\left(\sum_{i=0}^3 c_{2i} \lambda^i \right) y^{-1} = \pm(1 + O(\lambda^{-2})) = \lambda^3 y^{-1},$$

hence

$$(9.28) \quad c_{23} = 1, \quad c_{22} = 0.$$

For $d\Omega_3$ the asymptotic condition reads

$$\left(\sum_{i=0}^5 c_{3i} \lambda^i \right) y^{-1} = \pm(12\lambda^2 + O(\lambda^{-2})) = 12\lambda^2(1 + O(\lambda^{-4})) = 12\lambda^2 \cdot \lambda^3(1 + \hat{v}_1 \lambda^{-2}) y^{-1},$$

and matching powers of λ gives

$$(9.29) \quad c_{35} = 12, \quad c_{34} = 0, \quad c_{33} = 12\hat{v}_1, \quad c_{32} = 0.$$

Method B — direct series expansion. Expand the rational expressions in powers of λ^{-1} using the explicit series (9.5) and compare coefficients.

For $d\Omega_2$ compute

$$(9.30) \quad \begin{aligned} \left(\sum_{i=0}^3 c_{2i} \lambda^i \right) y^{-1} &= (c_{23} \lambda^3 + c_{22} \lambda^2 + c_{21} \lambda + c_{20}) \lambda^{-3} (1 - \hat{v}_1 \lambda^{-2} + \dots) \\ &= c_{23} + c_{22} \lambda^{-1} + (c_{21} - c_{23} \hat{v}_1) \lambda^{-2} + \dots \end{aligned}$$

Matching with $\pm(1 + O(\lambda^{-2}))$ shows directly that $c_{23} = 1$ and $c_{22} = 0$, in agreement with (9.28).

For $d\Omega_3$ compute

$$(9.31) \quad \begin{aligned} \left(\sum_{i=0}^5 c_{3i} \lambda^i \right) y^{-1} &= (c_{35} \lambda^5 + c_{34} \lambda^4 + c_{33} \lambda^3 + c_{32} \lambda^2 + \dots) \lambda^{-3} (1 - \hat{v}_1 \lambda^{-2} + \dots) \\ &= c_{35} \lambda^2 + c_{34} \lambda + (c_{33} - \hat{v}_1 c_{35}) + (c_{32} - \hat{v}_1 c_{34}) \lambda^{-1} + \dots \end{aligned}$$

Matching with $(12\lambda^2 + O(\lambda^{-2}))$ yields $c_{35} = 12$, $c_{34} = 0$, and $c_{33} - \hat{v}_1 c_{35} = 0$, $c_{32} - \hat{v}_1 c_{34} = 0$, hence

$$c_{35} = 12, \quad c_{34} = 0, \quad c_{33} = 12\hat{v}_1, \quad c_{32} = 0,$$

which coincides with (9.29).

Thus both routes produce the same values for the top coefficients. Collecting the results gives the following forms

$$(9.32) \quad d\Omega_2(P) = \frac{\lambda^3 + c_{21} \lambda + c_{20}}{y(\lambda)} d\lambda,$$

and

$$(9.33) \quad d\Omega_3(P) = \frac{12\lambda^5 + 12\hat{v}_1\lambda^3 + c_{31}\lambda + c_{30}}{y(\lambda)} d\lambda,$$

where the remaining coefficients $c_{21}, c_{20}, c_{31}, c_{30}$ are fixed by the a -period normalization conditions.

Week 10: Algebro-Geometric Solutions for the mKdV Equation (10)

2025-11-25

Objectives: Focus on Case 1. Compute $c_{2i}, c_{3,i}$ and $\Omega_j^{(1)}(P)$.

The meromorphic differentials take the form

$$(10.1) \quad \begin{aligned} d\Omega_2^{(1)}(P) &= \frac{\lambda^3 + c_{21}\lambda + c_{20}}{y(\lambda)} d\lambda, \\ d\Omega_3^{(1)}(P) &= \frac{12\lambda^5 - 3v\lambda^3 + c_{31}\lambda + c_{30}}{y(\lambda)} d\lambda, \end{aligned}$$

with

$$(10.2) \quad v = 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2).$$

The constants $c_{21}, c_{20}, c_{31}, c_{30}$ are determined by imposing vanishing of the a -periods,

$$(10.3) \quad \oint_{a_i} d\Omega_2^{(1)} = \oint_{a_i} d\Omega_3^{(1)} = 0, \quad i = 1, 2.$$

For $d\Omega_2^{(1)}$ the conditions (10.3) yield the linear system

$$(10.4) \quad \begin{pmatrix} \oint_{a_1} \frac{\lambda}{y} d\lambda & \oint_{a_1} \frac{d\lambda}{y} \\ \oint_{a_2} \frac{\lambda}{y} d\lambda & \oint_{a_2} \frac{d\lambda}{y} \end{pmatrix} \begin{pmatrix} c_{21} \\ c_{20} \end{pmatrix} = - \begin{pmatrix} \oint_{a_1} \frac{\lambda^3}{y} d\lambda \\ \oint_{a_2} \frac{\lambda^3}{y} d\lambda \end{pmatrix} = - \oint_{a_1} \frac{\lambda^3}{y} d\lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Using the symmetry of the chosen a -cycles the period matrix on the left of (10.4) decomposes as

$$\begin{pmatrix} \oint_{a_1} \frac{\lambda}{y} d\lambda & \oint_{a_1} \frac{d\lambda}{y} \\ \oint_{a_2} \frac{\lambda}{y} d\lambda & \oint_{a_2} \frac{d\lambda}{y} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \oint_{a_1} \frac{\lambda}{y} d\lambda & 0 \\ 0 & \oint_{a_1} \frac{d\lambda}{y} \end{pmatrix},$$

so the system (10.4) is readily solved. The solution is

$$(10.5) \quad \begin{pmatrix} c_{21} \\ c_{20} \end{pmatrix} = - \oint_{a_1} \frac{\lambda^3}{y} d\lambda \begin{pmatrix} \left(\oint_{a_1} \frac{\lambda}{y} d\lambda \right)^{-1} & 0 \\ 0 & \left(\oint_{a_1} \frac{d\lambda}{y} \right)^{-1} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = - \left(\oint_{a_1} \frac{\lambda^3}{y} d\lambda \left(\oint_{a_1} \frac{\lambda}{y} d\lambda \right)^{-1} \right).$$

Using the previously computed values of the a_1 -periods,

$$(10.6) \quad \mathcal{Y}_{a_1}^{[3]} = - \frac{2i(\lambda_3^2 K_2^{(1)} + (\lambda_1^2 - \lambda_3^2) E_2^{(1)})}{\sqrt{\lambda_1^2 - \lambda_3^2}}, \quad \mathcal{Y}_{a_1}^{[1]} = - \frac{2i K_2^{(1)}}{\sqrt{\lambda_1^2 - \lambda_3^2}},$$

one obtains the closed form

$$(10.7) \quad c_{21} = \frac{(\lambda_3^2 - \lambda_1^2) E_2^{(1)}}{K_2^{(1)}} - \lambda_3^2, \quad c_{20} = 0.$$

The same procedure applied to $d\Omega_3^{(1)}$ (matching its a -periods to zero) gives

$$(10.8) \quad c_{30} = 0, \quad c_{31} = 4v_1 - \lambda_3^2 v - v(\lambda_1^2 - \lambda_3^2) \frac{E_2^{(1)}}{K_2^{(1)}},$$

where

$$(10.9) \quad v_1 = \lambda_1^2 \lambda_2^2 + \lambda_1^2 \lambda_3^2 + \lambda_2^2 \lambda_3^2,$$

and v is given by (10.2). Thus the constants appearing in (10.1) are determined by (10.7) and (10.8).

Thus

$$(10.10) \quad \begin{aligned} \Omega_2^{(1)}(P) &= \int_{(\lambda_3, 0)}^P d\Omega_2^{(1)} = \int_{\lambda_3}^{\lambda} \frac{\mu^3 + \left((\lambda_3^2 - \lambda_1^2) \frac{E_2^{(1)}}{K_2^{(1)}} - \lambda_3^2 \right) \mu}{y(\mu)} d\mu \\ &= \frac{1}{i\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_2^{(1)}}^v \left(\frac{\lambda_3^2 - \lambda_1^2}{\operatorname{sn}^2(u, k_2^{(1)})} + (\lambda_3^2 - \lambda_1^2) \frac{E_2^{(1)}}{K_2^{(1)}} - (\lambda_3^2 - \lambda_1^2) \right) du \\ &= -i\sqrt{\lambda_1^2 - \lambda_3^2} \int_{K_2^{(1)}}^v \left(-\frac{1}{\operatorname{sn}^2(u, k_2^{(1)})} - \frac{E_2^{(1)}}{K_2^{(1)}} + 1 \right) du, \end{aligned}$$

where

$$(10.11) \quad v = F\left(\sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, k_2^{(1)}\right), \quad \operatorname{sn}^2(v, k_2^{(1)}) = \frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}, \quad \lambda^2 = \frac{\lambda_3^2 - \lambda_1^2 \operatorname{cn}^2(v, k_2^{(1)})}{\operatorname{sn}^2(v, k_2^{(1)})}.$$

Define the Jacobi Zeta function by

$$(10.12) \quad Z(u, k) = \frac{\partial}{\partial u} \ln \Theta_4\left(\frac{i\pi u}{K(k)}\right) = \int_0^u (\operatorname{dn}^2(z, k) - \frac{E(k)}{K(k)}) dz,$$

where $K(k)$ and $E(k)$ are the complete elliptic integrals of the first and second kind.

Writing $k = k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}$ and denoting the elliptic argument by x , substitution into (10.10) gives

$$(10.13) \quad \begin{aligned} \Omega_2^{(1)}(P) &= -i\sqrt{\lambda_1^2 - \lambda_3^2} \int_{K_2^{(1)}}^v \left(\operatorname{dn}^2(x, k) + k^2 \operatorname{sn}^2(x, k) - \frac{E_2^{(1)}}{K_2^{(1)}} - \frac{1}{\operatorname{sn}^2(x, k)} \right) dx \\ &= -i\sqrt{\lambda_1^2 - \lambda_3^2} \left[Z(x, k) + \frac{\operatorname{cn}(x, k) \operatorname{dn}(x, k)}{\operatorname{sn}(x, k)} \right]_{x=K_2^{(1)}}^{x=v}. \end{aligned}$$

Since the boundary term at $x = K_2^{(1)}$ vanishes, (10.13) becomes

$$(10.14) \quad \Omega_2^{(1)}(P) = -i\sqrt{\lambda_1^2 - \lambda_3^2} \left(Z(v, k) + \frac{\operatorname{cn}(v, k) \operatorname{dn}(v, k)}{\operatorname{sn}(v, k)} \right).$$

Using (10.11), one has

$$\operatorname{sn}(v, k) = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, \quad \operatorname{cn}(v, k) = \sqrt{\frac{\lambda_3^2 - \lambda^2}{\lambda_1^2 - \lambda^2}}, \quad \operatorname{dn}(v, k) = \sqrt{\frac{\lambda_2^2 - \lambda^2}{\lambda_1^2 - \lambda^2}},$$

and then

$$(10.15) \quad \frac{\operatorname{cn}(v, k) \operatorname{dn}(v, k)}{\operatorname{sn}(v, k)} = \sqrt{\frac{(\lambda_3^2 - \lambda^2)(\lambda_2^2 - \lambda^2)}{(\lambda_1^2 - \lambda^2)(\lambda_1^2 - \lambda_3^2)}}.$$

Similarly,

$$(10.16) \quad \Omega_3^{(1)}(P) = \int_{\lambda_3}^{\lambda} \frac{12\mu^5 - 4v\mu^3 + 4v_1\mu}{y(\mu)} d\mu + v\Omega_2^{(1)}(P) = 4y + v\Omega_2^{(1)}(P),$$

where v and v_1 are defined by (10.2) and (10.9).

Furthermore,

$$(10.17) \quad d\Omega_1^{(1)} = \frac{\lambda^2 + c_{10}}{y(\lambda)} d\lambda, \quad c_{10} = -\lambda_1^2 \Pi\left(\frac{\lambda_2^2 - \lambda_1^2}{\lambda_2^2}, k_1^{(1)}\right) / K_1^{(1)},$$

and the corresponding elliptic parametrization is

$$(10.18) \quad \nu = F\left(\sqrt{\frac{\lambda_1^2(\lambda_1^2 - \lambda_3^2)}{\lambda_3^2(\lambda_1^2 - \lambda_2^2)}}, k_1^{(1)}\right), \quad \lambda^2 = \frac{\lambda_1^2 \lambda_3^2 \operatorname{sn}^2(\nu, k_1^{(1)})}{\lambda_1^2 - \lambda_3^2 \operatorname{cn}^2(\nu, k_1^{(1)})}.$$

Hence

$$(10.19) \quad \Omega_1^{(1)}(P) = \int_{\lambda_3}^{\lambda} \frac{\mu^2 + c_{10}}{y(\mu)} d\mu.$$

One has

$$(10.20) \quad \Pi(u, \alpha^2, k) = \frac{\operatorname{sn}(a, k)}{\operatorname{cn}(a, k) \operatorname{dn}(a, k)} \left(\frac{1}{2} \ln \frac{\vartheta_1\left(\frac{i\pi(a+u)}{K}, \tau\right)}{\vartheta_1\left(\frac{i\pi(a-u)}{K}, \tau\right)} - u Z(a, k) \right), \quad \alpha = \frac{1}{\operatorname{sn}(a, k)}$$

and

$$(10.21) \quad \Omega_1^{(1)}(P) = \frac{-i}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1^{(1)}}^{\nu} \frac{\lambda_1^2 \lambda_3^2 \operatorname{sn}^2(x, k_1^{(1)})}{\lambda_1^2 - \lambda_3^2 \operatorname{cn}^2(x, k_1^{(1)})} - \frac{\lambda_1^2}{K_1^{(1)}} \Pi\left(\frac{\lambda_2^2 - \lambda_1^2}{\lambda_2^2}, k_1^{(1)}\right) dx.$$

Supplementary computations:

Computation of Type 1 Integrals ($\mathcal{Y}_{a_i}^{[2]}, \mathcal{Y}_{b_i}^{[2]}$).

We introduce the auxiliary function Λ (cf. (7.42)) by

$$(10.22) \quad \Lambda(z) = \frac{\lambda_1^2 z^2}{z^2 - \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} = \frac{\lambda_1^2 \lambda_3^2 z^2}{\lambda_3^2 z^2 + (\lambda_1^2 - \lambda_3^2)} = \frac{\lambda_1^2 \lambda_3^2 z^2}{\lambda_1^2 - \lambda_3^2 (1 - z^2)}.$$

Recall the relation between z and the elliptic argument ν ,

$$(10.23) \quad \nu = F(z, k_1^{(1)}), \quad z = \operatorname{sn}(\nu, k_1^{(1)}).$$

We will use the imaginary-period shift identities for Jacobi elliptic functions,

$$(10.24) \quad \operatorname{sn}(\nu + iK_1', k_1^{(1)}) = \frac{1}{k_1^{(1)} \operatorname{sn}(\nu, k_1^{(1)})}, \quad \operatorname{cn}(\nu + iK_1', k_1^{(1)}) = -i \frac{\operatorname{dn}(\nu, k_1^{(1)})}{k_1^{(1)} \operatorname{sn}(\nu, k_1^{(1)})}.$$

Set

$$(10.25) \quad \nu = \nu_1 + iK_1',$$

so that $\operatorname{sn}(\nu, k_1^{(1)})$ and $\operatorname{cn}(\nu, k_1^{(1)})$ in (10.22) may be expressed in terms of ν_1 . Substituting $z = \operatorname{sn}(\nu, k_1^{(1)})$ into (10.22) and using (10.24) yields

$$(10.26) \quad \begin{aligned} \Lambda &= \frac{\lambda_1^2 \lambda_3^2 \operatorname{sn}^2(\nu, k_1^{(1)})}{\lambda_1^2 - \lambda_3^2 \operatorname{cn}^2(\nu, k_1^{(1)})} = \frac{\lambda_1^2 \lambda_3^2}{\lambda_1^2 (k_1^{(1)})^2 \operatorname{sn}^2(\nu_1, k_1^{(1)}) + \lambda_3^2 \operatorname{dn}^2(\nu_1, k_1^{(1)})} \\ &= \frac{\lambda_1^2 \lambda_3^2}{\lambda_3^2 + (k_1^{(1)})^2 (\lambda_1^2 - \lambda_3^2) \operatorname{sn}^2(\nu_1, k_1^{(1)})}. \end{aligned}$$

Using the identity

$$(k_1^{(1)})^2 = \frac{(\lambda_1^2 - \lambda_2^2) \lambda_3^2}{(\lambda_1^2 - \lambda_3^2) \lambda_2^2},$$

the expression (10.26) can be rewritten in the equivalent forms

$$(10.27) \quad \Lambda = \frac{\lambda_1^2 \lambda_3^2}{\lambda_3^2 + \frac{\lambda_3^2}{\lambda_2^2} (\lambda_1^2 - \lambda_2^2) \operatorname{sn}^2(\nu_1, k_1^{(1)})} = \frac{\lambda_1^2}{1 - \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2} \operatorname{sn}^2(\nu_1, k_1^{(1)})}.$$

The last form highlights that Λ becomes a simple rational function of $\text{sn}^2(\nu_1, k_1^{(1)})$, which is convenient for integrating.

We now compute $\mathcal{Y}_{a_1}^{[2]}$. Starting from the integral in (type 1) and changing variables to the elliptic parameter ν (with the shift $\nu = \nu_1 + iK_1'$) one obtains

$$(10.28) \quad \mathcal{Y}_{a_1}^{[2]} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda^2 d\lambda}{y(\lambda)} = \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1 + iK_1'}^{iK_1'} \frac{d\nu}{1 - \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2} \text{sn}^2(\nu - iK_1', k_1^{(1)})}.$$

The integrand is an elementary function of sn^2 , hence the integral is an incomplete elliptic integral of the third kind. Writing the integral in terms of $\Pi(\alpha_1^2, u, k)$, with characteristic $\alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}$, modulus $k = k_1^{(1)}$, and upper limit u , we have

$$(10.29) \quad \mathcal{Y}_{a_1}^{[2]} = \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi(\alpha_1^2, u, k_1^{(1)}) \Big|_{u=K_1}^{u=0}, \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}.$$

Evaluating the difference of the incomplete elliptic integrals between $u = K_1$ and $u = 0$ gives the complete elliptic integral of the third kind $\Pi(\alpha_1^2, k)$. Therefore

$$(10.30) \quad \mathcal{Y}_{a_1}^{[2]} = \frac{-2i\lambda_1^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi(\alpha_1^2, k_1^{(1)}), \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}.$$

By symmetry of the contour (or by the corresponding sign change in the integration limits), one obtains the companion relation

$$(10.31) \quad \mathcal{Y}_{a_2}^{[2]} = -\mathcal{Y}_{a_1}^{[2]}.$$

Similarly,

$$(10.32) \quad \begin{aligned} \mathcal{Y}_{b_2}^{[2]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda^2 d\lambda}{y(\lambda)} = \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1}^{K_1 + iK_1'} \frac{d\nu}{1 - \alpha_1^2 \text{sn}^2(\nu - iK_1', k_1^{(1)})} \\ &= \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \Pi(\alpha_1^2, u, k_1^{(1)}) \Big|_{u=K_1 - iK_1'}^{u=K_1}, \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}. \end{aligned}$$

We use the standard Jacobi identities

$$(10.33) \quad \text{sn}(z + K, k) = \frac{\text{cn}(z, k)}{\text{dn}(z, k)}, \quad \text{sn}(iz, k) = i \frac{\text{sn}(z, k')}{\text{cn}(z, k')},$$

$$(10.34) \quad \text{cn}(iz, k) = \frac{1}{\text{cn}(z, k')}, \quad \text{dn}(iz, k) = \frac{\text{dn}(z, k')}{\text{cn}(z, k')},$$

which imply

$$(10.35) \quad \text{dn}(z - K_1', k_1') = \frac{k_1}{\text{dn}(z, k_1')}, \quad \text{sn}(iz + K_1, k) = \frac{1}{\text{dn}(z, k')}.$$

Hence,

$$(10.36) \quad \text{sn}(iu + K_1 - iK_1', k_1^{(1)}) = \frac{1}{\text{dn}(u - K_1', k_1^{(1)'})} = \frac{\text{dn}(u, k_1^{(1)'})}{k_1^{(1)}}.$$

Let $v = iu + K_1$. Then

$$\begin{aligned}
\mathcal{Y}_{b_2}^{[2]} &= \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_0^{K'_1} \frac{i du}{1 - \alpha_1^2 \operatorname{sn}^2(iu + K_1 - iK'_1, k_1^{(1)})} \\
(10.37) \quad &= \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_0^{K'_1} \frac{i du}{1 - \alpha_1^2 \left(\frac{1}{(k_1^{(1)})^2} - \left(\frac{k_1^{(1)'}}{k_1^{(1)}} \right)^2 \operatorname{sn}^2(u, k_1^{(1)'}) \right)} \\
&= \frac{-2\lambda_3^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_0^{K'_1} \frac{du}{1 - \alpha_2^2 \operatorname{sn}^2(u, k_1^{(1)'})} = \frac{-2\lambda_3^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi(\alpha_2^2, k_1^{(1)'}), \quad \alpha_2^2 = \frac{\lambda_2^2 - \lambda_3^2}{\lambda_2^2}.
\end{aligned}$$

One has

$$\begin{aligned}
\mathcal{Y}_{b_1}^{[2]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda^2 d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{\lambda^2 d\lambda}{y(\lambda)} \\
(10.38) \quad &= \mathcal{Y}_{b_2}^{[2]} + \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{iK'_1}^{-iK'_1} \frac{dv}{1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)})}.
\end{aligned}$$

We simplify the elliptic integrand using standard shifts of Jacobi functions. For complementary modulus k' one has

$$\operatorname{sn}(u - K', k') = -\frac{\operatorname{cn}(u, k')}{\operatorname{dn}(u, k')}, \quad \operatorname{cn}(u - K', k') = \frac{k \operatorname{sn}(u, k')}{\operatorname{dn}(u, k')}.$$

From these identities it follows that,

$$\operatorname{sn}(iu - iK'_1, k_1^{(1)}) = i \frac{\operatorname{sn}(u - K'_1, k_1^{(1)'})}{\operatorname{cn}(u - K'_1, k_1^{(1)'})} = -\frac{i}{k_1^{(1)}} \frac{\operatorname{cn}(u, k_1^{(1)'})}{\operatorname{sn}(u, k_1^{(1)'})}.$$

Now set $v = iu$ and substitute the last expression into the integrand. Using

$$\alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}, \quad (k_1^{(1)})^2 = \frac{(\lambda_1^2 - \lambda_2^2)\lambda_3^2}{(\lambda_1^2 - \lambda_3^2)\lambda_2^2},$$

we compute the denominator appearing in the integrand

$$\begin{aligned}
1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)}) &= 1 - \frac{\lambda_1^2 - \lambda_2^2}{\lambda_2^2} \frac{1}{(k_1^{(1)})^2} \frac{\operatorname{cn}^2(u, k_1^{(1)'})}{\operatorname{sn}^2(u, k_1^{(1)'})} = 1 - \frac{\lambda_1^2 - \lambda_3^2}{\lambda_3^2} \frac{\operatorname{cn}^2(u, k_1^{(1)'})}{\operatorname{sn}^2(u, k_1^{(1)'})} \\
(10.39) \quad &= \frac{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2) \operatorname{cn}^2(u, k_1^{(1)'})}{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'})} = \frac{\lambda_1^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2)}{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'})}.
\end{aligned}$$

This expression for the denominator is used to rewrite the integral in (10.38) in terms of elliptic functions with argument u . From (10.39) we obtain the partial-fraction-type decomposition

$$\begin{aligned}
\frac{1}{1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)})} &= \frac{\lambda_3^2}{\lambda_1^2} + \frac{\frac{\lambda_3^2}{\lambda_1^2} (\lambda_1^2 - \lambda_3^2)}{\lambda_1^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2)} \\
(10.40) \quad &= \frac{\lambda_3^2}{\lambda_1^2} - \frac{\lambda_3^2}{\lambda_1^2} \frac{1}{1 - \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2} \operatorname{sn}^2(u, k_1^{(1)'})}.
\end{aligned}$$

Hence the relation between $\mathcal{Y}_{b_1}^{[2]}$ and $\mathcal{Y}_{b_2}^{[2]}$ follows:

$$\begin{aligned}
(10.41) \quad \mathcal{Y}_{b_1}^{[2]} &= \mathcal{Y}_{b_2}^{[2]} + \frac{2\lambda_3^2}{-i\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1'}^{-K_1'} \left(1 - \frac{1}{1 - \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2} \operatorname{sn}^2(u, k_1^{(1)'})} \right) i du \\
&= \mathcal{Y}_{b_2}^{[2]} + \frac{4\lambda_3^2 K_1'}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} - \frac{4\lambda_3^2}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \Pi\left(\frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2}, k_1^{(1)'}\right) \\
&= \mathcal{Y}_{b_2}^{[2]} + 2\lambda_3^2 \mathcal{Y}_{b_1}^{[0]} - \frac{4\lambda_3^2}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \Pi\left(\alpha_3^2, k_1^{(1)'}\right), \quad \alpha_3^2 = \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2},
\end{aligned}$$

where the middle line uses the definition of the complete elliptic integral of the third kind and the last line expresses the K_1' -term via $\mathcal{Y}_{b_1}^{[0]}$.

Computation of Type 2 Integrals ($\mathcal{Y}_{a_i}^{[3]}, \mathcal{Y}_{b_i}^{[3]}$).

For integrals of type 2, we use the parametrization

$$(10.42) \quad \Lambda = \lambda_1^2 + \frac{\lambda_3^2 - \lambda_1^2}{z^2}, \quad z = \operatorname{sn}(\nu, k_2^{(1)}), \quad (k_2^{(1)})^2 = \frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda_3^2}.$$

Writing $\nu = \nu_1 + iK'$, this expression admits several equivalent forms. Indeed,

$$\Lambda = \lambda_1^2 + (\lambda_3^2 - \lambda_1^2) (k_2^{(1)})^2 \operatorname{sn}^2(\nu_1, k_2^{(1)}) = \lambda_1^2 + (\lambda_3^2 - \lambda_1^2) (1 - \operatorname{dn}^2(\nu_1, k_2^{(1)})),$$

and hence

$$\Lambda = \lambda_3^2 + (\lambda_1^2 - \lambda_3^2) \operatorname{dn}^2(\nu_1, k_2^{(1)}).$$

Using this representation, we compute

$$(10.43) \quad \mathcal{Y}_{a_1}^{[3]} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda^3 d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_2 + iK_2'}^{iK_2'} \left(\lambda_3^2 + (\lambda_1^2 - \lambda_3^2) \operatorname{dn}^2(\nu - iK_2', k_2^{(1)}) \right) d\nu.$$

The integral separates into a constant term and a term involving dn^2 . Evaluating each contribution yields

$$\mathcal{Y}_{a_1}^{[3]} = -\frac{2i\lambda_3^2 K_2^{(1)}}{\sqrt{\lambda_1^2 - \lambda_3^2}} + \frac{2i(\lambda_1^2 - \lambda_3^2)}{\sqrt{\lambda_1^2 - \lambda_3^2}} E(u, k_2^{(1)}) \Big|_{u=K_2}^{u=0} = -\frac{2i(\lambda_3^2 K_2^{(1)} + (\lambda_1^2 - \lambda_3^2) E_2^{(1)})}{\sqrt{\lambda_1^2 - \lambda_3^2}}.$$

By symmetry of the a -cycles,

$$(10.44) \quad \mathcal{Y}_{a_2}^{[3]} = \mathcal{Y}_{a_1}^{[3]}.$$

We next turn to the b -cycle integrals. Starting from

$$(10.45) \quad \mathcal{Y}_{b_2}^{[3]} = 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda^3 d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-K_2 + iK_2'}^{-K_2} \left(\lambda_3^2 + (\lambda_1^2 - \lambda_3^2) \operatorname{dn}^2(\nu - iK_2', k_2^{(1)}) \right) d\nu,$$

we employ the standard shift identities

$$(10.46) \quad \operatorname{dn}(z - K, k) = \frac{k'}{\operatorname{dn}(z, k)}, \quad \operatorname{cn}(z - K', k') = \frac{k \operatorname{sn}(z, k')}{\operatorname{dn}(z, k')}.$$

From these formulas one obtains

$$\begin{aligned}
 \operatorname{dn}(iu - K_2 - iK'_2, k_2^{(1)}) &= \frac{k_2^{(1)'}}{\operatorname{dn}(i(u - K'_2), k_2^{(1)})} \\
 (10.47) \quad &= k_2^{(1)' } \frac{\operatorname{cn}(u - K'_2, k_2^{(1)'})}{\operatorname{dn}(u - K'_2, k_2^{(1)'})} = k_2^{(1)' } \operatorname{sn}(u, k_2^{(1)'}).
 \end{aligned}$$

Setting $v = iu - K_2$, the integral becomes

$$\begin{aligned}
 \mathcal{Y}_{b_2}^{[3]} &= \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K'_2}^0 \left(\lambda_3^2 + (\lambda_1^2 - \lambda_3^2) (k_2^{(1)'})^2 \operatorname{sn}^2(u, k_2^{(1)'}) \right) i du \\
 (10.48) \quad &= \frac{2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_0^{K'_2} \left(\lambda_1^2 - (\lambda_1^2 - \lambda_3^2) \operatorname{dn}^2(u, k_2^{(1)'}) \right) du = \frac{2\lambda_1^2 K'_2 - 2(\lambda_1^2 - \lambda_3^2) E'_2}{\sqrt{\lambda_1^2 - \lambda_3^2}}.
 \end{aligned}$$

Again by symmetry of the b -cycles,

$$(10.49) \quad \mathcal{Y}_{b_1}^{[3]} = \mathcal{Y}_{b_2}^{[3]}.$$

Week 11: Algebraic-Geometric Solutions for the mKdV Equation (11)

2025-12-02

Objectives: Focus on Case 1. Determine the parameters $C_{u_0}^{(1)}$ and $D^{(1)}$.

We set

$$(11.1) \quad D = \mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}), \quad \Delta = \mathcal{A}_{P_0}(\infty^+) - \mathcal{A}_{P_0}(\infty^-) = \mathcal{A}_{\infty^-}(\infty^+), \quad \mu_0 = \mu(x_0, t_0).$$

The Abel images of the divisors D_μ and D_{μ^*} are related by

$$(11.2) \quad \alpha_{P_0}(D_{\mu^*}) = \alpha_{P_0}(D_\mu) + \Delta.$$

Moreover, the evolution of $\alpha_{P_0}(D_\mu)$ along the flow is given by

$$(11.3) \quad \alpha_{P_0}(D_\mu) = \alpha_{P_0}(D_{\mu_0}) - i\widehat{U}(x - x_0) - i\widehat{V}(t - t_0) = \alpha_{P_0}(D_{\mu_0}) - i\widehat{U}(x + vt),$$

where we have fixed $(x_0, t_0) = (0, 0)$ and used the relation

$$\widehat{V} = v\widehat{U}, \quad v = 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2).$$

Combining these identities with the definition of D yields the following shifted forms:

$$(11.4) \quad D + \Delta = \mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0}),$$

$$(11.5) \quad D + \Delta + i\widehat{U}(x + vt) = \mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu),$$

$$(11.6) \quad D + i\widehat{U}(x + vt) = \mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu) = \mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}),$$

and

$$(11.7) \quad D - \Delta + i\widehat{U}(x + vt) = \mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}).$$

We now introduce the constant

$$(11.8) \quad C_{u_0} = \frac{\Theta(D)}{\Phi_{1,0}^+ \Theta(D + \Delta)}.$$

With this notation, the representation (4.10) can be rewritten as

$$\begin{aligned}
 u(x, t) &= \frac{\Theta(D) \Theta(D + \Delta + i\widehat{U}(x + vt))}{\Phi_{1,0}^+ \Theta(D + \Delta) \Theta(D + i\widehat{U}(x + vt))} e^{2i\omega_2 x + 2i\omega_3 t} \\
 (11.9) \quad &= C_{u_0} \frac{\Theta(D + \Delta + i\widehat{U}(x + vt))}{\Theta(D + i\widehat{U}(x + vt))} e^{2i\omega_2 x + 2i\omega_3 t}.
 \end{aligned}$$

Equivalently, starting from (4.12), one obtains

$$(11.10) \quad u(x, t) = \frac{-4 \Theta(D - \Delta + i\widehat{U}(x + vt))}{\omega_1^2 C_{u_0} \Theta(D + i\widehat{U}(x + vt))} e^{-2i\omega_2 x - 2i\omega_3 t}.$$

For Case 1.

Since

$$(11.11) \quad \widehat{U} = (U_1, U_2) = (2d_{11}, 2d_{21}), \quad d_{21} = d_{11} = -\frac{\pi \sqrt{\lambda_1^2 - \lambda_3^2}}{2K_2^{(1)}} \in \mathbb{R}^-,$$

it follows that $U_1 = U_2$. Consequently,

$$(11.12) \quad \omega_2 = -\frac{i}{4\pi} \sum_{j=1}^2 U_j \mathcal{Y}_{a_j}^{[2]} = -\frac{i}{4\pi} (U_1 \mathcal{Y}_{a_1}^{[2]} + U_2 \mathcal{Y}_{a_2}^{[2]}) = 0,$$

where we have used $\mathcal{Y}_{a_1}^{[2]} = -\mathcal{Y}_{a_2}^{[2]}$. Hence

$$(11.13) \quad \omega_3 = v \omega_2 = 0.$$

Moreover,

$$(11.14) \quad \Delta = \Delta^{(1)} = \begin{pmatrix} \pi i + \nu^{(1)} \\ \pi i - \nu^{(1)} \end{pmatrix}, \quad \nu^{(1)} = \frac{\pi i}{K_1^{(1)}} F\left(\frac{\lambda_2}{\lambda_3}, k_1^{(1)}\right) \in i\mathbb{R}.$$

From this explicit form one readily finds

$$(11.15) \quad \Delta^* = -\Delta, \quad \sigma_1 \Delta = \begin{pmatrix} \pi i - \nu^{(1)} \\ \pi i + \nu^{(1)} \end{pmatrix} = \begin{pmatrix} 2\pi i \\ 2\pi i \end{pmatrix} - \Delta, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Using (9.16) and (9.17), the period matrix satisfies

$$(11.16) \quad \sigma_1 B^{(1)} \sigma_1 = B^{(1)}, \quad \sigma_1^\top = \sigma_1, \quad \sigma_1^2 = \mathbb{I}.$$

Letting $m = \sigma_1 n$, we compute

$$(11.17) \quad \begin{aligned} \Theta(B^{(1)}, \sigma_1 z) &= \sum_{n \in \mathbb{Z}^2} \exp\left\{\frac{1}{2} n^\top B^{(1)} n + n^\top \sigma_1 z\right\} \\ &= \sum_{n \in \mathbb{Z}^2} \exp\left\{\frac{1}{2} (\sigma_1 n)^\top \sigma_1 B^{(1)} \sigma_1 n + (\sigma_1 n)^\top z\right\} \\ &= \sum_{m \in \mathbb{Z}^2} \exp\left\{\frac{1}{2} m^\top B^{(1)} m + m^\top z\right\} = \Theta(B^{(1)}, z). \end{aligned}$$

Using the identities

$$\Theta^*(z) = \Theta(z^*), \quad \Theta(\sigma_1 z) = \Theta(z), \quad \Theta(-z) = \Theta(z),$$

and the representation

$$(11.18) \quad u(x, t) = C_{u_0} \frac{\Theta(D + \Delta + i\widehat{U}(x + vt))}{\Theta(D + i\widehat{U}(x + vt))},$$

we obtain

$$(11.19) \quad \begin{aligned} u^*(x, t) &= C_{u_0}^* \frac{\Theta(D^* + \Delta^* - i\widehat{U}(x + vt))}{\Theta(D^* - i\widehat{U}(x + vt))} \\ &= C_{u_0}^* \frac{\Theta(\sigma_1 D^* + \sigma_1 \Delta^* - i\widehat{U}(x + vt))}{\Theta(\sigma_1 D^* - i\widehat{U}(x + vt))} \\ &= C_{u_0}^* \frac{\Theta(-\sigma_1 D^* - \Delta + i\widehat{U}(x + vt))}{\Theta(-\sigma_1 D^* + i\widehat{U}(x + vt))}. \end{aligned}$$

Requiring $u(x, t) \in \mathbb{R}$ and comparing (11.10) with (11.19), we obtain the constraint

$$(11.20) \quad D = -\sigma_1 D^* + 2\pi i n + B^{(1)} m, \quad D = \begin{pmatrix} D_1 \\ D_2 \end{pmatrix}, \quad n, m \in \mathbb{Z}^2.$$

Equivalently, this condition can be written componentwise as

$$(11.21) \quad \begin{aligned} D_1 + D_2^* &= 2\pi i n_1 + B_{11} m_1 + B_{12} m_2, \\ D_1^* + D_2 &= 2\pi i n_2 + B_{21} m_1 + B_{22} m_2. \end{aligned}$$

Using the symmetry properties of the period matrix,

$$(11.22) \quad B_{11} = B_{22}, \quad B_{21} = B_{12}, \quad B_{ij} \in \mathbb{R},$$

and noting that $B_{11} \neq B_{12}$, we conclude that

$$(11.23) \quad n_1 = -n_2, \quad m_1 = m_2.$$

Substituting these relations back into (11.21) yields

$$(11.24) \quad D_1 + D_2^* = 2\pi i n_1 + (B_{11} + B_{12}) m_1.$$

On the other hand, rewriting $u(x, t)$ in (11.10) in the form

$$(11.25) \quad u(x, t) = \frac{-4\Theta(\sigma_1 D - \sigma_1 \Delta + i\widehat{U}(x + vt))}{\omega_1^2 C_{u_0} \Theta(D + i\widehat{U}(x + vt))} = \frac{-4\Theta(\sigma_1 D + \Delta + i\widehat{U}(x + vt))}{\omega_1^2 C_{u_0} \Theta(D + i\widehat{U}(x + vt))},$$

and comparing with (11.18), we impose the condition

$$(11.26) \quad D = \sigma_1 D + 2\pi i n', \quad n' \in \mathbb{Z}^2.$$

This is equivalent to the pair of relations

$$(11.27) \quad D_1 - D_2 = 2\pi i n'_1, \quad D_2 - D_1 = 2\pi i n'_2,$$

and hence

$$(11.28) \quad n'_2 = -n'_1, \quad D_1 - D_2 = 2\pi i n'_1.$$

Under the condition (11.26), comparison of (11.10) with (11.18) yields

$$(11.29) \quad C_{u_0}^2 = \frac{-4\Theta(D - \Delta + i\widehat{U}(x + vt))}{\omega_1^2 \Theta(D + \Delta + i\widehat{U}(x + vt))} = \frac{-4\Theta(\sigma_1 D + \Delta + i\widehat{U}(x + vt))}{\omega_1^2 \Theta(D + \Delta + i\widehat{U}(x + vt))} = \frac{-4}{\omega_1^2},$$

where the σ_1 -symmetry of the theta function has been used. Consequently,

$$(11.30) \quad C_{u_0} = \frac{2i}{\omega_1}.$$

Combining (11.24) with (11.28), we obtain the constraints

$$(11.31) \quad D_1 - D_2 = 2\pi i n'_1, \quad D_1 + D_2^* = 2\pi i n_1 + (B_{11} + B_{12}) m_1.$$

These imply

$$(11.32) \quad \operatorname{Re} D_1 = \operatorname{Re} D_2 = \frac{1}{2} m_1 (B_{11} + B_{12}), \quad \operatorname{Im} D_1 = \operatorname{Im} D_2 + 2\pi n_1.$$

Equivalently, the vector D admits the parameterization

$$(11.33) \quad \begin{pmatrix} D_1 \\ D_2 \end{pmatrix} = \frac{m_1}{2} B^{(1)} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ 1 \end{pmatrix} \alpha + \begin{pmatrix} 2\pi i n_1 \\ 0 \end{pmatrix}, \quad \alpha \in \mathbb{R}.$$

By (11.20), we have

$$(11.34) \quad -\sigma_1 D^* = D - 2\pi i n - B^{(1)} m.$$

Under this condition, the complex conjugate of $u(x, t)$ in (11.19) can be written as

$$\begin{aligned}
 u^*(x, t) &= C_{u_0}^* \frac{\Theta(D - 2\pi i n - B^{(1)}m - \Delta + i\hat{U}(x + vt))}{\Theta(D - 2\pi i n - B^{(1)}m + i\hat{U}(x + vt))} \\
 (11.35) \quad &= C_{u_0}^* \frac{\Theta(D - \Delta + i\hat{U}(x + vt)) \exp\left\{\langle D - \Delta + i\hat{U}(x + vt), m \rangle + \frac{1}{2}(B_{11}m_1 + B_{22}m_2)\right\}}{\Theta(D + i\hat{U}(x + vt)) \exp\left\{\langle D + i\hat{U}(x + vt), m \rangle + \frac{1}{2}(B_{11}m_1 + B_{22}m_2)\right\}} \\
 &= C_{u_0}^* \frac{\Theta(D - \Delta + i\hat{U}(x + vt)) \exp\{-\langle \Delta, m \rangle\}}{\Theta(D + i\hat{U}(x + vt))}.
 \end{aligned}$$

Here $m_2 = m_1$, and therefore

$$\begin{aligned}
 (11.36) \quad \exp\{-\langle \Delta, m \rangle\} &= \exp\left\{-(\pi i + v^{(1)})m_1 - (\pi i - v^{(1)})m_1\right\} \\
 &= \exp(-2\pi i m_1) = 1.
 \end{aligned}$$

It then follows from (11.10) and (11.35) that

$$(11.37) \quad |C_{u_0}|^2 = \frac{-4}{\omega_1^2}.$$

Week 12: Algebro-Geometric Solutions for the mKdV Equation (12)

2025-12-09

Objectives: Focus on Case 1. Introduce the effective integration method.

Recall that

$$(12.1) \quad L(\lambda; x, t) = \frac{1}{4} V(\lambda; Q) + \alpha_1 U(\lambda; Q),$$

whose entries are given explicitly by

$$\begin{aligned}
 (12.2) \quad L_{11}(\lambda) &= -i(\lambda^3 + \lambda(\alpha_1 - \frac{1}{2}u^2)), \quad L_{22}(\lambda) = -L_{11}(\lambda), \\
 L_{12}(\lambda) &= u\left(\lambda^2 + i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2}\right) = u(\lambda - \mu_1)(\lambda - \mu_2), \\
 L_{21}(\lambda) &= -u\left(\lambda^2 - i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2}\right) = -u(\lambda - \mu_1^*)(\lambda - \mu_2^*).
 \end{aligned}$$

The effective integration method amounts to exploiting the fact that the x -part of (12.3) is an ordinary differential equation,

$$u_{xxx} + 6u^2u_x - 4\alpha_1u_x = 0,$$

which is encoded in the equations

$$(12.3) \quad L_x = [U(\lambda, Q), L], \quad L_t = [V(\lambda, Q), L].$$

We compute the determinant of L . Using the structure of the matrix,

$$\begin{aligned}
 (12.4) \quad \det L &= -L_{11}^2 - L_{12}L_{21} \\
 &= \lambda^2(\lambda^2 + \alpha_1 - \frac{1}{2}u^2)^2 + u^2\left[\left(\lambda^2 - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2}\right)^2 + \frac{u_x^2}{4u^2}\lambda^2\right].
 \end{aligned}$$

A direct expansion yields the polynomial form

$$\begin{aligned}
 (12.5) \quad \det L &= \lambda^6 + 2\alpha_1\lambda^4 + \left(\frac{1}{4}(u_x^2 - 2uu_{xx} - 3u^4) + \alpha_1u^2 + \alpha_1^2\right)\lambda^2 + \left(\frac{u_{xx}}{4} - \alpha_1u + \frac{u^3}{2}\right)^2 \\
 &= \lambda^6 + 2\alpha_1\lambda^4 + s_2\lambda^2 + s_0.
 \end{aligned}$$

Identifying coefficients in (12.5) gives, at order $O(1)$,

$$(12.6) \quad u_{xx} - 4\alpha_1 u + 2u^3 = 4\sqrt{s_0},$$

and, at order $O(\lambda^2)$,

$$(12.7) \quad u_x^2 - 2uu_{xx} - 3u^4 + 4\alpha_1 u^2 + 4\alpha_1^2 = 4s_2.$$

Substituting (12.6) into (12.7) eliminates u_{xx} and yields

$$(12.8) \quad u_x^2 - 2u(4\sqrt{s_0} + 4\alpha_1 u - 2u^3) - 3u^4 + 4\alpha_1 u^2 + 4\alpha_1^2 = 4s_2,$$

which simplifies to

$$(12.9) \quad u_x^2 - 8u\sqrt{s_0} - 8\alpha_1 u^2 + u^4 + 4\alpha_1 u^2 + 4\alpha_1^2 = 4s_2.$$

Equivalently,

$$(12.10) \quad \begin{aligned} u_x^2 &= -(u^4 - 4\alpha_1 u^2 - 8u\sqrt{s_0} + 4\alpha_1^2 - 4s_2) \\ &= -R(u) = -\prod_{i=1}^4 (u - u_i), \end{aligned}$$

from which it follows immediately that

$$(12.11) \quad \sum_{i=1}^4 u_i = 0.$$

Multiplying (12.6) by u_x yields

$$(12.12) \quad u_x u_{xx} = 4\alpha_1 u u_x + 2u^3 u_x = 4\sqrt{s_0} u_x.$$

Integrating once with respect to x , we obtain

$$(12.13) \quad u_x^2 = -(-4\alpha_1 u^2 + u^4 - 8\sqrt{s_0} u) + c_0, \quad c_0 = 4s_2 - 4\alpha_1^2,$$

which is precisely equivalent to (12.10). From a different perspective, the conserved energy is

$$(12.14) \quad E = 4s_2 - 4\alpha_1^2 = u_x^2 + (u^4 - 4\alpha_1 u^2 - 8\sqrt{s_0} u).$$

Since $u_x^2 = -R(u)$, we may write equivalently

$$(12.15) \quad \frac{du}{dx} = \pm \sqrt{-R(u)}.$$

Separating variables gives

$$(12.16) \quad \int \frac{du}{\sqrt{-R(u)}} = \pm x + C.$$

Note that u and u_x are real-valued.

Case I. $s_0 \neq 0$, $u_i \in \mathbb{R}$, $u_1 > u_2 > u_3 > u_4$. In this case there are two types of real solutions, corresponding to the admissible intervals

$$(12.17) \quad u_2 \leq u^{[1]} \leq u_1, \quad u_4 \leq u^{[2]} \leq u_3.$$

Case II. $s_0 \neq 0$, $u_{1,3} \in \mathbb{R}$, $u_1 > u_3$, and $u_{2,4} \in \mathbb{C} \setminus \mathbb{R}$. Then the real solution satisfies

$$(12.18) \quad u_3 \leq u^{[3]} \leq u_1.$$

Case III. $s_0 = 0$, $u_4 = -u_1$, $u_3 = -u_2$ (genus one).

We now describe the correspondence between the branch points λ_i and the roots u_i . If $R(u_i) = 0$, then $\mu_1^* = \mu_1$ and $\mu_2^* = \mu_2$, and the determinant of L takes the form

$$\begin{aligned}
 \det L &= \lambda^2 \left(\lambda^2 + \alpha_1 - \frac{u_i^2}{2} \right)^2 + u_i^2 (\lambda - \mu_1)^2 (\lambda - \mu_2)^2 \\
 (12.19) \quad &= \left(\lambda^3 + \lambda \left(\alpha_1 - \frac{u_i^2}{2} \right) + i u_i (\lambda - \mu_1) (\lambda - \mu_2) \right) \\
 &\quad \times \left(\lambda^3 + \lambda \left(\alpha_1 - \frac{u_i^2}{2} \right) - i u_i (\lambda - \mu_1) (\lambda - \mu_2) \right).
 \end{aligned}$$

We wish to decompose $\det L = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)$ into two cubic polynomials of the above form. This requires that the two factors have opposite coefficients in the quadratic and constant terms. In the Case 1, all λ_j lie on the imaginary axis, $\lambda_j \in i\mathbb{R}$. In the Case 2, one has $\lambda_1 \in \mathbb{C} \setminus (\mathbb{R} \cup i\mathbb{R})$, $\lambda_2 \in i\mathbb{R}$, and $\lambda_3 = -\lambda_1^*$.

For distinct $\lambda_i, \lambda_j, \lambda_k$, the identities

$$\begin{aligned}
 (12.20) \quad &(\lambda^2 - \lambda_i^2)(\lambda - \lambda_j) = \lambda^3 - \lambda_j \lambda^2 - \lambda_i^2 \lambda + \lambda_i^2 \lambda_j, \\
 &(\lambda^2 - \lambda_k^2)(\lambda + \lambda_j) = \lambda^3 + \lambda_j \lambda^2 - \lambda_k^2 \lambda - \lambda_k^2 \lambda_j
 \end{aligned}$$

show that the constant terms cannot be matched in this manner.

Instead, consider $S_i, S_j, S_k \in \{\pm 1\}$ and distinct $\lambda_i, \lambda_j, \lambda_k$. Then

$$(12.21) \quad (\lambda - S_i \lambda_i)(\lambda - S_j \lambda_j)(\lambda - S_k \lambda_k) = \lambda^3 - (S_i \lambda_i + S_j \lambda_j + S_k \lambda_k) \lambda^2 + \cdots - S_i S_j S_k \lambda_i \lambda_j \lambda_k,$$

and

$$(12.22) \quad (\lambda + S_i \lambda_i)(\lambda + S_j \lambda_j)(\lambda + S_k \lambda_k) = \lambda^3 + (S_i \lambda_i + S_j \lambda_j + S_k \lambda_k) \lambda^2 + \cdots + S_i S_j S_k \lambda_i \lambda_j \lambda_k.$$

Comparing coefficients with (12.19), we obtain the relations

$$\begin{aligned}
 (12.23) \quad &i u_1 = \lambda_1 + \lambda_2 + \lambda_3, \quad i u_2 = -(\lambda_1 + \lambda_2 - \lambda_3), \\
 &i u_3 = -(\lambda_1 - \lambda_2 + \lambda_3), \quad i u_4 = -(-\lambda_1 + \lambda_2 + \lambda_3).
 \end{aligned}$$

Supplementary computations:

Set $\sqrt{s_0} = i \lambda_1 \lambda_2 \lambda_3$. With this choice, the quartic equation admits four roots given explicitly by

$$u_1 = -i(\lambda_1 + \lambda_2 + \lambda_3), \quad u_2 = i(\lambda_1 + \lambda_2 - \lambda_3), \quad u_3 = i(\lambda_1 - \lambda_2 + \lambda_3), \quad u_4 = i(-\lambda_1 + \lambda_2 + \lambda_3).$$

Pairing the factors gives

$$(u - u_1)(u - u_2) = (u + i(\lambda_1 + \lambda_2 + \lambda_3))(u - i(\lambda_1 + \lambda_2 - \lambda_3)) = u^2 + 2i\lambda_3 u + (\lambda_1 + \lambda_2)^2 - \lambda_3^2.$$

$$(u - u_3)(u - u_4) = (u - i(\lambda_1 - \lambda_2 + \lambda_3))(u - i(-\lambda_1 + \lambda_2 + \lambda_3)) = u^2 - 2i\lambda_3 u + (\lambda_1 - \lambda_2)^2 - \lambda_3^2.$$

Hence

$$\begin{aligned}
 \prod_{i=1}^4 (u - u_i) &= (u^2 + 2i\lambda_3 u + (\lambda_1 + \lambda_2)^2 - \lambda_3^2)(u^2 - 2i\lambda_3 u + (\lambda_1 - \lambda_2)^2 - \lambda_3^2) \\
 &= u^4 + 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)u^2 - 8i\lambda_1 \lambda_2 \lambda_3 u + ((\lambda_1^2 + \lambda_2^2 + \lambda_3^2)^2 - 4(\lambda_1^2 \lambda_2^2 + \lambda_1^2 \lambda_3^2 + \lambda_2^2 \lambda_3^2)).
 \end{aligned}$$

Using $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 = -2\alpha_1$, $\lambda_1^2 \lambda_2^2 + \lambda_1^2 \lambda_3^2 + \lambda_2^2 \lambda_3^2 = s_2$, and $\sqrt{s_0} = i \lambda_1 \lambda_2 \lambda_3$, we obtain

$$\prod_{i=1}^4 (u - u_i) = u^4 - 4\alpha_1 u^2 - 8\sqrt{s_0} u + 4\alpha_1^2 - 4s_2 = R(u),$$

so each u_i is a root of $R(u) = 0$.

Objectives: Derive explicit expressions for u by integrating the first-order equation $u_x^2 = -R(u)$ and reducing the resulting integrals to standard elliptic forms.

We begin with

$$(13.1) \quad u_x^2 = -R(u) = -\prod_{i=1}^4 (u - u_i).$$

We solve this equation by direct integration, distinguishing cases according to the location of the roots.

Case 1. All roots are real, $u_i \in \mathbb{R}$ (equivalently $\lambda_i \in i\mathbb{R}$), ordered as

$$u_1 > u_2 > u_3 > u_4.$$

We consider the initial value problem on the interval $u_4 \leq u \leq u_3$,

$$(13.2) \quad \begin{cases} \frac{du}{dx} = \pm \sqrt{-R(u)}, \\ u(0) = u_3. \end{cases}$$

Setting

$$y = u_x, \quad z = u,$$

the corresponding algebraic curve takes the form

$$(13.3) \quad y^2 = -\prod_{i=1}^4 (z - u_i).$$

Integrating (13.2) yields

$$(13.4) \quad \int_{u_3}^u \frac{dv}{\sqrt{-R(v)}} = \pm \int_0^x dy = \pm x.$$

We recall the incomplete elliptic integral of the first kind,

$$(13.5) \quad v = F(z, k) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \quad z = \text{sn}(v, k).$$

Differentiating with respect to v gives

$$1 = \frac{1}{\sqrt{(1-z^2)(1-k^2z^2)}} \frac{dz}{dv},$$

and hence

$$(13.6) \quad \frac{dz}{dv} = \sqrt{(1-z^2)(1-k^2z^2)} = \text{cn}(v, k) \text{dn}(v, k),$$

where

$$\text{cn}(v, k) = \sqrt{1-z^2}, \quad \text{dn}(v, k) = \sqrt{1-k^2z^2}.$$

To reduce the integral (13.4) to the standard form, we introduce a fractional linear transformation $u \mapsto t$, followed by the substitution $t = z^2$, such that

$$u_3 \leftrightarrow 0, \quad u_4 \leftrightarrow 1, \quad u_2 \leftrightarrow \infty.$$

An explicit choice is

$$(13.7) \quad t = \frac{u - u_3}{u - u_2} \frac{u_4 - u_2}{u_4 - u_3}.$$

Solving for u gives

$$(13.8) \quad u = u_2 + \frac{(u_3 - u_2)(u_4 - u_2)}{u_4 - u_2 + (u_3 - u_4)t}, \quad t = z^2.$$

From (13.8), straightforward algebra yields

$$(13.9) \quad u - u_1 = \frac{(u_3 - u_1)(u_4 - u_2) + (u_2 - u_1)(u_3 - u_4)t}{u_4 - u_2 + (u_3 - u_4)t},$$

$$(13.10) \quad u - u_3 = \frac{(u_2 - u_3)(u_3 - u_4)t}{u_4 - u_2 + (u_3 - u_4)t},$$

and

$$(13.11) \quad u - u_4 = \frac{(u_2 - u_4)(u_4 - u_3)(1 - t)}{u_4 - u_2 + (u_3 - u_4)t}.$$

Combining these expressions, we find

$$(13.12) \quad \prod_{i=1}^4 (u - u_i) = \frac{-(u_2 - u_3)^2(u_2 - u_4)^2(u_3 - u_4)^2 t(1 - t) \left((u_1 - u_3)(u_2 - u_4) + (u_2 - u_1)(u_3 - u_4)t \right)}{(u_4 - u_2 + (u_3 - u_4)t)^4}.$$

Consequently,

$$(13.13) \quad \sqrt{-R(u)} = \frac{(u_2 - u_3)(u_2 - u_4)(u_3 - u_4)}{(u_4 - u_2 + (u_3 - u_4)t)^2} \sqrt{(u_1 - u_3)(u_2 - u_4)} \sqrt{t(1 - t)(1 - k^2 t)},$$

where the elliptic modulus is given by

$$(13.14) \quad k^2 = \frac{(u_1 - u_2)(u_3 - u_4)}{(u_1 - u_3)(u_2 - u_4)}.$$

Moreover, differentiating (13.8) with respect to t yields

$$(13.15) \quad du = \frac{(u_2 - u_4)(u_2 - u_3)(u_4 - u_3)}{(u_4 - u_2 + (u_3 - u_4)t)^2} dt.$$

Substituting this expression together with (13.13) and setting $t = z^2$, we obtain

$$(13.16) \quad \frac{du}{\sqrt{-R(u)}} = \frac{-dt}{\sqrt{(u_1 - u_3)(u_2 - u_4)} \sqrt{t(1 - t)(1 - k^2 t)}}$$

$$(13.17) \quad = \frac{-2dz}{\sqrt{(u_1 - u_3)(u_2 - u_4)} \sqrt{(1 - z^2)(1 - k^2 z^2)}}, \quad z \in (0, 1).$$

Consequently, for $u \in [u_4, u_3]$,

$$(13.18) \quad \int_{u_3}^u \frac{dv}{\sqrt{-R(v)}} = -\frac{2}{\sqrt{(u_1 - u_3)(u_2 - u_4)}} \int_0^z \frac{dt}{\sqrt{(1 - t^2)(1 - k^2 t^2)}}$$

$$(13.19) \quad = -\frac{2}{\sqrt{(u_1 - u_3)(u_2 - u_4)}} F(z, k) = \pm x.$$

Inverting this relation gives

$$(13.20) \quad z^2 = \text{sn}^2(\alpha x, k), \quad \alpha^2 = \frac{(u_1 - u_3)(u_2 - u_4)}{4}.$$

Substituting back into (13.8) yields the explicit solution

$$(13.21) \quad u(x) = u_2 + \frac{(u_3 - u_2)(u_4 - u_2)}{(u_4 - u_2) + (u_3 - u_4) \text{sn}^2(\alpha x, k)}, \quad x \in \mathbb{R}, \quad u \in [u_4, u_3].$$

Case 2. Assume

$$u_1 > u_3 \in \mathbb{R}, \quad u_2 = u_4^* \in \mathbb{C} \setminus (\mathbb{R} \cup i\mathbb{R}).$$

Then

$$(13.22) \quad \begin{aligned} -R(u) &= -(u - u_3)(u - u_1)(u - u_2)(u - u_2^*) \\ &= -(u - u_3)(u - u_1)((u - u_{2R})^2 + u_{2I}^2), \end{aligned}$$

where $u_2 = u_{2R} + iu_{2I} = a + ib$.

Using the identity $\text{cn}'(x, k) = -\text{sn}(x, k) \text{dn}(x, k)$, we set

$$z = \text{cn}(x, k).$$

Then

$$(13.23) \quad \left(\frac{dz}{dx}\right)^2 = \text{sn}^2(x, k) \text{dn}^2(x, k) = (1 - z^2)(k'^2 + k^2 z^2),$$

where $k' = \sqrt{1 - k^2}$. It follows that

$$(13.24) \quad \frac{dz}{\sqrt{(1 - z^2)(k'^2 + k^2 z^2)}} = -dx.$$

Integrating from $x = 0$ (so that $z = 1$) to x gives

$$(13.25) \quad -\int_1^z \frac{dt}{\sqrt{(1 - t^2)(k'^2 + k^2 t^2)}} = \int_0^x dy = x.$$

To relate z to u , we introduce a Möbius transformation mapping

$$u_1 \leftrightarrow 1, \quad u_3 \leftrightarrow -1.$$

Let

$$(13.26) \quad \delta \frac{u - u_1}{u - u_3} = \frac{z - 1}{z + 1}.$$

Solving for u gives

$$(13.27) \quad u = \frac{u_1 - \frac{z - 1}{\delta(z + 1)} u_3}{1 - \frac{z - 1}{\delta(z + 1)}} = u_1 + \frac{(u_1 - u_3) \frac{z - 1}{\delta(z + 1)}}{1 - \frac{z - 1}{\delta(z + 1)}}.$$

Equivalently, this can be written as

$$(13.28) \quad u = u_1 + \frac{(u_3 - u_1)(1 - z)}{(1 - z) + \delta(1 + z)} = u_1 + \frac{(u_3 - u_1)(1 - z)}{1 + \delta + (\delta - 1)z}.$$

Supplementary computations:

To simplify the computations, we employ the complex projective line notation on $\mathbb{CP}^1$. We write a Möbius transformation as

$$\frac{az + bw}{cz + dw} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix}, \quad \begin{pmatrix} z \\ w \end{pmatrix} = (z : w) = \frac{z}{w} \in \mathbb{CP}^1.$$

For any nonzero scalar f , this representation is homogeneous:

$$\begin{pmatrix} fa & fb \\ fc & fd \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} = f \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix}.$$

Moreover, if two matrices share the same second row, then

$$\begin{pmatrix} a_1 & b_1 \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} + \begin{pmatrix} a_2 & b_2 \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} = \begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ c & d \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix}.$$

From (13.26) we have the matrix identity

$$(13.29) \quad \begin{pmatrix} 1 & -u_1 \\ 1 & -u_3 \end{pmatrix} \begin{pmatrix} u \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ \delta & \delta \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

Multiplying by the inverse of the left matrix yields

$$(13.30) \quad \begin{pmatrix} u \\ 1 \end{pmatrix} = \frac{1}{u_1 - u_3} \begin{pmatrix} -u_3 & u_1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ \delta & \delta \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix} = \begin{pmatrix} \delta u_1 - u_3 & \delta u_1 + u_3 \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

Equivalently,

$$(13.31) \quad u = \frac{(\delta u_1 - u_3)z + \delta u_1 + u_3}{(\delta - 1)z + \delta + 1} = \frac{\delta u_1(z + 1) + u_3(1 - z)}{\delta(z + 1) + 1 - z}.$$

Subtracting $\begin{pmatrix} u_2 \\ 1 \end{pmatrix}$ from (13.30) gives

$$(13.32) \quad \begin{pmatrix} u \\ 1 \end{pmatrix} - \begin{pmatrix} u_2 \\ 1 \end{pmatrix} = \begin{pmatrix} \delta u_1 - u_3 & \delta u_1 + u_3 \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix} - \begin{pmatrix} u_2(\delta - 1) & u_2(\delta + 1) \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix} \\ = \begin{pmatrix} \delta(u_1 - u_2) + u_2 - u_3 & \delta(u_1 - u_2) + u_3 - u_2 \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

Recall that multiplication in $\mathbb{CP}^1$ is represented componentwise:

$$\begin{pmatrix} z_1 \\ w_1 \end{pmatrix} \cdot \begin{pmatrix} z_2 \\ w_2 \end{pmatrix} = \begin{pmatrix} z_1 z_2 \\ w_1 w_2 \end{pmatrix} = \frac{z_1 z_2}{w_1 w_2}.$$

Similarly,

$$(13.33) \quad \begin{pmatrix} u - u_2^* \\ 1 \end{pmatrix} = \begin{pmatrix} \delta(u_1 - u_2^*) + u_2^* - u_3 & \delta(u_1 - u_2^*) + u_3 - u_2^* \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

It follows that the denominator of $(u - u_2)(u - u_2^*)$ is

$$(13.34) \quad ((\delta - 1)z + \delta + 1)^2,$$

while the numerator is

$$(13.35) \quad (z - 1) \begin{pmatrix} \delta(u_1 - u_2) + u_2 - u_3 \\ \delta(u_1 - u_2) + u_3 - u_2 \end{pmatrix} \begin{pmatrix} \delta(u_1 - u_2^*) + u_2^* - u_3 & \delta(u_1 - u_2^*) + u_3 - u_2^* \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}.$$

Requiring the coefficient of the linear term in z to vanish leads to

$$(13.36) \quad \operatorname{Re}((\delta(u_1 - u_2) + u_2 - u_3)(\delta(u_1 - u_2^*) + u_3 - u_2^*)) = 0.$$

Equivalently,

$$(13.37) \quad \delta^2 |u_1 - u_2|^2 + \operatorname{Re}((u_2 - u_3)(u_1 - u_2^*) - (u_2^* - u_3)(u_1 - u_2)) \delta - |u_2 - u_3|^2 = 0.$$

Using

$$\operatorname{Re}((u_2 - u_3)(u_1 - u_2^*) - (u_2^* - u_3)(u_1 - u_2)) = \operatorname{Re}(2i \operatorname{Im}(u_2 - u_3)(u_1 - u_2)) = 0,$$

we obtain

$$(13.38) \quad \delta^2 = \frac{|u_2 - u_3|^2}{|u_1 - u_2|^2}, \quad \delta = \frac{|u_2 - u_3|}{|u_1 - u_2|}.$$

Hence

$$(13.39) \quad (u - u_2)(u - u_2^*) = \frac{|\delta(u_1 - u_2) + u_2 - u_3|^2 z^2 + |\delta(u_1 - u_2) + u_3 - u_2|^2}{((\delta - 1)z + \delta + 1)^2}.$$

Moreover,

$$\begin{pmatrix} u - u_1 \\ 1 \end{pmatrix} = \begin{pmatrix} u_1 - u_3 & u_3 - u_1 \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix}, \quad \begin{pmatrix} u - u_3 \\ 1 \end{pmatrix} = \begin{pmatrix} \delta(u_1 - u_3) & \delta(u_1 - u_3) \\ \delta - 1 & \delta + 1 \end{pmatrix} \begin{pmatrix} z \\ 1 \end{pmatrix},$$

so that

$$(13.40) \quad u - u_1 = \frac{(u_1 - u_3)(z - 1)}{(\delta - 1)z + \delta + 1}, \quad u - u_3 = \frac{\delta(u_1 - u_3)(z + 1)}{(\delta - 1)z + \delta + 1}.$$

Substituting (13.39) and (13.40) into (13.22), we obtain

$$(13.41) \quad -R(u) = \frac{\delta(u_1 - u_3)^2(1 - z)(1 + z)}{((\delta - 1)z + \delta + 1)^4} \left(|\delta(u_1 - u_2) + u_2 - u_3|^2 z^2 + |\delta(u_1 - u_2) + u_3 - u_2|^2 \right).$$

Using the identity, valid for all $z_1, z_2 \in \mathbb{C}$,

$$(13.42) \quad |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2,$$

we find

$$(13.43) \quad \begin{aligned} & |\delta(u_1 - u_2) + u_2 - u_3|^2 + |\delta(u_1 - u_2) + u_3 - u_2|^2 \\ &= 2\delta^2|u_1 - u_2|^2 + 2|u_2 - u_3|^2 = 4|u_2 - u_3|^2, \end{aligned}$$

where we used $\delta^2|u_1 - u_2|^2 = |u_2 - u_3|^2$.

Consequently,

$$(13.44) \quad \sqrt{-R(u)} = \frac{2\sqrt{\delta}(u_1 - u_3)|u_2 - u_3|}{((\delta - 1)z + \delta + 1)^2} \sqrt{(1 - z^2)(k'^2 + k^2 z^2)},$$

where

$$(13.45) \quad k = \frac{|\delta(u_1 - u_2) + u_2 - u_3|}{2|u_2 - u_3|}, \quad k' = \sqrt{1 - k^2}.$$

Differentiating (13.31) with respect to z gives

$$(13.46) \quad \begin{aligned} du &= \frac{(\delta u_1 - u_3)((\delta - 1)z + \delta + 1) - ((\delta u_1 - u_3)z + \delta u_1 + u_3)(\delta - 1)}{((\delta - 1)z + \delta + 1)^2} dz \\ &= \frac{2\delta(u_1 - u_3)}{((\delta - 1)z + \delta + 1)^2} dz. \end{aligned}$$

Combining this with (13.44), we obtain

$$(13.47) \quad \frac{du}{\sqrt{-R(u)}} = \frac{\sqrt{\delta}}{|u_2 - u_3|} \frac{dz}{\sqrt{(1 - z^2)(k'^2 + k^2 z^2)}} = \frac{1}{\alpha} \frac{dz}{\sqrt{(1 - z^2)(k'^2 + k^2 z^2)}},$$

where

$$(13.48) \quad \alpha = \frac{|u_2 - u_3|}{\sqrt{\delta}} = \sqrt{|u_1 - u_2||u_2 - u_3|}.$$

For $u_3 \leq u \leq u_1$, integration yields

$$(13.49) \quad \int_{u_1}^u \frac{dv}{\sqrt{-R(v)}} = \frac{1}{\alpha} \int_1^z \frac{dt}{\sqrt{(1 - t^2)(k'^2 + k^2 t^2)}} = \pm x.$$

Since cn is an even function, inversion gives

$$(13.50) \quad z = \text{cn}(\alpha x, k).$$

Substituting this expression into (13.31), we arrive at the explicit solution

$$(13.51) \quad u(x) = u_1 + \frac{(u_3 - u_1)(1 - \text{cn}(\alpha x, k))}{1 + \delta + (\delta - 1)\text{cn}(\alpha x, k)}, \quad x \in \mathbb{R}, \quad u \in [u_3, u_1].$$

The parameters are given by

$$(13.52) \quad \delta = \frac{|u_2 - u_3|}{|u_1 - u_2|}, \quad \alpha^2 = |(u_1 - u_2)(u_2 - u_3)|, \quad k^2 = \frac{|\delta(u_1 - u_2) + u_2 - u_3|^2}{4|u_2 - u_3|^2}.$$

Week 14: Algebraic-Geometric Solutions for the mKdV Equation (14)

2025-12-23

Objectives: Consider Case 1 with $\lambda_j \in i\mathbb{R}$. Verify by direct computation that the two explicit expressions for the solution u coincide.

Recall from (13.21) that for $x \in \mathbb{R}$ and $u \in [u_4, u_3]$ one has

$$(14.1) \quad u(x, t) = u_2 + \frac{(u_3 - u_2)(u_4 - u_2)}{(u_4 - u_2) + (u_3 - u_4) \text{sn}^2(\alpha(x + vt), k)},$$

where the parameters are given by

$$(14.2) \quad k^2 = \frac{(u_1 - u_2)(u_3 - u_4)}{(u_1 - u_3)(u_2 - u_4)}, \quad \alpha^2 = \frac{(u_1 - u_3)(u_2 - u_4)}{4}.$$

The constants u_j are expressed in terms of λ_j as

$$(14.3) \quad \begin{aligned} u_1 &= -i(\lambda_1 + \lambda_2 + \lambda_3), & u_2 &= i(\lambda_1 + \lambda_2 - \lambda_3), \\ u_3 &= i(\lambda_1 - \lambda_2 + \lambda_3), & u_4 &= i(-\lambda_1 + \lambda_2 + \lambda_3). \end{aligned}$$

On the other hand, the theta-functional representation (11.18) reads

$$(14.4) \quad u(x, t) = C_{u_0}^{(1)} \frac{\Theta(D^{(1)} + \Delta^{(1)} + iU^{(1)}(x + vt))}{\Theta(D^{(1)} + iU^{(1)}(x + vt))},$$

with

$$U^{(1)} = \kappa^{(1)} \mathbf{1}, \quad \Delta^{(1)} = (i\pi + \nu^{(1)} \sigma_3) \mathbf{1}, \quad \mathbf{1} = [1 \ 1]^\top, \quad D^{(1)} = \mathbf{0},$$

and

$$\kappa^{(1)} = \frac{\pi(\lambda_1^2 - \lambda_3^2)^{1/2}}{K_2^{(1)}}, \quad k_2^{(1)} = \frac{(\lambda_1^2 - \lambda_2^2)^{1/2}}{(\lambda_1^2 - \lambda_3^2)^{1/2}}, \quad k_1^{(1)} = k_2^{(1)} \frac{\lambda_3}{\lambda_2}, \quad \nu^{(1)} = \frac{i\pi}{K_1^{(1)}} F\left(\frac{\lambda_2}{\lambda_3}, k_1^{(1)}\right).$$

Using (7.17) and (7.24), the moduli $k_1^{(1)}$ and $k_2^{(1)}$ can be written equivalently as

$$(14.5) \quad k_1^{(1)} = \frac{\lambda_3}{\lambda_2} \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}, \quad k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}.$$

One verifies directly that

$$k = k_2^{(1)}, \quad \alpha = \kappa^{(1)} \frac{K_2^{(1)}}{\pi}.$$

Applying the theta-function decomposition (9.22),

$$(14.6) \quad \Theta(B^{(1)}; z) = \vartheta_3(z_1 - z_2, 2\tau_1^{(1)}) \vartheta_3(z_1 + z_2, 2\tau_2^{(1)}) + \vartheta_2(z_1 - z_2, 2\tau_1^{(1)}) \vartheta_2(z_1 + z_2, 2\tau_2^{(1)}),$$

we first write out explicitly the numerator and denominator of (14.4). Using $D^{(1)} = \mathbf{0}$, $\Delta^{(1)} = (i\pi + \nu^{(1)} \sigma_3) \mathbf{1}$, and $U^{(1)} = \kappa^{(1)} \mathbf{1}$, we obtain

$$(14.7) \quad \begin{aligned} \Theta(D^{(1)} + \Delta^{(1)} + iU^{(1)}(x + vt)) &= \Theta((i\pi + \nu^{(1)} \sigma_3) \mathbf{1} + i\kappa^{(1)} \mathbf{1}(x + vt)) \\ &= \vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)}) \vartheta_3(2i\kappa^{(1)}(x + vt) + 2\pi i, 2\tau_2^{(1)}) + \vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)}) \vartheta_2(2i\kappa^{(1)}(x + vt) + 2\pi i, 2\tau_2^{(1)}) \\ &= \vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)}) \vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)}) - \vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)}) \vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)}), \end{aligned}$$

and

$$(14.8) \quad \begin{aligned} \Theta(D^{(1)} + iU^{(1)}(x + vt)) &= \Theta(i\kappa^{(1)} \mathbf{1}(x + vt)) \\ &= \vartheta_3(0, 2\tau_1^{(1)}) \vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)}) + \vartheta_2(0, 2\tau_1^{(1)}) \vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)}). \end{aligned}$$

Substituting (14.7) and (14.8) into (14.4) and dividing numerator and denominator by $\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})$, the ratio simplifies to

$$(14.9) \quad u(x, t) = C_{u_0}^{(1)} \frac{\vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)}) - \vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)}) \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}{\vartheta_3(0, 2\tau_1^{(1)}) + \vartheta_2(0, 2\tau_1^{(1)}) \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}.$$

Equivalently, factoring out the theta-constants $\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})$ and $\vartheta_2(0, 2\tau_1^{(1)})$ from the numerator and denominator, respectively, yields

$$(14.10) \quad u(x, t) = C_{u_0}^{(1)} \frac{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(0, 2\tau_1^{(1)})} \frac{\frac{\vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})} - \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}{\frac{\vartheta_3(0, 2\tau_1^{(1)})}{\vartheta_2(0, 2\tau_1^{(1)})} + \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}.$$

Introduce the parameters $\tilde{v}_1$ and $\tilde{v}_2$ by

$$(14.11) \quad \tilde{v}_1 = \frac{i\pi}{K_2^{(1)}} F\left(\frac{(u_2 - u_4)^{1/2}}{(u_3 - u_4)^{1/2}}, k_2^{(1)}\right), \quad \tilde{v}_2 = \frac{i\pi}{K_2^{(1)}} F\left(\frac{(u_3(u_2 - u_4))^{1/2}}{(u_2(u_3 - u_4))^{1/2}}, k_2^{(1)}\right).$$

By (14.1), the solution can be rewritten as

$$(14.12) \quad u(x, t) = u_2 \frac{\frac{u_3(u_2 - u_4)}{u_2(u_3 - u_4)} - \text{sn}^2(\alpha(x + vt), k)}{\frac{(u_2 - u_4)}{(u_3 - u_4)} - \text{sn}^2(\alpha(x + vt), k)}.$$

Using the definitions (14.11), the constant ratios in (14.12) admit the representations

$$\frac{u_2 - u_4}{u_3 - u_4} = \text{sn}^2(\tilde{v}_1 K_2^{(1)} / (i\pi), k_2^{(1)}), \quad \frac{u_3(u_2 - u_4)}{u_2(u_3 - u_4)} = \text{sn}^2(\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)}).$$

Substituting these expressions yields

$$(14.13) \quad u(x, t) = u_2 \frac{\text{sn}^2(\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)}) - \text{sn}^2(\alpha(x + vt), k_2^{(1)})}{\text{sn}^2(\tilde{v}_1 K_2^{(1)} / (i\pi), k_2^{(1)}) - \text{sn}^2(\alpha(x + vt), k_2^{(1)})}.$$

Lemma 1.

$$(a) \quad \frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} = \frac{\vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})}.$$

$$(b) \quad \frac{\vartheta_2(2\tilde{v}_1, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_1, 2\tau_2^{(1)})} = -\frac{\vartheta_3(0, 2\tau_1^{(1)})}{\vartheta_2(0, 2\tau_1^{(1)})}.$$

Proof. We begin with the three-term bilinear identities

$$(14.14) \quad \begin{aligned} 2\vartheta_3(u + v, 2\tau)\vartheta_3(u - v, 2\tau) &= \vartheta_3(u, \tau)\vartheta_3(v, \tau) + \vartheta_4(u, \tau)\vartheta_4(v, \tau), \\ 2\vartheta_2(u + v, 2\tau)\vartheta_3(u - v, 2\tau) &= \vartheta_2(u, \tau)\vartheta_2(v, \tau) - \vartheta_1(u, \tau)\vartheta_1(v, \tau). \end{aligned}$$

Setting $u = 2\tilde{v}_2$ and $v = 0$, these identities reduce to

$$(14.15) \quad \begin{aligned} 2\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})\vartheta_3(0, 2\tau_2^{(1)}) &= \vartheta_3(2\tilde{v}_2, \tau_2^{(1)})\vartheta_3(0, \tau_2^{(1)}) + \vartheta_4(2\tilde{v}_2, \tau_2^{(1)})\vartheta_4(0, \tau_2^{(1)}), \\ 2\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})\vartheta_3(0, 2\tau_2^{(1)}) &= \vartheta_2(2\tilde{v}_2, \tau_2^{(1)})\vartheta_2(0, \tau_2^{(1)}) - \vartheta_1(2\tilde{v}_2, \tau_2^{(1)})\vartheta_1(0, \tau_2^{(1)}). \end{aligned}$$

Since $\vartheta_1(0, \tau_2^{(1)}) = 0$, the last term in the second identity vanishes. Dividing the second identity by the first, we obtain

$$(14.16) \quad \frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} = \frac{\vartheta_2(2\tilde{v}_2, \tau_2^{(1)})\vartheta_2(0, \tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, \tau_2^{(1)})\vartheta_3(0, \tau_2^{(1)}) + \vartheta_4(2\tilde{v}_2, \tau_2^{(1)})\vartheta_4(0, \tau_2^{(1)})}.$$

We now express the right-hand side in terms of Jacobi elliptic functions. Using the standard relations

$$(14.17) \quad \begin{aligned} \operatorname{cn}(u, k) &= \frac{\vartheta_4(0, \tau) \vartheta_2(\pi i u / K, \tau)}{\vartheta_2(0, \tau) \vartheta_4(\pi i u / K, \tau)}, & \operatorname{dn}(u, k) &= \frac{\vartheta_4(0, \tau) \vartheta_3(\pi i u / K, \tau)}{\vartheta_3(0, \tau) \vartheta_4(\pi i u / K, \tau)}, \\ k &= \frac{\vartheta_2^2(0, \tau)}{\vartheta_3^2(0, \tau)}, & k' &= \frac{\vartheta_4^2(0, \tau)}{\vartheta_3^2(0, \tau)}, \end{aligned}$$

with $\tau = \tau_2^{(1)}$ and $u = 2\tilde{v}_2 K_2^{(1)} / (i\pi)$, we obtain

$$\frac{\vartheta_2(2\tilde{v}_2, \tau_2^{(1)}) \vartheta_2(0, \tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, \tau_2^{(1)}) \vartheta_3(0, \tau_2^{(1)})} = k_2^{(1)} \frac{\operatorname{cn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)})}{\operatorname{dn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)})},$$

and

$$\frac{\vartheta_4(2\tilde{v}_2, \tau_2^{(1)}) \vartheta_4(0, \tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, \tau_2^{(1)}) \vartheta_3(0, \tau_2^{(1)})} = \frac{k_2^{(1)'}}{\operatorname{dn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)})}.$$

Substituting these expressions into (14.16) yields

$$(14.18) \quad \frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} = \frac{k_2^{(1)} \operatorname{cn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)})}{\operatorname{dn}(2\tilde{v}_2 K_2^{(1)} / (i\pi), k_2^{(1)}) + k_2^{(1)'}}.$$

Applying the double-angle identities

$$(14.19) \quad \operatorname{cn}(2u) = \frac{\operatorname{cn}^2(u) - \operatorname{sn}^2(u) \operatorname{dn}^2(u)}{1 - k^2 \operatorname{sn}^4(u)}, \quad \operatorname{dn}(2u) = \frac{\operatorname{dn}^2(u) - k^2 \operatorname{sn}^2(u) \operatorname{cn}^2(u)}{1 - k^2 \operatorname{sn}^4(u)},$$

together with the definition of $\tilde{v}_2$ and (14.3), the expression (14.18) simplifies to

$$\frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} = \frac{(\lambda_1^2 - \lambda_2^2 - \lambda_3^2) k_1^{(1)'} - (\lambda_1^2 - \lambda_2^2 + \lambda_3^2)}{k_1^{(1)} (\lambda_1^2 + \lambda_2^2 - \lambda_3^2)}.$$

An analogous computation starting from $\nu^{(1)}$ yields the same value for

$$\frac{\vartheta_3(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})},$$

which proves the claim. □

By Lemma 1, (14.10) can be rewritten as

$$(14.20) \quad u(x, t) = C_{u_0}^{(1)} \frac{\vartheta_2(2\nu^{(1)}, 2\tau_1^{(1)})}{\vartheta_2(0, 2\tau_1^{(1)})} \frac{\frac{\vartheta_2(2\tilde{v}_2, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_2, 2\tau_2^{(1)})} - \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}{-\frac{\vartheta_2(2\tilde{v}_1, 2\tau_2^{(1)})}{\vartheta_3(2\tilde{v}_1, 2\tau_2^{(1)})} + \frac{\vartheta_2(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}{\vartheta_3(2i\kappa^{(1)}(x + vt), 2\tau_2^{(1)})}}.$$

From (14.20) it follows that the poles of $u(x, t)$ are located at

$$(14.21) \quad 2i\kappa^{(1)}(x + vt) = \pm 2\tilde{v}_1 + 4in\pi + 4im\pi\tau_2^{(1)}, \quad n, m \in \mathbb{Z},$$

while its zeros are given by

$$(14.22) \quad 2i\kappa^{(1)}(x + vt) = \pm 2\tilde{v}_2 + 4in\pi + 4im\pi\tau_2^{(1)}, \quad n, m \in \mathbb{Z}.$$

Objectives: Introduce some basics of elliptic functions and elliptic integrals.

For an ellipse, a convenient parametrization is

$$(15.1) \quad \begin{cases} x = a \cos \theta, \\ y = b \sin \theta, \end{cases} \quad 0 \leq \theta < 2\pi, \quad a \geq b.$$

Differentiating with respect to θ yields

$$(15.2) \quad \begin{aligned} ds &= ((dx)^2 + (dy)^2)^{1/2} = (a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{1/2} d\theta \\ &= a \left(1 - \frac{a^2 - b^2}{a^2} \cos^2 \theta \right)^{1/2} d\theta. \end{aligned}$$

Introduce the parameter

$$(15.3) \quad k^2 = \frac{a^2 - b^2}{a^2}.$$

The arc length can then be written as

$$(15.4) \quad s = a \int (1 - k^2 \cos^2 \theta)^{1/2} d\theta.$$

With the substitution $t = \cos \theta$, one has $dt = -\sin \theta d\theta = -\sqrt{1-t^2} d\theta$, and hence

$$(15.5) \quad s = -a \int \frac{\sqrt{1-k^2 t^2}}{\sqrt{1-t^2}} dt = a \int \frac{k^2 t^2 - 1}{\sqrt{(1-t^2)(1-k^2 t^2)}} dt.$$

The elliptic integral of the first kind is defined by

$$(15.6) \quad u = F(z, k) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k^2 t^2)}}, \quad z = \operatorname{sn}(u, k).$$

Setting $t = \sin \varphi$, this integral takes the form

$$(15.7) \quad u = F(z, k) = \int_0^{\arcsin z} \frac{d\varphi}{\sqrt{1-k^2 \sin^2 \varphi}} = \int_0^\theta \frac{d\varphi}{\sqrt{1-k^2 \sin^2 \varphi}}, \quad \theta = \operatorname{am}(u, k).$$

By definition of the Jacobi amplitude, the Jacobi elliptic functions satisfy

$$(15.8) \quad \operatorname{sn}(u, k) = \sin(\operatorname{am}(u, k)), \quad \operatorname{cn}(u, k) = \cos(\operatorname{am}(u, k)).$$

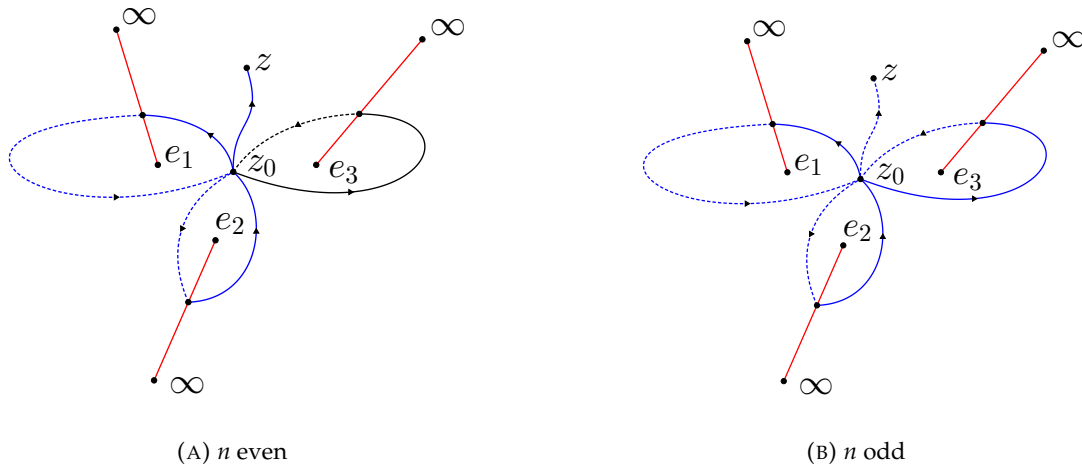


FIGURE 7. Integration paths for $F(z)$ winding a total of n times around branch points.

Consider the elliptic integral

$$(15.9) \quad F(z) = \int_{z_0}^z \frac{dt}{\sqrt{(t-e_1)(t-e_2)(t-e_3)}} = \int_{z_0}^z \frac{dt}{y(t)},$$

where

$$y(t) = \sqrt{(t-e_1)(t-e_2)(t-e_3)},$$

and the branch cuts of $y(t)$ are chosen as rays connecting each branch point e_j to ∞ .

For each branch point e_j , introduce the quantities

$$(15.10) \quad E_j = \int_{z_0}^{e_j} \frac{dt}{y(t)}, \quad I = \int_{\vec{z_0 z}} \frac{dt}{y(t)},$$

where $\vec{z_0 z}$ denotes the chosen path from z_0 to z that does not wind around any branch point. Let $C_{e_j}^\varepsilon$ be a sufficiently small positively oriented circle around e_j . The integral over the closed contour $\vec{z_0 e_j} + C_{e_j}^\varepsilon + \vec{e_j z_0}$ then satisfies

$$(15.11) \quad \int_{\vec{z_0 e_j} + C_{e_j}^\varepsilon + \vec{e_j z_0}} \frac{dt}{y(t)} = 2E_j.$$

Suppose now that the integration path defining $F(z)$ winds around branch points a total of n times. The resulting value of $F(z)$ depends only on the parity of n . If n is even, the contributions from the successive loops combine to give

$$(15.12) \quad F(z) = 2(E_\alpha - E_\beta) + 2(E_\gamma - E_\delta) + \cdots + 2(E_\kappa - E_\rho) + I.$$

If n is odd, the contribution of the final loop changes sign, and one obtains

$$(15.13) \quad F(z) = 2(E_\alpha - E_\beta) + 2(E_\gamma - E_\delta) + \cdots + 2(E_\mu - E_\rho) + 2E_\mu - I.$$

Introduce the quantities

$$(15.14) \quad \omega_1 = E_1 - E_3, \quad \omega_2 = E_2 - E_3, \quad \omega_3 = 0.$$

In particular,

$$(15.15) \quad E_1 - E_2 = \omega_1 - \omega_2.$$

With this notation, the multivaluedness of $F(z)$ can be described as follows. If n is even,

$$(15.16) \quad F(z) = I + 2m_1\omega_1 + 2m_2\omega_2, \quad m_1, m_2 \in \mathbb{Z}.$$

If n is odd,

$$(15.17) \quad F(z) = 2E_\mu - I + 2m'_1\omega_1 + 2m'_2\omega_2, \quad m'_1, m'_2 \in \mathbb{Z}.$$

Moreover, the periods ω_1 and ω_2 are linearly independent over $\mathbb{R}$, since

$$(15.18) \quad \operatorname{Im}\left(\frac{\omega_2}{\omega_1}\right) \neq 0.$$

Let $f(z)$ be a single-valued meromorphic function admitting two periods 2ω and $2\omega'$, with

$$(15.19) \quad \operatorname{Im}\left(\frac{\omega}{\omega'}\right) > 0.$$

Then

$$(15.20) \quad f(z + 2\omega) = f(z), \quad f(z + 2\omega') = f(z).$$

For integers m_1, n_1 , define the lattice

$$(15.21) \quad W = 2m_1\omega + 2n_1\omega', \quad m_1, n_1 \in \mathbb{Z}.$$

The function f is invariant under translations by any lattice vector, $f(z + W) = f(z)$. Functions with this property are called elliptic functions. An elliptic function is a doubly periodic meromorphic function whose only singularities are poles. The order of an elliptic function is defined as the total number of its poles, counted with multiplicities, contained in a fundamental period parallelogram.

Proposition 4. *The derivative of an elliptic function is again an elliptic function.*

Proposition 5. *Let $f(z)$ be an elliptic function and let $\mathcal{P}$ be a fundamental period parallelogram. Then*

$$(15.22) \quad \int_{\partial\mathcal{P}} f(z) dz = 2\pi i \sum_j \operatorname{Res}_{z=z_j} f(z) = 0,$$

where the sum is taken over all poles z_j of f in $\mathcal{P}$.

As a consequence, a nonconstant elliptic function must have order at least 2.

The Weierstrass $\wp$ -function associated with a lattice is defined by

$$(15.23) \quad \wp(z) = \frac{1}{z^2} + \sum_{w \neq 0} \left(\frac{1}{(z-w)^2} - \frac{1}{w^2} \right),$$

where the sum extends over all nonzero lattice points w . It satisfies the differential equation

$$(15.24) \quad (\wp'(z))^2 = 4\wp^3(z) - g_2\wp(z) - g_3,$$

with invariants

$$(15.25) \quad g_2 = 60 \sum_{w \neq 0} w^{-4}, \quad g_3 = 140 \sum_{w \neq 0} w^{-6}.$$

The Jacobi theta functions satisfy the quasi-periodicity relations

$$(15.26) \quad \vartheta_2(z+1) = -\vartheta_2(z), \quad \vartheta_3(z+1) = \vartheta_3(z),$$

and

$$(15.27) \quad \vartheta_2(z+\tau) = q^{-1} e^{-2\pi i z} \vartheta_2(z), \quad \vartheta_3(z+\tau) = q^{-1} e^{-2\pi i z} \vartheta_3(z), \quad q = e^{i\pi\tau}.$$

They are even functions of the argument,

$$(15.28) \quad \vartheta_{2,3}(z) = \vartheta_{2,3}(-z).$$

The zeros of the Jacobi theta functions $\vartheta_k(z)$ are located at

$$(15.29) \quad z = m + m'\tau + V_k, \quad m, m' \in \mathbb{Z},$$

where the shifts V_k are given by

$$(15.30) \quad V_1 = 0, \quad V_2 = \frac{1}{2}, \quad V_3 = \frac{1}{2} + \frac{\tau}{2}, \quad V_4 = \frac{\tau}{2}.$$

To compare the two representations (14.13) and (14.20), introduce the elliptic function

$$(15.31) \quad f(z) = \frac{\vartheta_2\left(z, 2\tau_2^{(1)}\right)}{\vartheta_3\left(z, 2\tau_2^{(1)}\right)}.$$

By the quasi-periodicity and parity properties of the Jacobi theta functions, f is doubly periodic and even. In particular,

$$(15.32) \quad f(z+\omega) = f(z), \quad \omega = 2m_1 + 2n_1\tau_2^{(1)}, \quad f(z) = f(-z),$$

for all $m_1, n_1 \in \mathbb{Z}$. Moreover, $f(z)$ is an elliptic function of order 2, since $\vartheta_3\left(z, 2\tau_2^{(1)}\right)$ has exactly two zeros in a fundamental period parallelogram associated with $f(z)$. It therefore follows that the zeros and poles of $u(x, t)$ are precisely those described in (14.22) and (14.21).

On the other hand, the zeros and poles of the solution $u(x, t)$ given by (14.13) are determined by the corresponding properties of the Jacobi elliptic functions. More precisely, the zeros occur at

$$(15.33) \quad \alpha(x+vt) = \frac{\kappa^{(1)}K_2^{(1)}}{\pi}(x+vt) = \pm \frac{\tilde{v}_2K_2^{(1)}}{i\pi} + 2nK_2^{(1)} + 2imK_2^{(1)'}, \quad n, m \in \mathbb{Z},$$

while the poles are located at

$$(15.34) \quad \alpha(x+vt) = \frac{\kappa^{(1)}K_2^{(1)}}{\pi}(x+vt) = \pm \frac{\tilde{v}_1K_2^{(1)}}{i\pi} + 2nK_2^{(1)} + 2imK_2^{(1)'}, \quad n, m \in \mathbb{Z}.$$

Hence the solution $u(x, t)$ in (14.13) possesses the same zeros and poles, with the same multiplicities, as the representation (14.20). Since two elliptic functions of the same order that share identical zeros and poles can differ only by a multiplicative constant, evaluating the two expressions at a single convenient point suffices to conclude that (14.13) and (14.20) are identical.

References

- [1] Eugene D. Belokolos, Alexander I. Bobenko, Viktor Z. Enolskii, Alexander R. Its, and Vladimir B. Matveev. *Algebro-Geometric Approach to Nonlinear Integrable Equations*. Springer Series in Nonlinear Dynamics. Springer, 1994.
- [2] Marco Bertola. Riemann surfaces and theta functions. Lecture notes. URL: <https://mypage.concordia.ca/mathstat/bertola/ThetaCourse/ThetaCourse.pdf>.
- [3] Alexander Bobenko. Compact Riemann surfaces. Lecture notes. URL: <https://page.math.tu-berlin.de/~bobenko/Lehre/Skripte/RS.pdf>.
- [4] Fritz Gesztesy and Helge Holden. *Soliton Equations and Their Algebro-Geometric Solutions*, volume 1 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2003.
- [5] Liming Ling and Xuan Sun. Focusing mKdV equation: Two-phase solutions and their stability analysis. 2025. URL: <https://arxiv.org/abs/2510.12073>.
- [6] Barry Simon. *Basic Complex Analysis*. A Comprehensive Course in Analysis, Part 2A. American Mathematical Society, 2015.
- [7] Matthew Taylor. Notes on compact Riemann surfaces. Lecture notes. URL: <https://mtaylor.web.unc.edu/wp-content/uploads/sites/16915/2018/04/RSURF.pdf>.
- [8] Zhu-Xi Wang and Dun-Ren Guo. *Special Functions*. World Scientific, 1989.

SCHOOL OF MATHEMATICS, SOUTH CHINA UNIVERSITY OF TECHNOLOGY, GUANGZHOU
 Email address: zxdengmath@163.com